A Copula-Based Framework for Structural Causal Modelling:
Interventionist Representation and the Marginal Causal Effect

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Abstract

1 Motivation

A primary goal of causal inference is to estimate the **Average Causal Effect** (**ACE**), defined as the change in the expected outcome Y resulting from an intervention on the treatment X, after removing confounding by Z. In observational studies, the naive association between X and Y is typically biased; the ACE formalizes the causal contrast that would ideally be measured in a randomized experiment. Consequently, the proposed framework offers a more credible estimate of the impact of an intervention in observational settings, where randomized experiments are infeasible due to ethical or practical constraints. This provides valuable information for policymakers and scientific fields that rely on non-experimental data.

According to Pearl (2009): classical statistics is primarily concerned with the joint distribution of observed quantities—with estimation, association, and prediction—but remains silent on a fundamentally different class of questions: what would happen to an outcome if we were to forcibly set a cause to a particular value, regardless of how that cause is naturally determined in the population? This distinction is not merely philosophical. As the present work makes precise, the observational conditional expectation $\mathbb{E}[Y|X = x]$ conflates the direct effect of X on Y with spurious contributions arising from confounders Z that jointly influence both variables, so that naive association is not causation in any operational sense. The framework of Structural Causal Models addresses this by representing the data-generating process not as a single joint distribution but as an ordered system of structural mechanisms—a directed acyclic graph whose joint density factorizes according to autonomous causal modules, each encoding how a variable is determined by its direct causes (see Section 1.3).

Within this framework, causal reasoning requires a shift from *seeing* the world (conditioning on observed values) to *intervening* in it (Bareinboim et al., 2022): the do-operator $\text{do}(X = x)$ encodes a hypothetical experiment in which the natural mechanism determining X is surgically replaced by an external assignment, formally severing the arrow $Z \to X$ and yielding a distinct post-intervention density $f^{(X\downarrow)}_{X,Y,Z}$ (see Theorem 2.1). The local Average Causal Effect/Marginal Causal Effect—defined as the derivative of the interventional mean $\mu(x) = \mathbb{E}^{(X\downarrow)}[Y|X = x]$ with respect to x—is thus the quantity that would be identified in a randomized experiment, and the copula-based backdoor adjustment developed here provides its closed-form identification from purely observational data whenever a sufficient adjustment set can be justified on structural grounds.

Causal graphical models and copula theory provide two complementary frameworks for constructing multivariate probability distributions, hence copulas provide a natural mathematical language for representing such interventions and deriving the ACE. Structural causal models encode asymmetric relations through directed acyclic graphs and factorize joint densities into

conditional mechanisms (Pearl, 2009). Copula theory, on the other hand, decomposes joint distributions into marginal distributions and a dependence structure, allowing flexible modeling of multivariate dependence (Nelsen, 2006; Joe, 2014).

At first glance, these approaches appear unrelated. However, this chapter shows that copulas provide a natural and explicit representation of causal interventions. In particular, do-calculus operations correspond to precise algebraic transformations of copula densities. The results presented here synthesize and streamline the main findings of Chapter 5 of Justus (2024), embedding them into a unified copula-based framework that naturally extends to the multivariate setting studied later in this work.

1.1 Copula preliminaries

The definition of a 2-dimensional copula is presented below. Copulas are multivariate cumulative distribution functions that fully characterize the joint distribution of a random vector given only its marginal distributions and dependence structure (Nelsen, 2006; Joe, 2014)—providing the key mathematical tool for our analysis by separating these components.

Definition 1.1. *A 2-copula is a function* $C : [0,1]^2 \to [0,1]$ *with the following properties,*

i. $C(u,0) = C(0,v) = 0$, *and* $C(u,1) = u, C(1,v) = v$, $\forall u, v : u, v \in [0,1]$;

ii. for every $u_i, v_i \in [0,1], i = 1,2$, *such that* $u_1 \leq u_2$ *and* $v_1 \leq v_2$,

$$C(u_2, v_2) - C(u_2, v_1) - C(u_1, v_2) + C(u_1, v_1) \geq 0.$$

Every 2-copula is a bivariate cumulative distribution function on $[0,1]^2$ with standard-uniform marginals, so the copula encodes dependence structure independently of marginal behaviour.

Theorem 1.1 (Sklar's theorem (Sklar, 1959; Nelsen, 2006))**.** *Let* $H_{X,Y}$ *be a joint distribution function with continuous marginals* F_X *and* F_Y*. Then there exists a unique* 2*-copula* $C_{X,Y}$ *such that*

$$H_{X,Y}(x,y) \;=\; C_{X,Y}\big(F_X(x),\, F_Y(y)\big), \qquad (x,y) \in \mathbb{R}^2. \tag{1.1}$$

Conversely, for any copula C *and marginals* F*,* G*, the function* $H(x,y) = C(F(x), G(y))$ *defines a valid joint distribution.*

When $C_{X,Y}$ is absolutely continuous its mixed partial derivative $c_{X,Y}(u,v) = \partial^2 C_{X,Y}/\partial u\, \partial v$ exists, and the joint density factors as

$$f_{X,Y}(x,y) \;=\; c_{X,Y}\big(F_X(x),\, F_Y(y)\big)\, f_X(x)\, f_Y(y). \tag{1.2}$$

(For any 2-copula the partial derivative $\partial C/\partial u$ exists for almost all u, which underpins the h-functions used in the D-vine construction below; see Nelsen (2006, Theorem 2.2.7).) Equation (1.2) — separating marginal behaviour from rank-based dependence — is the algebraic identity on which Theorem 2.1 rests.

1.2 Technical (Global) assumptions

Throughout this paper we maintain the following regularity conditions:

R1. Absolute continuity The joint law of (X, Y, Z) admits measurable conditional CDFs $F_{Y|X}(\cdot|x)$, $F_{Z|X}(\cdot|x)$ and marginal F_Z. For almost every x, the conditional distributions $Y|X = x$ and $Z|X = x$ admit densities $f_{Y|X}(\cdot|x)$ and $f_{Z|X}(\cdot|x)$ (i.e., the conditional CDFs are absolutely continuous in their first argument). The mixed partial $c_{Y,Z|X}(u,w|x) = \frac{\partial^2 C_{Y,Z|X}(u,w|x)}{\partial u \partial w}$ exists almost everywhere.

R2. Copula regularity. For almost every x, the conditional copula $C_{Y,Z|X}(\cdot,\cdot|x)$ is Lipschitz continuous on $[0,1]^2$ with integrable partial derivatives. The copula density $c_{Y,Z|X}(u,w|x)$ satisfies $0 \le c_{Y,Z|X}(u,w|x) < \infty$ for almost every $(u,w) \in [0,1]^2$, and

$$\int_0^1 c_{Y,Z|X}(u,w|x)\,du = 1 \quad \text{for almost every } w \in [0,1].$$

R3. Differentiability for Marginal Causal Effect Pearl (2009). For almost every $(y,z) \in \mathbb{R}^2$, the map

$$x \longmapsto y\, f_{Y|Z}(y \mid z)\, f_{X|Z}(x \mid z)\, c_{X,Y|Z}\big(F_{X|Z}(x \mid z), F_{Y|Z}(y \mid z) \mid z\big)$$

is differentiable in x. Moreover, there exists a function $g \in L^1(\mathbb{R}^2)$ such that

$$\Big|\partial_x\big[y\, f_{Y|Z}(y \mid z)\, f_{X|Z}(x \mid z)\, c_{X,Y|Z}(F_{X|Z}(x \mid z), F_{Y|Z}(y \mid z) \mid z)\big]\Big| \le g(y,z)$$

for all x in a neighborhood of interest and for almost every (y,z).

R4. Interventional distribution regularity. For every x, the interventional conditional CDF $F^{(X\downarrow)}_{Y|X}(\cdot \mid x)$ is absolutely continuous in y with density $f^{(X\downarrow)}_{Y|X}(\cdot \mid x)$, and the mixed partial derivatives $\partial_x\partial_y F^{(X\downarrow)}_{Y|X}(y \mid x)$ and $\partial_y\partial_x F^{(X\downarrow)}_{Y|X}(y \mid x)$ exist and coincide a.e. in y. Moreover, $\partial_x F^{(X\downarrow)}_{Y|X}(y \mid x) \to 0$ as $y \to \pm\infty$.

1.3 Graphical Models and Copulas

Our object of study is the two graphs below, assuming that (X,Y,Z) is an absolutely continuous vector, whose density functions are denoted by $f_{X,Y,Z}$ and $f^{(X\downarrow)}_{X,Y,Z}$, respectively

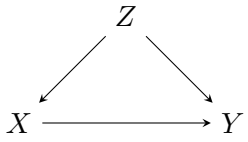


Figure 1: Confounding graph (observational)

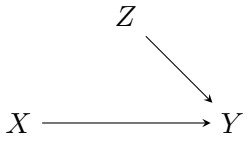


Figure 2: Interventionist graph (do(X))

Figure 3: (Left) Observational causal graph. (Right) Graph under do($X = x$) (edge $Z \to X$ removed), hence the same causal structure but under a intervention fixing $X = x$.

Let

$$F_{Y|X}(y \mid x) = \Pr(Y \le y \mid X = x), \qquad F_{Z|X}(z \mid x) = \Pr(Z \le z \mid X = x),$$

and let F_Z denote the marginal distribution of Z. When densities are invoked we assume they exist almost everywhere (see Assumption R1 from Section 7.7).

In Figure 1 we consider the standard confounding structure where X denotes a treatment, Y an outcome, and Z a confounder. The associated joint density factorizes as

$$f_{X,Y,Z}(x,y,z) = f_{Y|X,Z}(y \mid x,z)\, f_{X|Z}(x \mid z)\, f_Z(z). \tag{1.3}$$

The intervention do($X = x$) (here denoted as $(X\downarrow)$ removes the arrow $Z \to X$, leading to the post-intervention density, represented in Figure 2

$$f^{(X\downarrow)}_{X,Y,Z}(x,y,z) = f_{Y|X,Z}(y \mid x,z)\, f_X(x)\, f_Z(z). \tag{1.4}$$

Modeling premise: modularity. Let $G = (V, E)$ be a directed acyclic graph (DAG) with nodes $V = \{1, \dots, d\}$. For each $i \in V$, define the set of parents of node i as

$$\mathrm{Pa}_i := \{j \in V : (j, i) \in E\}.$$

We assume that the joint distribution of $X = (X_1, \dots, X_d)$ factorizes according to G as

$$P(X) = \prod_{i=1}^{d} P\big(X_i \mid X_{\mathrm{Pa}_i}\big), \tag{1.5}$$

where $X_{\mathrm{Pa}_i} := (X_j)_{j \in \mathrm{Pa}_i}$. Each conditional distribution $P(X_i \mid X_{\mathrm{Pa}_i})$ is called a *causal mechanism* (cf. Pearl (2009); Peters et al. (2017)).

Principle 1 (Independent mechanisms)**.** *For any intervention* $\mathrm{do}(X_j := x)$*, the post-intervention distribution satisfies*

$$P^{(X_j\downarrow)}(X_i \mid X_{\mathrm{Pa}_i}) = P(X_i \mid X_{\mathrm{Pa}_i}), \qquad \textit{for all } i \neq j,$$

while the mechanism $P(X_j \mid X_{\mathrm{Pa}_j})$ *may change arbitrarily.*

Or, informally, as stated in Peters et al. (2017, p. 19), the causal generative process of a system's variables is composed of autonomous modules that do not inform or influence each other. In the probabilistic case, this means that the conditional distribution of each variable given its causes (its *mechanism*) does not inform or influence the other conditional distributions.

Thus, modularity means that interventions act locally on a single conditional distribution, leaving the remaining components of the factorization unchanged.

2 Copula representation of causal interventions

Theorem 2.1 (Copula representation of $\mathrm{do}(X)$)**.** *Assume the global regularity conditions R1–R4 hold for* (X, Y, Z)*. In particular (R1–R2),* (X, Y, Z) *is an absolutely continuous random vector with joint density* $f_{X,Y,Z}$*, and the copula* $C_{X,Z}$ *is absolutely continuous with density* $c_{X,Z}(F_X(x), F_Z(z)) > 0$ *almost everywhere. Then the joint density under the intervention* $\mathrm{do}(X)$[1] *is given by*

$$f^{(X\downarrow)}_{X,Y,Z}(x, y, z) = \frac{f_{X,Y,Z}(x, y, z)}{c_{X,Z}(F_X(x), F_Z(z))}. \tag{2.1}$$

Proof. By R1 (absolute continuity of the joint law), from (1.3) the observational factorization holds:

$$f_{X,Y,Z}(x, y, z) = f_{Y|X,Z}(y \mid x, z)\, f_{X,Z}(x, z),$$

where densities $f_{X,Y,Z}$, $f_{Y|X,Z}$, and $f_{X,Z}$ exist a.e., with measurable CDFs F_X, F_Z.

By Sklar's theorem (justified by R1: (X, Z) absolutely continuous),

$$f_{X,Z}(x, z) = c_{X,Z}(F_X(x), F_Z(z))\, f_X(x)\, f_Z(z).$$

Here, R2 (copula regularity) guarantees $C_{X,Z}$ admits density $c_{X,Z}$, and $c_{X,Z} > 0$ a.e. ensures positivity/overlap ($f_{X|Z} > 0$).

[1] We denote the intervention on X by $(X\downarrow)$ to emphasize the surgical removal of the incoming causal influences on X in the sense of structural causal models. This notation highlights that the intervention operates at the level of the data-generating mechanism for X, severing the dependence between X and its parents Z. In contrast, the more common notation $\mathrm{do}(X = x)$ refers to the deterministic intervention that sets X to the constant value x. The representation derived in this theorem corresponds to the former type of intervention—removal of the structural mechanism linking Z to X—from which the standard post-intervention quantities $f_{Y|\mathrm{do}(X=x)}$ can be obtained by conditioning on $X = x$. Hence, the intervention $\mathrm{do}(X = x)$ replaces the structural equation for X with $X = x$, yielding a new post-intervention distribution $P(Y, Z \mid \mathrm{do}(X = x))$.

Substituting yields the interventional factorization (modularity invariance of $f_{Y|X,Z}$):

$$f^{(X\downarrow)}_{X,Y,Z}(x,y,z) = f_{Y|X,Z}(y \mid x,z)\, f_X(x)\, f_Z(z) = \frac{f_{X,Y,Z}(x,y,z)}{c_{X,Z}(F_X(x), F_Z(z))}.$$

which proves the result.[2] □

Interpretation. Equation (2.1) shows that the intervention do(X) acts by removing the dependence between X and Z, quantified exactly by the copula density $c_{X,Z}$. If X and Z are independent, then $c_{X,Z} \equiv 1$ and the intervention leaves the joint distribution unchanged.

Importantly, the copula representation of the post-intervention density should not be interpreted as defining the intervention through a purely statistical manipulation of the observational distribution. Rather, the identity in Equation (2.1) follows from the semantics of structural causal models together with the modularity (or invariance) assumption that underlies the definition of the do-operator. In the SCM framework, the intervention do($X = x$) is defined by the surgical replacement of the structural equation determining X with the constant assignment x, while leaving all other structural mechanisms unchanged. Under this intervention, the influence of the parents Z on X is removed, while the mechanism generating Y given (X, Z) remains invariant. The division by the copula density $c_{X,Z}(F_X(x), F_Z(z))$ therefore provides an algebraic representation—expressed in copula coordinates—of this surgical operation, removing the dependence between X and its parents while preserving the remaining structural relations. Consequently, the copula-based expression should be understood as an analytical representation of the intervention implied by the underlying causal model, rather than as an independent statistical definition of the do-operator. This interpretation is consistent with the manipulation-neutral semantics of interventions emphasized by Pearl (2018), where causal effects are defined through hypothetical structural modifications of the data-generating process rather than through the specific means by which a variable may be manipulated.

3 Factorized representations

Vine-copula representation. For an absolutely continuous triple (X, Y, Z) the joint density admits the usual pair-copula (vine) factorization into univariate marginals and 2-copulas. In particular one may write

$$\begin{aligned} f_{X,Y,Z}(x,y,z) =& c_{X,Y|Z}(F_{X|Z}(x|z), F_{Y|Z}(y|z)|z) \times c_{X,Z}(F_X(x), F_Z(z)) \times c_{Y,Z}(F_Y(y), F_Z(z)) \\ & \times f_X(x) f_Y(y) f_Z(z); \end{aligned} \tag{3.1}$$

$$\begin{aligned} f_{X,Y,Z}(x,y,z) =& c_{Y,Z|X}(F_{Y|X}(y|x), F_{Z|X}(z|x)|x) \times c_{X,Y}(F_X(x), F_Y(y)) \times c_{X,Z}(F_X(x), F_Z(z)) \\ & \times f_X(x) f_Y(y) f_Z(z); \end{aligned} \tag{3.2}$$

and analogous representations are obtained by permuting the variable order. These factorizations express the full trivariate dependence in terms of (conditional) bivariate copulas and marginals and form the basis for higher-dimensional vine constructions; see Czado and Nagler (2022) for a systematic treatment.

Remark 3.1. *Remark.* In many applied settings the *simplifying assumption* is invoked, namely that conditional pair-copulas are invariant with respect to the conditioning variable (e.g. $c_{X,Y|Z}(\cdot,\cdot \mid z) \equiv c_{X,Y|Z}(\cdot,\cdot)$). This is a modeling restriction introduced to reduce dimensionality and facilitate estimation in vine constructions (Czado and Nagler, 2022). By contrast, Assumption R2 is

[2] R3–R4 are not required here but enable differentiation for quantities like MCE(x)

a regularity condition ensuring existence and integrability of the relevant copula densities and derivatives; it imposes no restriction on the dependence of conditional copulas on the conditioning value. All results in this paper therefore hold without invoking the simplifying assumption.

Corollary 3.1 (Vine representations of the interventional law)**.** *Under the assumptions of Theorem 2.1, the interventional joint density of* (X, Y) *admits the equivalent representations*

$$f_{X,Y}^{(X^{\downarrow})}(x,y) = f_X(x) f_Y(y) \int_{-\infty}^{\infty} c_{X,Y|Z}(F_{X|Z}(x|z), F_{Y|Z}(y|z)|z) \times c_{Y,Z}(F_Y(y), F_Z(z)) \times f_Z(z) dz; \tag{3.3}$$

$$f_{X,Y}^{(X^{\downarrow})}(x,y) = f_{X,Y}(x,y) \int_{-\infty}^{\infty} c_{Y,Z|X}(F_{Y|X}(y|x), F_{Z|X}(z|x)|x) f_Z(z) dz. \tag{3.4}$$

Proof. Apply Theorem 2.1 to standard vine copula factorizations of $f_{X,Y,Z}$ (Equations (3.1) and (3.2)) and integrate out the confounder Z. The resulting expressions yield (3.3) and (3.4). □

Causal interpretation. Equation (3.3) decomposes the interventional density into a direct deviation from independence—measured by the conditional copula $C_{X,Y|Z}$—whereas Equation (3.4) quantifies the deviation from the observational X-Y association via $C_{Y,Z|X}$.

Since the conditional copula representation in (3.3) offers greater interpretability for causal practitioners—explicitly isolating the direct treatment-outcome dependence after conditioning on Z—all subsequent derivations employ this vine factorization. The two forms are mathematically equivalent by construction (Corollary 3.1), yielding identical results.

4 Causal target: The interventional mean and the Marginal Causal Effect

Having established the copula representation of interventions, we now turn to the central causal quantities of interest. The fundamental problem in causal inference is that association is not causation: in observational data, the conditional expectation $\mathbb{E}[Y|X = x]$ conflates the direct effect of X on Y with confounding by Z.

To isolate the causal effect, we ask a different question: *What would happen if we could set* $X = x$ *for all units, regardless of their natural value of* Z*?* This hypothetical operation is denoted by the *intervention* $\mathrm{do}(X = x)$ (Pearl, 2009). The corresponding **interventional mean**

$$\mu(x) = \mathbb{E}^{(X^{\downarrow})}[Y|X = x]$$

averages the outcome over the marginal distribution of Z, effectively breaking the confounding link $Z \to X$. In a randomized experiment, where X is assigned independently of Z, we have $\mathbb{E}^{(X^{\downarrow})}[Y|X = x] = \mathbb{E}[Y|X = x]$; in observational settings, the two generally differ.

The **Average Causal Effect (ACE)** quantifies how $\mu(x)$ changes with the treatment level. For a discrete contrast between two levels x_1 and x_2, it is defined as

$$\mathrm{ACE}(x_1, x_2) = \mu(x_1) - \mu(x_2).$$

For a continuous treatment, we define the incremental causal effect (or marginal causal effect) as

$$\mathrm{MCE}(x) = \frac{d}{dx}\mu(x),$$

which measures the instantaneous rate of change in the expected outcome under an intervention on X. We avoid referring to $\mu'(x)$ as an "average causal effect" since it depends on the

chosen scale of X. This formulation is particularly valuable for policy and business applications, where the *marginal impact* of an intervention informs cost-benefit analysis and optimal resource allocation (Heckman and Vytlacil, 2005; Boardman, 2011). When X and Z are independent, $\mathrm{MCE}(x)$ coincides with the derivative of the observational regression curve; otherwise, confounding induces a discrepancy that the copula representation explicitly corrects.

Traditional methods for estimating ACE (or MCE), such as regression adjustment, stratification, or propensity-score weighting (Imbens and Rubin, 2015; Rosenbaum and Rubin, 2023), implicitly attempt to reconstruct the interventional distribution by adjusting for Z. The copula-based approach presented here provides a unified, dependence-aware representation of standard backdoor-adjusted causal quantities, yielding closed-form expressions for $\mu(x)$ and $\mathrm{MCE}(x)$ in tractable copula families (e.g. Gaussian), and facilitating their evaluation—analytically where possible and numerically in more general non-Gaussian and multivariate settings.

In the remainder of this section, we derive these expressions using the copula machinery developed above.

4.1 Copula representation: From the interventional density to the MCE

In observational studies, as illustrated in Figure 1, the confounder Z induces a spurious association between treatment X and outcome Y. To identify the causal effect of X on Y, it is essential to eliminate this confounding. This can be achieved through external intervention on X, as depicted in Figure 2, which breaks the confounding link by setting $X = x$ exogenously.

In observational studies, as illustrated in Figure 1, the confounder Z induces a spurious association between treatment X and outcome Y. To identify the causal effect of X on Y, it is essential to eliminate this confounding, which can be achieved through an external intervention on X that breaks the $Z \to X$ link by setting $X = x$ exogenously. Much applied work still reports scalar causal summaries such as average treatment effects or marginal treatment effects (Imbens and Rubin, 2015; Cunningham, 2021). At the same time, there is a substantial econometric literature on *distributional* and *quantile* treatment effects, including models for the full distributions of potential outcomes and effects on conditional and unconditional quantiles (e.g. Chernozhukov and Hansen, 2005, 2013; Abbring and Heckman, 2007; Blundell et al., 2007; Firpo et al., 2009). Our contribution is complementary to this body of work: the copula-based framework developed here provides explicit closed-form expressions for the entire interventional distribution $F_{Y|X}^{(X\downarrow)}(y \mid x)$ generated by $\mathrm{do}(X = x)$ for continuous treatments, from which scalar, quantile, and other distributional estimands can be obtained as functionals.

The derivation follows a structured path. First, the interventional cumulative distribution function $F_{Y|X}^{(X\downarrow)}(y|x)$ is obtained. This function provides a complete characterization of the outcome under intervention and serves as the foundation for all subsequent causal quantities. Second, the interventional mean $\mu(x)$ is derived from this CDF via its standard tail-integral representation. Third, the incremental causal effect is defined as the derivative of the interventional mean ($\mathrm{MCE}(x)$), quantifying the instantaneous rate of change in the expected outcome with respect to the treatment level.

This sequence ensures the interventional CDF remains the primary object. Any other causal estimand can be computed directly from it. Thus, the copula formulation supports a complete distributional causal inference, generalizing beyond the average effect.

Having motivated the central role of the interventional CDF, we now provide its formal representation within the copula framework. Let $f_{Y|X}^{(X\downarrow)}(\cdot|x)$ denote the interventional conditional density of Y under $\mathrm{do}(X = x)$.

$$f_{Y|X}^{(X\downarrow)}(y; x) = f_Y(y) \int_{-\infty}^{\infty} c_{X,Y|Z}(F_{X|Z}(x|z), F_{Y|Z}(y|z) \mid z)\, c_{Y,Z}(F_Y(y), F_Z(z))\, f_Z(z)\, dz. \quad (4.1)$$

By definition, the interventional conditional CDF of Y given $X = x$ under the intervention that

removes the arrow $Z \to X$ is

$$F_{Y|X}^{(X^{\downarrow})}(y \mid x) = \int_{-\infty}^{y} f_{Y|X}^{(X^{\downarrow})}(t \mid x)\, dt. \tag{4.2}$$

Using the copula-based representation of the interventional joint density from Corollary 3.1 and the definition of conditional density, one obtains a double-integral expression for $F_{Y|X}^{(X^{\downarrow})}$ in terms of f_Y, f_Z and the pair-copulas involving X, Y, Z. The next proposition shows that this expression admits a closed-form representation in terms of the conditional copula $C_{X,Y|Z}$. Recall that, under Assumptions R1–R2, the conditional distribution of (X, Y) given $Z = z$ admits a copula representation

$$F_{X,Y|Z}(x, y \mid z) = C_{X,Y|Z}\big(F_{X|Z}(x \mid z),\, F_{Y|Z}(y \mid z) \mid z\big)\,,$$

with absolutely continuous marginals in the first argument and a conditional copula $C_{X,Y|Z}(\cdot, \cdot \mid z)$ that is Lipschitz with integrable partial derivatives. By Sklar's theorem, this implies the existence of a conditional copula density $c_{X,Y|Z}(u, v \mid z)$ and of the partial derivative $\partial_u C_{X,Y|Z}(u, v \mid z) = \frac{\partial}{\partial u} C_{X,Y|Z}(u, v \mid z)$ for almost every z, which justifies the use of Fubini's theorem and of the change of variables based on $F_{Y|Z}(\cdot \mid z)$ in the proof below.

Proposition 4.1 (Interventional CDF via copula representation)**.** *Under Assumptions R1–R2, and assuming that the conditional law $f_{Y|X,Z}$ is invariant to the intervention that removes the arrow $Z \to X$, the interventional CDF after* $\mathrm{do}(X = x)$ *is*

$$F_{Y|X}^{(X^{\downarrow})}(y \mid x) = \int_{-\infty}^{\infty} \partial_u C_{X,Y|Z}\big(F_{X|Z}(x \mid z),\, F_{Y|Z}(y \mid z) \mid z\big)\; f_Z(z)\, dz, \tag{4.3}$$

where

$$\partial_u C_{X,Y|Z}(u, v \mid z) \;:=\; \frac{\partial}{\partial u} C_{X,Y|Z}(u, v \mid z)\Big|_{v = F_{Y|Z}(y|z)}$$

denotes the partial derivative of the conditional copula $C_{X,Y|Z}(\cdot, \cdot \mid z)$ with respect to its first argument.

Proof. Under the intervention that severs the edge $Z \to X$, the structural equation for X is replaced by the constant assignment $X = x$, while the mechanisms generating Z and Y given (X, Z) remain unchanged (modularity of SCMs). By the truncated factorization, the interventional conditional density of Y given $X = x$ can therefore be written as

$$f_{Y|X}^{(X^{\downarrow})}(y \mid x) = \int_{-\infty}^{\infty} f_{Y|X,Z}(y \mid x, z)\; f_{Z|X}^{(X^{\downarrow})}(z \mid x)\, dz.$$

Because the intervention does not change the marginal distribution of Z, and X is independent of Z under the intervention, we have $f_{Z|X}^{(X^{\downarrow})}(z \mid x) = f_Z(z)$. Hence

$$f_{Y|X}^{(X^{\downarrow})}(y \mid x) = \int_{-\infty}^{\infty} f_{Y|X,Z}(y \mid x, z)\; f_Z(z)\, dz.$$

Integrating both sides in y and using Fubini's theorem (justified by Assumption R2),

$$F_{Y|X}^{(X^{\downarrow})}(y \mid x) = \int_{-\infty}^{\infty} \left(\int_{-\infty}^{y} f_{Y|X,Z}(t \mid x, z)\, dt \right) f_Z(z)\, dz = \int_{-\infty}^{\infty} F_{Y|X,Z}(y \mid x, z)\; f_Z(z)\, dz. \tag{4.4}$$

It remains to express $F_{Y|X,Z}(y \mid x, z)$ via the conditional copula. By Sklar's theorem and Assumptions R1–R2, the conditional density of (X, Y) given $Z = z$ admits the representation

$$f_{X,Y|Z}(x, t \mid z) = c_{X,Y|Z}\big(F_{X|Z}(x \mid z), F_{Y|Z}(t \mid z) \mid z\big)\; f_{X|Z}(x \mid z)\, f_{Y|Z}(t \mid z),$$

so that

$$f_{Y|X,Z}(t \mid x, z) = \frac{f_{X,Y|Z}(x, t \mid z)}{f_{X|Z}(x \mid z)} = c_{X,Y|Z}\big(F_{X|Z}(x \mid z), F_{Y|Z}(t \mid z) \mid z\big)\ f_{Y|Z}(t \mid z).$$

With the change of variables $v = F_{Y|Z}(t \mid z)$ (so $dv = f_{Y|Z}(t \mid z)\, dt$), we obtain

$$\begin{aligned} F_{Y|X,Z}(y \mid x, z) &= \int_{-\infty}^{y} f_{Y|X,Z}(t \mid x, z)\, dt \\ &= \int_{0}^{F_{Y|Z}(y|z)} c_{X,Y|Z}\big(F_{X|Z}(x \mid z),\, v \mid z\big)\ dv. \end{aligned}$$

By the relation between a copula and its partial derivative,

$$\int_0^v c_{X,Y|Z}(u, s \mid z)\, ds = \frac{\partial}{\partial u} C_{X,Y|Z}(u, v \mid z),$$

so, setting $u = F_{X|Z}(x \mid z)$ and $v = F_{Y|Z}(y \mid z)$,

$$F_{Y|X,Z}(y \mid x, z) = \partial_u C_{X,Y|Z}\big(F_{X|Z}(x \mid z),\, F_{Y|Z}(y \mid z) \mid z\big)\,.$$

Substituting this identity into (4.4) yields (4.3), which completes the proof. □

Remark 4.1 (Backdoor Adjustment Interpretation of the Interventional Density). The reformulation of the interventional conditional density in Equation (4.1),

$$f_{Y|X}^{(X\downarrow)}(y; x) = f_Y(y) \int_{-\infty}^{\infty} c_{X,Y|Z}(F_{X|Z}(x|z), F_{Y|Z}(y|z) \mid z)\, c_{Y,Z}(F_Y(y), F_Z(z))\, f_Z(z)\, dz,$$

provides an explicit copula-theoretic articulation of Pearl's *backdoor adjustment principle* (Pearl, 2009), the foundational identification strategy for causal effects under confounding.

The interventional CDF derived in Proposition 4.1 and the standard observational conditional CDF are related by a single copula factor. The next results make that relationship precise, obtaining expressions for the observational versions of the previously deduced results: (i) interventional CDF (Proposition 4.1), (ii) interventional mean and MCE (Corollary 4.2), and (iii) observational CDF (Corollary 4.7 below).

Remark 4.2. It is a direct consequence from vine copula factorization (3.1) that:

$$f_{Y|X}(y|x) = f_Y(y) \int_{-\infty}^{\infty} c_{X,Y|Z}(F_{X|Z}(x|z), F_{Y|Z}(y|z)|z) \times c_{X,Z}(F_X(x), F_Z(z)) \times c_{Y,Z}(F_Y(y), F_Z(z)) \tag{4.5}$$

$$\times\ f_Z(z)\, dz. \tag{4.6}$$

Corollary 4.1. *[Non-Interventional CDF] Under assumptions and notation of Proposition 4.1,*

$$F_{Y|X}(y \mid x) = \int_{-\infty}^{\infty} \partial_u C_{X,Y|Z}\big(F_{X|Z}(x \mid z),\, F_{Y|Z}(y \mid z)\big)\ c_{X,Z}(F_X(x), F_Z(z)) f_Z(z)\, dz. \tag{4.7}$$

Proof. Note that

$$\begin{aligned} f_{Y|X}(y|x) &= \int_{-\infty}^{\infty} f_{Y,Z|X}(y, z|x)\, dz \\ &= \int_{-\infty}^{\infty} \frac{f_{Y,Z|X}(y, z|x)}{f_{Z|X}(z|x)} f_{Z|X}(z|x)\, dz \\ &= \int_{-\infty}^{\infty} f_{Y|X,Z}(y|x, z) f_{Z|X}(z|x)\, dz. \end{aligned}$$

Therefore, by the same steps as in proof of Proposition 4.1,

$$F_{Y|X}(y|x) = \int_{-\infty}^{\infty} F_{Y|X,Z}(y|x,z) f_{Z|X}(z|x)\, dz.$$

In proof of Proposition 4.1, we showed that

$$F_{Y|X,Z}(y \mid x,z) = \partial_u C_{X,Y|Z}\left(F_{X|Z}(x \mid z),\, F_{Y|Z}(y \mid z) \mid z\right).$$

Taking into account that $f_{Z|X}(z|x) = c_{X,Z}(F_X(x), F_Z(z)) f_Z(z)$, we obtain

$$F_{Y|X}(y \mid x) = \int_{-\infty}^{\infty} \partial_u C_{X,Y|Z}\big(F_{X|Z}(x \mid z),\, F_{Y|Z}(y \mid z) \mid z\big)\; c_{X,Z}(F_X(x), F_Z(z)) f_Z(z)\, dz, \quad (4.8)$$

as we wished to prove. □

Remark 4.3. Note that, if we set $c_{X,Z}(F_X(x), F_Z(z)) = 1$ in Equations 4.5 and 4.7, we obtain, respectively, the conditional density induced by Equation 4.1 and Equation 4.3. Therefore, $F^{(X\downarrow)}_{Y|X}(y \mid x) = F_{Y|X}(y \mid x)$ and $f^{(X\downarrow)}_{Y|X}(y \mid x) = f_{Y|X}(y \mid x)$.

Remark 4.4. Expressions for the conditional expectation and MCE are obtained in the same way as Corollary 4.2, replacing $f^{(X\downarrow)}_{Y|X}(y \mid x)$ by $f_{Y|X}(y \mid x)$ and replacing $F^{(X\downarrow)}_{Y|X}(y \mid x)$ by $F_{Y|X}(y \mid x)$.

Proposition 4.1 and Corollary 4.1 allow to see the difference between $F^{(X\downarrow)}_{Y|X}(y \mid x)$ and $F_{Y|X}(y \mid x)$, which is characterized by the copula density of (X, Z). This makes clear the central role of the dependence between the confounder and the treatment.

Connection to Pearl's Backdoor Formula. When Z satisfies the backdoor criterion relative to (X, Y), Pearl's identification formula states

$$\Pr(Y = y \mid \mathrm{do}(X = x)) = \int_z \Pr(Y = y \mid X = x, Z = z)\, \Pr(Z = z)\, dz.$$

This integrates the conditional treatment effect over the marginal distribution of the confounder, thereby removing confounding. The reformulation (4.1) translates this logic into dependence components: each factor in the integrand carries distinct causal meaning.

Decomposition of Causal Components. The integrand decomposes into four elements:

1. **Conditional copula** $c_{X,Y|Z}(F_{X|Z}(x|z), F_{Y|Z}(y|z) \mid z)$: Captures the conditional dependence between treatment and outcome given the confounder. After stratifying on Z, this copula represents the dependence between X and Y with backdoor paths blocked, in accordance with the backdoor criterion (Pearl, 2009, Ch. 5).

2. **Marginal copula** $c_{Y,Z}(F_Y(y), F_Z(z))$: Encodes the observational dependence between Y and Z. Its appearance arises from the copula factorization of the joint density, but it does not play a causal role in the backdoor adjustment itself, which depends only on $f_{Y|X,Z}$ and f_Z.

3. **Marginal density** $f_Z(z)$: The intervention $\mathrm{do}(X = x)$ severs only the edge $Z \to X$, leaving Z untouched (Figure 2). Thus Z retains its observational marginal distribution post-intervention.

4. **Fronting marginal** $f_Y(y)$: Ensures dimensional consistency and proper marginal scaling of the interventional conditional density.

Identifiability and Observability. All components are identifiable[3] from observational data: the conditional copula $c_{X,Y|Z}$ is observable via Sklar's theorem applied to the conditional distribution of $(X, Y) \mid Z = z$, and the marginal copula and distributions are directly observable. No intervention on Z is required, confirming that the backdoor criterion is sufficient for causal identification (Pearl, 2009, Th. 3.3.2). The formula thus demonstrates explicitly how do-calculus corresponds to reweighting observational conditional distributions to account for confounding removal.

Positivity, Modularity, and Causal Scope Causal effects in this work are defined within the framework of Structural Causal Models (SCMs), formalized as ordered probabilistic factorizations of the joint density $f_{X,Y,Z}$. Modularity (or autonomy) means that the conditional densities generating each variable—in particular, $f_{Y|X,Z}(y \mid x, z)$—remain invariant under interventions that replace the mechanism of X. Under $\mathrm{do}(X = x)$, the density generating X is surgically replaced by a Dirac delta at x, while all other conditional densities remain unchanged. The post-interventional joint density is thus obtained via the *truncated factorization*:

$$f^{(X\downarrow)}_{X,Y,Z}(x, y, z) = f_{Y|X,Z}(y \mid x, z) \cdot f_Z(z),$$

where the conditional density f_X has been deleted (truncated). This reflects the preservation of all non-intervened mechanisms under intervention.

A further assumption required for the validity of the copula-based intervention is **positivity** (or **overlap**). The interventional transformation involves division by the copula density $c_{X,Z}(u, v)$, which implicitly requires this quantity to be strictly positive almost everywhere. In causal inference, positivity guarantees that the identification formula defining the causal effect is well-defined, ensuring that denominators appearing in adjustment or transformation formulas do not vanish on regions of observational support. Formally, identification requires conditions of the form

$$f_Z(z) > 0 \;\Rightarrow\; f_{X|Z}(x \mid z) > 0 \quad \text{for all } x \text{ in the support of } X.$$

so that all treatment levels occur within relevant covariate strata. When this condition fails, the causal estimand itself becomes undefined rather than merely difficult to estimate. Recent work emphasizes that overlap should be understood primarily as a requirement for the *well-definedness of identification formulas*, particularly when causal quantities involve ratios or conditional densities (Hwang et al., 2024).

In the copula representation, Sklar's theorem implies

$$f_{X|Z}(x \mid z) = c_{X,Z}(F_X(x), F_Z(z))\; f_X(x),$$

provided that $f_X(x) > 0$, positivity of $f_{X|Z}(x|z)$ is equivalent to

$$c_{X,Z}(F_X(x), F_Z(z)) > 0 \quad \text{almost everywhere.}$$

Violations of overlap therefore correspond geometrically to regions of the copula domain where joint ranks of treatment and confounders are absent. In such regions, the copula ratio defining the intervention becomes undefined, directly paralleling the instability or failure of adjustment and inverse-probability-weighting estimators under lack of overlap. Positivity in the copula framework is thus not an additional assumption, but the continuous-density analogue of

[3]Lewbel (2019) defines point identification (the standard notion of "identification") as the parameter θ (or set of parameters) being uniquely determined from the observable population features P under the model M: no two distinct θ, θ' in the parameter space are "observationally equivalent" (i.e., there do not exist model realizations m, m' in M yielding the same P). In causal adjustment, the backdoor criterion ensures identifiability of $P(Y_{do(X=x)})$ as the adjustment functional $\sum_z P(Y|X = x, Z = z)P(Z = z)$—uniquely recovering the interventional mean (or distribution) from observables under conditional independence given the adjustment set Z (Pearl, 2009; Pearl et al., 2016).

the standard identification requirement in structural causal models (Imbens and Rubin, 2015; Hernán and Robins, 2020).

Under these assumptions, conditioning on $X = x$ in the stochastic intervention $(X \downarrow)$ yields

$$\mathbb{E}^{(X\downarrow)}[Y \mid X = x] = \mathbb{E}[Y \mid \mathrm{do}(X = x)],$$

since both are given by the same truncated factorization.

Finally, the theoretical results presented here are developed under an assumption of absolute continuity of the involved variables. The use of copula densities and derivative-based identities therefore applies directly to continuous treatment and outcome variables. Extensions to discrete or mixed settings require alternative constructions (e.g., subcopulas or measure-theoretic formulations) and lie beyond the scope of the present analysis.

It follows that under the hypotheses of Proposition 4.1, if $\mathbb{E}[|Y|] < \infty$, the interventional mean (causal mean) is well-defined and given by

$$\mu(x) = \mathbb{E}^{(X\downarrow)}[Y \mid X = x] = \int_{-\infty}^{\infty} y\, f_{Y|X}^{(X\downarrow)}(y|x)\, dy. \tag{4.9}$$

where $f_{Y|X}^{(X\downarrow)}(y|x)$ is the interventional density from (4.1).

Corollary 4.2 (Marginal Causal Effect (MCE))**.** *Under Assumptions R1–R4 and* $\mathbb{E}^{(X\downarrow)}[\,|Y|\mid X = x\,] < \infty$ *for every* x*, the Marginal Causal Effect*

$$\mathrm{MCE}(x) := \frac{d}{dx}\mathbb{E}^{(X\downarrow)}[Y \mid X = x] = \frac{d}{dx}\mu(x)$$

exists and is given by

$$\mathrm{MCE}(x) = -\int_{-\infty}^{\infty} \frac{\partial}{\partial x} F_{Y|X}^{(X\downarrow)}(y \mid x)\, dy.$$

Proof. Write $H(y, x) := F_{Y|X}^{(X\downarrow)}(y \mid x)$ and $h(y, x) := f_{Y|X}^{(X\downarrow)}(y \mid x)$ for brevity.

Step 1: Lebesgue representation of $\mu(x)$. By Equation (4.9) and the absolute continuity in y assumed in R4, $H(\cdot, x)$ is absolutely continuous for every x, with

$$h(y, x) = \frac{\partial}{\partial y} H(y, x) \quad \text{for a.e. } y.$$

Since $\mathbb{E}^{(X\downarrow)}[\,|Y|\mid X = x\,] < \infty$, the interventional mean admits the Lebesgue integral representation

$$\mu(x) = \mathbb{E}^{(X\downarrow)}[Y \mid X = x] = \int_{\mathbb{R}} y\, h(y, x)\, dy. \tag{4.10}$$

Step 2: Differentiation under the integral sign. From the copula representation of $f_{Y|X}^{(X\downarrow)}$ and Assumption R3, for almost every y the map $x \mapsto f(y, x)$ is differentiable and there exists $g \in L^1(\mathbb{R})$ such that

$$\big|y\, \frac{d}{dx} h(y, x)\big| \le g(y) \quad \text{for all } x \text{ and a.e. } y. \tag{4.11}$$

Thus the conditions of the Leibniz integral rule (differentiation under the integral sign) are satisfied, and we may differentiate (4.10) with respect to x inside the integral:

$$\frac{d}{dx}\mu(x) = \frac{d}{dx}\int_{\mathbb{R}} y\, h(y, x)\, dy = \int_{\mathbb{R}} y\, \frac{d}{dx} h(y, x)\, dy. \tag{4.12}$$

Step 3: Integration by parts in y. By R4, the mixed partial derivatives $\partial_x \partial_y H$ and $\partial_y \partial_x H$ exist and coincide a.e. in y, so

$$\partial_x h(y, x) = \partial_x \partial_y H(y, x) = \partial_y \partial_x H(y, x) \quad \text{for a.e. } y. \tag{4.13}$$

Define $k(y,x) := \partial_x H(y,x)$. Then $\partial_x h(y,x) = \partial_y k(y,x)$ a.e., and (4.12) becomes

$$\frac{d}{dx}\mu(x) = \int_{\mathbb{R}} y\, \partial_y k(y,x)\, dy. \tag{4.14}$$

We now integrate by parts in y over $(-\infty, \infty)$:

$$\int_{\mathbb{R}} y\, \partial_y k(y,x)\, dy = \Big[y\, k(y,x)\Big]_{y=-\infty}^{y=+\infty} - \int_{\mathbb{R}} 1 \cdot k(y,x)\, dy. \tag{4.15}$$

By R4 we have $k(y,x) = \partial_x H(y,x) \to 0$ as $|y| \to \infty$ for each fixed x, and $k(\cdot, x)$ is integrable in y. Hence

$$\lim_{|y|\to\infty} y\, k(y,x) = 0,$$

so the boundary term vanishes:

$$\Big[y\, k(y,x)\Big]_{y=-\infty}^{y=+\infty} = 0.$$

Hence (4.15) reduces to

$$\int_{\mathbb{R}} y\, \partial_y k(y,x)\, dy = -\int_{\mathbb{R}} k(y,x)\, dy = -\int_{\mathbb{R}} \partial_x H(y,x)\, dy. \tag{4.16}$$

Combining (4.14) and (4.16) yields

$$\frac{d}{dx}\mu(x) = -\int_{-\infty}^{\infty} \partial_x H(y,x)\, dy = -\int_{-\infty}^{\infty} \frac{\partial}{\partial x} F_{Y|X}^{(X\downarrow)}(y \mid x)\, dy,$$

which is the claimed representation of the Marginal Causal Effect. □

4.2 Closed-Form Copula Representation of the MCE

We now derive an explicit closed-form representation of the local Average Causal Effect (MCE) in terms of conditional copulas, analogous to the interventional distribution obtained in Proposition 4.1.

Recall from Corollary 4.2 that, under Assumptions R1–R3, the causal effect satisfies

$$\mathrm{MCE}(x) = \frac{d}{dx}\mu(x) = -\int_{-\infty}^{\infty} \frac{\partial}{\partial x} F_{Y|X}^{(X\downarrow)}(y \mid x)\, dy. \tag{4.17}$$

From Proposition 4.1, the interventional conditional distribution admits the copula representation

$$F_{Y|X}^{(X\downarrow)}(y \mid x) = \int \partial_u C_{X,Y|Z}\big(F_{X|Z}(x \mid z), F_{Y|Z}(y \mid z) \,\big|\, z\big)\, f_Z(z)\, dz,$$

where $\partial_u C$ denotes differentiation of the conditional copula with respect to its first argument.

Regularity Conditions. In addition to R1–R4, we assume throughout this section:

C1. Continuous differentiability and strict positivity of $F_{X|Z}(\cdot|z)$ and $F_{Y|Z}(\cdot|z)$.

C2. Twice continuous differentiability of $C_{X,Y|Z}(u,v|z)$ w.r.t. u on $(0,1)^2$ a.e. z.

C3. Dominated convergence: $\exists M(z) \in L^1$ s.t. $|\partial_u^2 C_{X,Y|Z}(u,v|z)| \le M(z)\ \forall (u,v)$.

C4. Finite L^1 conditional quantiles: $\int_0^1 |Q_{Y|Z}(v|z)|\, dv < \infty$ a.e. z.

These strengthen the smoothness/domination in R2–R3 to enable Leibniz differentiation under the integral and the canonical quantile representation.

Differentiation under the integral sign. Substituting (4.3) into (4.17) and invoking the above assumptions to exchange integration and differentiation yields

$$\mathrm{MCE}(x) = -\int f_Z(z) \left(\int \frac{\partial}{\partial x} \partial_u C_{X,Y|Z}\big(u(x,z), v(y,z) \mid z\big)\, dy \right) dz, \tag{4.18}$$

where we define

$$u(x,z) = F_{X|Z}(x \mid z), \qquad v(y,z) = F_{Y|Z}(y \mid z).$$

Since $v(y,z)$ does not depend on x, the chain rule gives

$$\frac{\partial}{\partial x} \partial_u C_{X,Y|Z}(u, v \mid z) = \partial_u^2 C_{X,Y|Z}(u, v \mid z)\, \frac{\partial}{\partial x} F_{X|Z}(x \mid z) = \partial_u^2 C_{X,Y|Z}(u, v \mid z)\, f_{X|Z}(x \mid z)$$

Hence,

$$\mathrm{MCE}(x) = -\int f_Z(z)\, f_{X|Z}(x \mid z) \left(\int \partial_u^2 C_{X,Y|Z}\big(F_{X|Z}(x \mid z), F_{Y|Z}(y \mid z) \mid z\big)\, dy \right) dz \tag{4.19}$$

Equation (4.19) provides a closed-form causal effect entirely determined by conditional copulas and conditional marginal distributions.

All integrations are taken over the supports $\mathcal{Z} = \mathrm{supp}(Z)$ and $\mathcal{Y}(z) = \mathrm{supp}(Y \mid Z = z)$. Under Assumptions R1–R3, these coincide with $\mathbb{R}$ and $\mathbb{R}$, respectively.

Canonical copula representation of the MCE. An equivalent and more compact representation of the Marginal Causal Effect is obtained by expressing the conditional mean directly in copula coordinates.

Using the conditional copula factorization,

$$f_{Y|X,Z}(y \mid x, z) = f_{Y|Z}(y \mid z)\, c_{X,Y|Z}(u, v \mid z),$$

where

$$u = F_{X|Z}(x \mid z), \qquad v = F_{Y|Z}(y \mid z),$$

the conditional expectation satisfies

$$\mathbb{E}[Y \mid X = x, Z = z] = \int_0^1 Q_{Y|Z}(v \mid z)\, c_{X,Y|Z}\big(F_{X|Z}(x \mid z), v \mid z\big)\, dv,$$

with

$$Q_{Y|Z}(v \mid z) = F_{Y|Z}^{-1}(v \mid z)$$

denoting the conditional quantile function.

Differentiating with respect to x yields

$$\frac{\partial}{\partial x} \mathbb{E}[Y \mid X = x, Z = z] = f_{X|Z}(x \mid z) \int_0^1 Q_{Y|Z}(v \mid z)\, \partial_u c_{X,Y|Z}\big(F_{X|Z}(x \mid z), v \mid z\big)\, dv.$$

Integrating over the distribution of Z gives the closed-form copula representation of the Marginal Causal Effect:

$$\mathrm{MCE}(x) = \int_{\mathcal{Z}} f_Z(z)\, f_{X|Z}(x \mid z) \int_0^1 Q_{Y|Z}(v \mid z)\, \partial_u c_{X,Y|Z}\big(F_{X|Z}(x \mid z), v \mid z\big)\, dv\, dz \tag{4.20}$$

Equation (4.21) shows that causal derivatives are fully determined by the conditional copula density together with conditional marginal distributions.

Structural interpretation in copula space. The canonical representation admits a particularly transparent interpretation in copula coordinates. Define the probability integral transform variables

$$U = F_{X|Z}(X \mid Z), \qquad V = F_{Y|Z}(Y \mid Z),$$

which are conditionally uniform on $[0,1]$ given Z. Dependence between exposure and outcome is therefore fully encoded by the conditional copula of (U, V).

Using the representation

$$\mathbb{E}[Y \mid X = x, Z = z] = \int_0^1 Q_{Y|Z}(v \mid z)\, c_{X,Y|Z}(F_{X|Z}(x \mid z), v \mid z)\, dv,$$

with $u = F_{X|Z}(x \mid z)$, the interventional mean may be written as

$$\begin{aligned}\mu(x) &= \int_{\mathcal{Z}} f_Z(z)\, \mathbb{E}\big[Q_{Y|Z}(V \mid z) \,\big|\, U = F_{X|Z}(x \mid z),\, Z = z\big]\, dz \\ &= \int_{\mathcal{Z}} f_Z(z) \int_0^1 Q_{Y|Z}(v \mid z)\, c_{X,Y|Z}(F_{X|Z}(x \mid z), v \mid z)\, dv\, dz.\end{aligned}$$

Differentiating with respect to x and applying the chain rule to the inner integral yields the structural characterization

$$\mathrm{MCE}(x) = \int_{\mathcal{Z}} f_Z(z)\, f_{X|Z}(x \mid z) \int_0^1 Q_{Y|Z}(v \mid z)\, \partial_u c_{X,Y|Z}(F_{X|Z}(x \mid z), v \mid z)\, dv\, dz. \tag{4.21}$$

An equivalent and more compact form follows by recognising that the inner integral is precisely

$$\frac{\partial}{\partial x}\mathbb{E}[Y \mid X = x, Z = z] = f_{X|Z}(x \mid z) \int_0^1 Q_{Y|Z}(v \mid z)\, \partial_u c_{X,Y|Z}\big(F_{X|Z}(x \mid z), v \mid z\big)\, dv.$$

Substituting this identity gives

$$\boxed{\mathrm{MCE}(x) = \int_{\mathcal{Z}} f_Z(z)\, \frac{\partial}{\partial x}\, \mathbb{E}[Y \mid X = x, Z = z]\, dz.} \tag{4.22}$$

Equation (4.22) admits a direct causal reading: the Marginal Causal Effect is a *confounder-weighted average of local conditional slopes*. The factor $f_Z(z)$ integrates over the marginal distribution of the confounder, implementing the backdoor adjustment; the factor $f_{X|Z}(x \mid z)$ reweights each stratum by the conditional density of the treatment, ensuring that strata in which $X = x$ is more probable contribute more to the aggregate effect. The two representations (4.21) and (4.22) are identical: the copula machinery produces the conditional slope $\partial_x\, \mathbb{E}[Y \mid X = x, Z = z]$ explicitly in terms of the dependence structure, while (4.22) expresses the same quantity in the language of regression, making the relationship to conventional regression-based estimators of the ACE immediately transparent.

Interpretation: From an applied perspective, Equation (4.22) quantifies how changes in exposure translate into changes in expected outcome, stratum by stratum, averaged over the population distribution of the confounder. Large causal effects arise when the conditional mean of Y is steeply increasing in x across most confounder strata; small effects correspond to settings in which the conditional mean is approximately flat regardless of the stratum. The copula-coordinate form (4.21) further reveals that this slope heterogeneity is driven by local variations in the dependence structure between the probability-integral-transform variables (U, V), separating the structural component of causal variation from marginal scale effects.

Relation to distributional and quantile treatment effects. Classical distributional causal analyses typically define effects in terms of contrasts between the marginal distributions or quantiles of potential outcomes $Y(1)$ and $Y(0)$ for binary or discrete treatments[4] leading to distributional treatment effects and quantile treatment effects (QTEs) (e.g. Chernozhukov and Hansen, 2005, 2013; Abbring and Heckman, 2007; Blundell et al., 2007). A related line of work studies *unconditional* quantile effects and policy effects, characterising how changes in the distribution of a covariate or treatment shift unconditional quantiles of Y via influence-function–based regressions and related tools (Firpo et al., 2009; Rothe, 2012; Alejo et al., 2025). Our copula representation fits naturally within this broader distributional framework. For each treatment level x, the copula-based backdoor formula identifies the entire interventional CDF $F_{Y|X}^{(X\downarrow)}(\cdot \mid x) = \Pr\{Y \leq y \mid \mathrm{do}(X = x)\}$ from observational data under a valid adjustment set Z. When X is binary, the marginals $F_{Y|X}^{(X\downarrow)}(\cdot \mid 0)$ and $F_{Y|X}^{(X\downarrow)}(\cdot \mid 1)$ coincide with the distributions of $Y(0)$ and $Y(1)$, so standard distributional and quantile treatment effects are recovered as differences of CDFs or quantiles computed from our copula factorization. For continuous X, the copula framework extends these ideas by providing closed-form expressions for $F_{Y|X}^{(X\downarrow)}(\cdot \mid x)$ and for its derivative with respect to x, yielding both mean effects () and continuous-treatment distributional and quantile effects as smooth functionals of the identified interventional law.

Advantages of the copula structural representation. Expressing the Average Causal Effect in copula coordinates provides a structural perspective on causal effects that separates dependence structure from marginal measurement scales. The probability integral transform maps exposure and outcome variables into relative positions within their marginal distributions, allowing causal relationships to be studied in terms of dependence rather than absolute magnitudes.

This representation offers several conceptual advantages. First, it separates causal mechanisms from marginal distributions: changes in units, rescalings, or monotone transformations of variables leave the copula—and therefore the dependence component of the model—unchanged. While the numerical value of the causal effect on the observed scale still depends on the marginal distributions, the copula representation isolates the structural component governing how exposure and outcome are linked within the system.

Second, the formulation clarifies the role of interventions. In copula space, an intervention modifies how movements across exposure ranks propagate into expected outcome ranks, conditional on background characteristics. The Average Causal Effect therefore captures how local shifts in exposure position influence outcome prospects within the dependence structure.

Third, the representation isolates dependence as a fundamental component of causal analysis. Classical regression-based approaches combine marginal variation and dependence structure within a single conditional mean, making interpretation sensitive to functional-form assumptions. In contrast, the copula formulation explicitly decomposes causal relationships into marginal distributions and a dependence component, allowing structural aspects of causal variation to be analyzed separately from scale effects.

Finally, this perspective facilitates comparisons across populations, datasets, or measurement systems. Because the dependence structure encoded by the copula is invariant to monotone transformations, the structural relationship between exposure and outcome can be compared even when variables are measured on different scales or units.

[4]In the structural causal model (SCM) framework adopted here, potential outcomes are defined via interventions on the treatment node. For a binary treatment $X \in \{0, 1\}$, $Y(1)$ denotes the value of Y in the hypothetical world where the structural equation for X is surgically repld by the constant assignment $X = 1$, i.e. under the intervention $\mathrm{do}(X = 1)$, while $Y(0)$ corresponds to $\mathrm{do}(X = 0)$; see Pearl (2009); Peters et al. (2017). Thus the distribution of $Y(x)$ coincides with the interventional law of Y under $\mathrm{do}(X = x)$ used in this paper, i.e. $Y(x) \stackrel{d}{=} Y \mid \mathrm{do}(X = x)$, and our copula-based representation provides an explicit factorization of these interventional distributions.

5 Illustrative Examples

5.1 The Gaussian Copula Model

Purpose and Scope This example serves three central objectives. First, it demonstrates that under Gaussian dependence, the copula-based backdoor adjustment formula

$$f_{Y|X}^{(X\downarrow)}(y;x) = f_Y(y) \int_{-\infty}^{\infty} c_{X,Y|Z}(F_{X|Z}(x|z), F_{Y|Z}(y|z)|z)\, c_{Y,Z}(F_Y(y), F_Z(z))\, f_Z(z)\, dz \tag{5.1}$$

admits closed-form evaluation. By working through each component—conditional copulas, marginal copulas, and conditional distributions—we will show that the Gaussian structure enables analytical computation of the entire interventional distribution without numerical approximation.

Second, this derivation provides a transparent decomposition of backdoor adjustment. Formula (5.1) explicitly separates:

- The conditional treatment-outcome copula $c_{X,Y|Z}$, which captures the direct causal relationship after stratifying on the confounder,
- The marginal outcome-confounder copula $c_{Y,Z}$, which encodes the confounding dependence structure,
- The integration over $f_Z(z)$, which averages out the confounder using its marginal (not conditional) distribution.

Third, and most importantly for practitioners, this example serves as a complete template for copula-based causal inference. By verifying each technical assumption explicitly and computing each quantity step-by-step, we provide readers with a blueprint for their own applications.

5.1.1 Construction Blueprint

To build this example, we make four sequential specification choices:

Choice 1: Marginal Distributions. We select uniform margins $F_X = F_Y = F_Z = \mathcal{U}(0,1)$, yielding densities $f_X(x) = f_Y(y) = f_Z(z) = 1$ on $(0,1)$. This simplifies (5.1) to:

$$f_{Y|X}^{(X\downarrow)}(y;x) = \int_0^1 c_{X,Y|Z}(F_{X|Z}(x|z), F_{Y|Z}(y|z)|z)\, c_{Y,Z}(y,z)\, dz, \tag{5.2}$$

since $f_Y(y) = 1$, $f_Z(z) = 1$, $F_Y(y) = y$, and $F_Z(z) = z$.

Choice 2: Copula Family. We select Gaussian copulas for all dependencies. The bivariate Gaussian copula with correlation ρ has density:

$$c(u,v;\rho) = \frac{1}{\sqrt{1-\rho^2}} \exp\left\{ \frac{2\rho\Phi^{-1}(u)\Phi^{-1}(v) - \rho^2[(\Phi^{-1}(u))^2 + (\Phi^{-1}(v))^2]}{2(1-\rho^2)} \right\}, \tag{5.3}$$

where Φ denotes the standard normal CDF and Φ^{-1} its inverse (the quantile function).

Choice 3: Correlation Parameters. We specify three of the following parameters:

- $\rho_{X,Y}$: correlation parameter of the *unconditional* copula $C_{X,Y}$.
- $\rho_{X,Z}$: correlation for $C_{X,Z}$
- $\rho_{Y,Z}$: correlation for $C_{Y,Z}$

These parameters must jointly ensure that the implied correlation matrix Σ is positive definite, satisfying the condition

$$\det(\Sigma) = 1 + 2\rho_{X,Y}\rho_{X,Z}\rho_{Y,Z} - \rho_{X,Y}^2 - \rho_{X,Z}^2 - \rho_{Y,Z}^2 > 0.$$

Once specified, these parameters determine the correlation $\rho_{X,Y|Z}$ of the *conditional* copula $C_{X,Y|Z}$.

Choice 4: Causal Structure Verification. We verify consistency with the confounding DAG (Z confounds $X \to Y$), typically requiring $\rho_{X,Z} \neq 0$, $\rho_{Y,Z} \neq 0$, and often $\rho_{X,Y} > \rho_{X,Z} \cdot \rho_{Y,Z}$ for a positive direct effect.

With these choices, all conditional distributions are determined by Sklar's theorem, and (5.2) becomes a concrete integral over Gaussian copula densities.

5.1.2 Verification of Technical Assumptions

We systematically verify that this construction satisfies all regularity requirements.

Assumption R1: Absolute Continuity. The Gaussian copula density (5.3) is strictly positive and continuous on $(0,1)^2$. For uniform margins, joint densities factor through copula densities, so conditional densities $f_{Y|X}(\cdot \mid x)$ and $f_{Z|X}(\cdot \mid x)$ exist for all $x \in (0,1)$, the conditional CDFs $F_{Y|X}(\cdot \mid x)$ and $F_{Z|X}(\cdot \mid x)$ are absolutely continuous in their first argument, and the mixed partial

$$c_{Y,Z|X}(u, w \mid x) = \frac{\partial^2}{\partial u\, \partial w} C_{Y,Z|X}(u, w \mid x)$$

exists almost everywhere; see, e.g., Nelsen (2006, p. 27).

Assumption R2: Copula Regularity. The Gaussian copula $C(\cdot, \cdot; \rho)$ is Lipschitz continuous (since Φ is globally Lipschitz), with partial derivatives $\partial C/\partial u$, $\partial C/\partial w$ existing and bounded by 1, hence integrable (Nelsen, 2006, Th. 2.2.7). The copula density satisfies $0 < c(u, w; \rho) < \infty$ and $\int_0^1 c(u, w; \rho)\, du = 1$ almost everywhere (Joe, 2014, Sec. 4.3). These properties hold for all marginal and conditional Gaussian copulas in our construction.

Assumption R3: Differentiability for MCE. For Gaussian copulas with uniform margins, the conditional density $f_{Y|X}(y \mid x)$ is a Gaussian density in y whose mean is affine in x and whose variance does not depend on x.[5] Consequently, its x-derivative $\partial_x f_{Y|X}(y \mid x)$ is a polynomial in y times the same Gaussian density, so $y\, \partial_x f_{Y|X}(y \mid x)$ is also a polynomial in y multiplied by a Gaussian density. Since multiplying a Gaussian density by any polynomial in y still yields an integrable function (the Gaussian tail decays like $\exp(-y^2/2)$ while the normal quantile $\Phi^{-1}(x)$ grows only on the order of $\sqrt{2\log(1/(1-x))}$ as $x \to 1$), there exists a function $g \in L^1(\mathbb{R})$ dominating $y\, \partial_x f_{Y|X}(y \mid x)$ uniformly in x. This verifies the domination condition in Assumption R3 and justifies differentiation under the integral in the definition of $\mu(x)$ by dominated convergence. Since $Y \sim \mathcal{U}(0,1)$ in this example, we have $\mathbb{E}[|Y|] = 1/2$, so the integrability requirement in Corollary 4.2 is also satisfied. Together, these observations check all technical conditions for R3 in the Gaussian case.

Corollary 4.2 Conditions. Since $Y \sim \mathcal{U}(0,1)$, we have $\mathbb{E}[|Y|] = 1/2 < \infty$.

All technical conditions are satisfied. This verification provides readers with a template for checking assumptions in other copula families.

[5] On the latent scale this is immediate from the linear–Gaussian representation; on the uniform scale it follows by monotone change of variables.

5.1.3 Derivation Strategy: Density to CDF Connection

Before computing (5.2) directly, we note an important relationship. From Proposition 4.1, the interventional CDF is:

$$F_{Y|X}^{(X\downarrow)}(y|x) = \int_{-\infty}^{\infty} \frac{\partial}{\partial u} C_{X,Y|Z}(u,w)\Big|_{u=F_{X|Z}(x|z),w=F_{Y|Z}(y|z)} f_Z(z)\,dz. \tag{5.4}$$

The interventional density is the derivative:

$$f_{Y|X}^{(X\downarrow)}(y|x) = \frac{\partial}{\partial y} F_{Y|X}^{(X\downarrow)}(y|x). \tag{5.5}$$

We will pursue a two-step strategy:

1. First, evaluate the CDF (5.4) using Gaussian integration identities (this yields a closed form),
2. Second, differentiate to obtain the density and verify consistency with (5.2).

This approach is pedagogically clearer than direct evaluation of (5.2), though we will show both routes lead to the same result. The CDF route leverages the well-known Gaussian integration formula:

$$\int_{-\infty}^{\infty} \Phi(\alpha+\beta t)\,\phi(t)\,dt = \Phi\left(\frac{\alpha}{\sqrt{1+\beta^2}}\right). \tag{5.6}$$

5.1.4 Computing the Interventional CDF

Step 1: Conditional CDFs. For a Gaussian copula, the conditional CDF is:

$$\frac{\partial C(u,w;\rho)}{\partial w} = \Phi\left(\frac{\Phi^{-1}(u) - \rho\Phi^{-1}(w)}{\sqrt{1-\rho^2}}\right).$$

With uniform margins where $C_{X,Z}(x,z) = \Phi_{\rho_{X,Z}}(\Phi^{-1}(x), \Phi^{-1}(z))$:

$$F_{X|Z}(x|z) = \Phi\left(\frac{\Phi^{-1}(x) - \rho_{X,Z}\Phi^{-1}(z)}{\sqrt{1-\rho_{X,Z}^2}}\right),$$

$$F_{Y|Z}(y|z) = \Phi\left(\frac{\Phi^{-1}(y) - \rho_{Y,Z}\Phi^{-1}(z)}{\sqrt{1-\rho_{Y,Z}^2}}\right).$$

For compact notation, define:

$$a_x := \Phi^{-1}(x), \quad a_y := \Phi^{-1}(y), \quad a_z := \Phi^{-1}(z).$$

Step 2: Conditional Copula Partial Derivative. Denote by $\rho_{X,Y|Z}$ the correlation parameter of the conditional Gaussian copula $C_{X,Y|Z}(\cdot,\cdot \mid z)$. The partial derivative with respect to the first argument at (u,v) is:

$$\frac{\partial C_{X,Y|Z}}{\partial u}(u,v) = \Phi\left(\frac{\Phi^{-1}(v) - \rho_{X,Y|Z}\Phi^{-1}(u)}{\sqrt{1-\rho_{X,Y|Z}^2}}\right).$$

Evaluated at $u = F_{X|Z}(x|z)$ and $v = F_{Y|Z}(y|z)$:

$$\begin{aligned}\frac{\partial C_{X,Y|Z}}{\partial u}\Big|_{(F_{X|Z}(x|z),F_{Y|Z}(y|z))} &= \Phi\left(\frac{\Phi^{-1}(F_{Y|Z}(y|z)) - \rho_{X,Y|Z}\Phi^{-1}(F_{X|Z}(x|z))}{\sqrt{1-\rho^2_{X,Y|Z}}}\right) \\ &= \Phi\left(\frac{A_{yz} - \rho_{X,Y|Z}A_{xz}}{\sqrt{1-\rho^2_{X,Y|Z}}}\right), \end{aligned} \tag{5.7}$$

where:

$$A_{xz} := \frac{a_x - \rho_{X,Z}a_z}{\sqrt{1-\rho^2_{X,Z}}}, \quad A_{yz} := \frac{a_y - \rho_{Y,Z}a_z}{\sqrt{1-\rho^2_{Y,Z}}}.$$

Step 3: Apply Gaussian Integration Formula. Substituting (5.7) into (5.4) with $f_Z(z) = 1$, and changing variables to $t = a_z$ (so $dz = \phi(t)\,dt$):

$$F^{(X\downarrow)}_{Y|X}(y|x) = \int_{-\infty}^{\infty} \Phi\left(\frac{A_{yt} - \rho_{X,Y|Z}A_{xt}}{\sqrt{1-\rho^2_{X,Y|Z}}}\right)\phi(t)\,dt,$$

where now:

$$A_{xt} = \frac{a_x - \rho_{X,Z}t}{\sqrt{1-\rho^2_{X,Z}}}, \quad A_{yt} = \frac{a_y - \rho_{Y,Z}t}{\sqrt{1-\rho^2_{Y,Z}}}.$$

The argument of Φ is linear in t:

$$\begin{aligned}\frac{A_{yt} - \rho_{X,Y|Z}A_{xt}}{\sqrt{1-\rho^2_{X,Y|Z}}} &= \frac{1}{\sqrt{1-\rho^2_{X,Y|Z}}}\left[\frac{a_y - \rho_{Y,Z}t}{\sqrt{1-\rho^2_{Y,Z}}} - \rho_{X,Y|Z}\frac{a_x - \rho_{X,Z}t}{\sqrt{1-\rho^2_{X,Z}}}\right] \\ &= \frac{1}{\sqrt{1-\rho^2_{X,Y|Z}}}\left[\frac{a_y}{\sqrt{1-\rho^2_{Y,Z}}} - \frac{\rho_{X,Y|Z}a_x}{\sqrt{1-\rho^2_{X,Z}}} + t\left(\frac{\rho_{X,Y|Z}\cdot\rho_{X,Z}}{\sqrt{1-\rho^2_{X,Z}}} - \frac{\rho_{Y,Z}}{\sqrt{1-\rho^2_{Y,Z}}}\right)\right] \\ &=: \alpha + \beta t, \end{aligned} \tag{5.8}$$

where:

$$\begin{aligned}\alpha &= \frac{1}{\sqrt{1-\rho^2_{X,Y|Z}}}\left[\frac{a_y}{\sqrt{1-\rho^2_{Y,Z}}} - \frac{\rho_{X,Y|Z}a_x}{\sqrt{1-\rho^2_{X,Z}}}\right], \\ \beta &= \frac{1}{\sqrt{1-\rho^2_{X,Y|Z}}}\left[\frac{\rho_{X,Y|Z}\cdot\rho_{X,Z}}{\sqrt{1-\rho^2_{X,Z}}} - \frac{\rho_{Y,Z}}{\sqrt{1-\rho^2_{Y,Z}}}\right].\end{aligned}$$

Applying (5.6):

$$F^{(X\downarrow)}_{Y|X}(y|x) = \Phi\left(\frac{\alpha}{\sqrt{1+\beta^2}}\right). \tag{5.9}$$

Step 4: Alternative Linear Representation. Equation (5.9) already provides a closed-form expression for the interventional CDF. Since α is linear in a_x and a_y, the argument of Φ can also be written as a linear combination

$$F^{(X\downarrow)}_{Y|X}(y|x) = \Phi\left(\frac{k_1 a_y + k_2 a_x}{s}\right), \tag{5.10}$$

for suitable constants k_1, k_2, s depending only on the correlation parameters. The constants are uniquely determined by the representation (5.9) after collecting the coefficients of a_y and a_x. Their explicit expressions are algebraically cumbersome and are not required for the subsequent derivations; only the linear form of the argument of Φ will be used below. For this reason, the representation (5.9) will be used in the sequel.

5.1.5 The Interventional Density

Differentiating (5.10) with respect to y using the chain rule:

$$\begin{aligned} f_{Y|X}^{(X\downarrow)}(y|x) &= \frac{\partial}{\partial y}\Phi\left(\frac{k_1 a_y + k_2 a_x}{s}\right) \\ &= \phi\left(\frac{k_1 a_y + k_2 a_x}{s}\right) \cdot \frac{k_1}{s} \cdot \frac{da_y}{dy} \\ &= \phi\left(\frac{k_1 a_y + k_2 a_x}{s}\right) \cdot \frac{k_1}{s} \cdot \frac{1}{\phi(a_y)}, \end{aligned} \tag{5.11}$$

where we used $da_y/dy = d\Phi^{-1}(y)/dy = 1/\phi(\Phi^{-1}(y)) = 1/\phi(a_y)$ by the inverse function theorem.

This density can be verified to integrate to 1 and to be consistent with the backdoor formula (5.2) (though we omit that verification for brevity).

5.1.6 The Interventional Mean and MCE

Interventional Mean. Using the tail-integral representation:

$$\mu(x) = \mathbb{E}^{(X\downarrow)}[Y|X=x] = \int_0^\infty [1 - F_{Y|X}^{(X\downarrow)}(y|x)]\, dy - \int_{-\infty}^0 F_{Y|X}^{(X\downarrow)}(y|x)\, dy.$$

For $Y \sim \mathcal{U}(0,1)$, this simplifies to:

$$\mu(x) = \int_0^1 [1 - F_{Y|X}^{(X\downarrow)}(y|x)]\, dy.$$

Substituting (5.10) and changing variables to $u = a_y$:

$$\begin{aligned} \mu(x) &= \int_0^1 \left[1 - \Phi\left(\frac{k_1 a_y + k_2 a_x}{s}\right)\right] dy \\ &= \int_{-\infty}^\infty \left[1 - \Phi\left(\frac{k_1 u + k_2 a_x}{s}\right)\right] \phi(u)\, du \\ &= \int_{-\infty}^\infty \Phi\left(-\frac{k_1 u + k_2 a_x}{s}\right) \phi(u)\, du. \end{aligned} \tag{5.12}$$

Using (5.6) with $\alpha = -k_2 a_x/s$ and $\beta = -k_1/s$:

$$\mu(x) = \Phi\left(\frac{-k_2 a_x/s}{\sqrt{1 + k_1^2/s^2}}\right) = \Phi\left(\frac{-k_2 a_x}{\sqrt{s^2 + k_1^2}}\right).$$

Define:

$$K := -\frac{k_2}{\sqrt{s^2 + k_1^2}}. \tag{5.13}$$

Then:

$$\mu(x) = \Phi(K \cdot a_x) = \Phi(K \cdot \Phi^{-1}(x)). \tag{5.14}$$

Average Causal Effect. Differentiating (5.14):

$$\begin{aligned}\mathrm{MCE}(x) = \frac{d\mu}{dx} &= \phi(K \cdot a_x) \cdot K \cdot \frac{da_x}{dx} \\ &= \phi(K \cdot \Phi^{-1}(x)) \cdot K \cdot \frac{1}{\phi(\Phi^{-1}(x))} \\ &= K \cdot \frac{\phi(K \cdot \Phi^{-1}(x))}{\phi(\Phi^{-1}(x))}. \end{aligned} \tag{5.15}$$

This shows that $\mathrm{MCE}(x)$ varies with x, even though the underlying dependence structure is constant in the Gaussian copula parameterization.

5.1.7 Numerical Illustration

We demonstrate how $\mathrm{MCE}(x)$ varies across treatment levels and parameter regimes.

We demonstrate how $\mathrm{MCE}(x)$ varies across treatment levels and parameter regimes. Below we keep the observable (unconditional) correlations denoted by $\rho_{X,Y}, \rho_{X,Z}, \rho_{Y,Z}$ and compute the causal scalar β (which equals K in the closed form for $\mu(x) = \Phi(K\Phi^{-1}(x))$) via the usual partial-correlation formula:

$$\beta = \frac{\rho_{X,Y} - \rho_{X,Z}\rho_{Y,Z}}{\sqrt{(1-\rho_{X,Z}^2)(1-\rho_{Y,Z}^2)}},$$

and we use $K := \beta$ as the effective interventional copula parameter in the closed-form expression $\mu(x) = \Phi(K\Phi^{-1}(x))$.

Case A: Moderate Confounding. Parameters:

$$\rho_{X,Z} = 0.5, \qquad \rho_{Y,Z} = 0.6, \qquad \rho_{X,Y} = 0.7.$$

Partial correlation:

$$\beta = \frac{0.7 - (0.5)(0.6)}{\sqrt{(1-0.5^2)(1-0.6^2)}} = \frac{0.4}{\sqrt{0.48}} \approx 0.577.$$

Table 1: Gaussian: interventional quantities (Case A: $K = 0.577$).

x	$\mu(x)$	$\mathrm{MCE}(x)$	comment
0.10	0.2297	0.998	lower tail
0.25	0.3485	0.672	first quartile
0.50	0.5000	0.577	median
0.75	0.6515	0.672	third quartile
0.90	0.7703	0.998	upper tail

Observations: $\mathrm{MCE}(x)$ is symmetric around $x = 0.5$ with $\mathrm{MCE}(0.5) = K$, exhibits tail amplification with substantially larger effects at the extremes than at the median, and attains its minimum at the median while increasing monotonically toward the extremes (as predicted by the Φ-ratio in (5.15)).

Case B: No Confounding. Set $\rho_{X,Z} = 0$ (keeping $\rho_{X,Y} = 0.7$, $\rho_{Y,Z} = 0.6$):

$$\beta = \frac{0.7}{\sqrt{1-0.36}} = \frac{0.7}{0.8} = 0.875.$$

Table 2: Gaussian: interventional quantities (Case B: $K = 0.875$).

x	$\mu(x)$	MCE(x)	comment
0.10	0.1311	1.061	lower tail
0.25	0.2775	0.923	first quartile
0.50	0.5000	0.875	median
0.75	0.7225	0.923	third quartile
0.90	0.8689	1.061	upper tail

Observations: MCE(x) maintains the same symmetric tail amplification pattern as Case A but with substantially larger effects throughout (MCE(0.5) = 0.875 > K_A). The confounding removal increases the overall effect magnitude, as reflected by the larger K. However, because K enters the ratio $\phi(K\Phi^{-1}(x))/\phi(\Phi^{-1}(x))$ nonlinearly, the amplification is *state-dependent*: comparing Table 2 with Table 1, the ratio of MCE values is approximately 1.52 at the median (0.875/0.577) but narrows to approximately 1.06 at the 10th percentile (1.061/0.998). The qualitative shape — minimum at the median, growing toward the tails — is preserved, but the degree of tail amplification is itself modulated by the structural parameter K.

Case C: Strong Confounding. Set $\rho_{X,Z} = 0.8$ (keep $\rho_{Y,Z} = 0.6$, $\rho_{X,Y} = 0.7$):

$$\beta = \frac{0.7 - (0.8)(0.6)}{\sqrt{(1 - 0.64)(1 - 0.36)}} = \frac{0.22}{\sqrt{0.2304}} \approx 0.458.$$

Table 3: Gaussian: interventional quantities (Case C: $K = 0.458$).

x	$\mu(x)$	MCE(x)	comment
0.10	0.2785	0.877	lower tail
0.25	0.3786	0.549	first quartile
0.50	0.5000	0.458	median
0.75	0.6214	0.549	third quartile
0.90	0.7215	0.877	upper tail

Observations: MCE(x) retains the characteristic Gaussian tail amplification but with attenuated effects (MCE(0.5) = 0.458 < K_A). Stronger confounding suppresses the causal signal across the treatment range while maintaining symmetry and the minimum-at-median property, attenuating the overall magnitude of the effect (as reflected by the smaller K) while preserving the same state-dependent shape.

Figures 4 summarize these numerical results. The panels were produced from simulations under dependence regimes in Case A–C. This figure shows interventional (i.e. do($X = x$)) quantities computed from the copula backdoor formula: the left panel plots the interventional mean $\mu(x) = \mathbb{E}^{(X\downarrow)}[Y|X]$, and the right panel plots the local Average Causal Effect MCE(x) = $\mathrm{d}\mu(x)/\mathrm{d}x$. The three curves display different latent causal-strength regimes as summarized by the scalar K appearing in the interventional mean representation (5.14). In the Gaussian benchmark, we set $K := \beta$, where $\beta = \rho_{Y^*,X^*|Z^*}$ is the latent partial correlation in the linear–Gaussian SEM. Because K is a structural (causal) parameter, the graphs compare *interventional* responses while holding the marginal confounding structure comparable across scenarios.

Two causal points are emphasized by the figure: First, although the underlying latent structural relation is linear with constant slope K, the observable marginal causal effect MCE(x) is *state-dependent*: it attains its minimum near the median $x \approx 0.5$ and increases symmetrically toward the boundaries. This non-constancy is induced entirely by the nonlinear probability

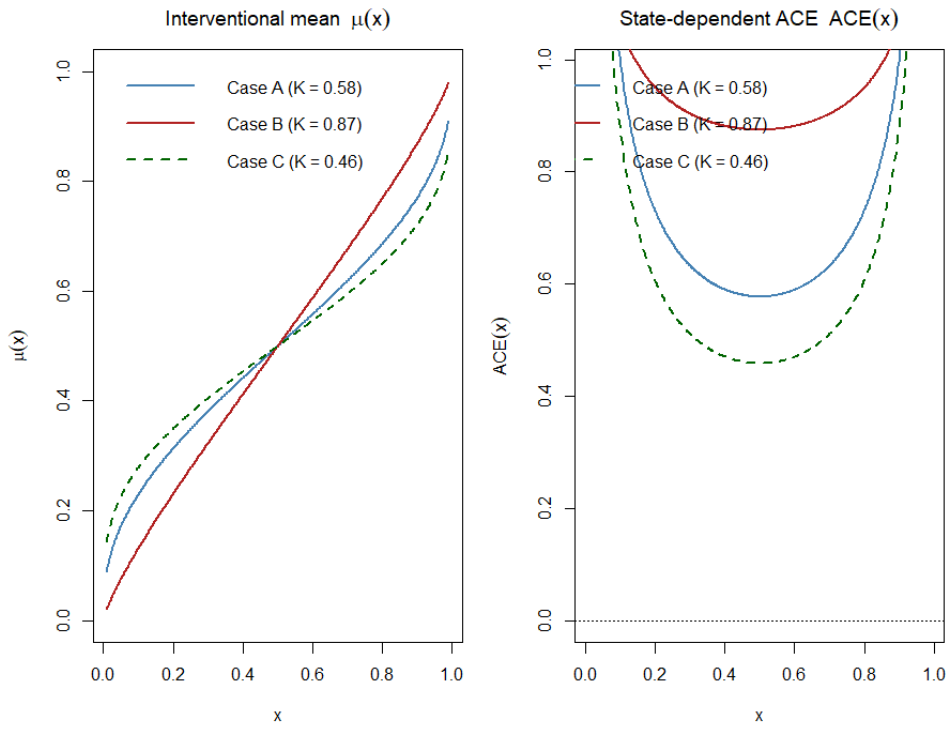


Figure 4: Interventional mean $\mu(x)$ (left) and state-dependent Average Causal Effect MCE(x) (right) under the Gaussian copula model with uniform margins. Curves correspond to three parameter scenarios: Case A (moderate confounding, $K \approx 0.58$), Case B (no confounding, $K \approx 0.87$), and Case C (strong confounding, $K \approx 0.46$). Plots show $x \in [0.01, 0.99]$ to avoid boundary artifacts.

integral transform (copula + marginal maps), not by heterogeneity in the latent structural coefficient. Second, the interventional mean $\mu(x)$ responds monotonically to K: larger K shifts $\mu(x)$ upward across all x, with the magnitude of change varying over the domain due to the nonlinear copula transform. Practically, this implies that the marginal return to increasing the treatment depends on the current treatment level — interventions targeted at extreme low or high values of X produce larger marginal changes in Y than equally sized changes near the median — even when the causal mechanism on the latent scale is homogeneous. All plotted quantities are interventional (identified via backdoor adjustment); they are not observational associations $\mathbb{E}[Y|X = x]$, and their interpretation is causal provided the usual backdoor-identification assumptions hold.

5.1.8 Parameter Sensitivity and Limiting Cases

The following cases describe limiting behaviour of the Gaussian copula parameterization in terms of partial correlations and dependence structure.

Limiting Case 1: Independence ($\rho_{X,Z} = 0$). Absence of X–Z dependence. When $\rho_{X,Z} = 0$, the dependence between X and Z vanishes, and the partial correlation simplifies to $\beta = \rho_{X,Y}/\sqrt{1 - \rho_{Y,Z}^2}$. If additionally $\rho_{Y,Z} = 0$, then $\beta = \rho_{X,Y}$, so the partial and marginal correlations coincide.

Limiting Case 2: Degenerate X–Z Dependence ($\rho_{X,Z} \to 1$). The denominator in (**??**) approaches zero, making β singular unless $\rho_{X,Y} = \rho_{Y,Z}$. The partial correlation becomes ill-

defined as $\rho_{X,Z} \to 1$, reflecting degeneracy in the Gaussian copula parameterization.

Limiting Case 3: Indirect Dependence Only ($\rho_{X,Y} = \rho_{Y,Z}\rho_{X,Z}$). The numerator vanishes: $\beta = 0$. The entire marginal association is explained by the dependence induced through Z. We have $K = 0$, $\mu(x) = 0.5$ for all x, and $\text{MCE}(x) = 0$.

Parameter Sensitivity. The MCE exhibits these monotonicity properties:

- $\partial\beta/\partial\rho_{X,Y} > 0$: Increasing $\rho_{X,Y}$ increases the partial correlation β.
- $\partial\beta/\partial\rho_{X,Z} < 0$ (when $\rho_{Y,Z} > 0$): Increasing $\rho_{X,Z}$ reduces the partial correlation β.
- $\partial\beta/\partial\rho_{Y,Z} < 0$ (when $\rho_{X,Z} > 0$): Increasing $\rho_{Y,Z}$ reduces the partial correlation β.

5.1.9 Interpretation and Key Insights

1. Closed-form Tractability. Gaussian dependence yields analytical expressions for the interventional distribution, density, mean, and MCE. This makes the Gaussian case invaluable as both a theoretical benchmark and a practical tool when Gaussian dependence is approximately valid.

2. Relationship between K and Partial Correlation. The scalar K appearing in $\mu(x) = \Phi(K \cdot \Phi^{-1}(x))$ coincides numerically with the partial correlation $\beta = \rho_{X,Y|Z}$ under the Gaussian copula parameterization used in this example. This relationship arises from the algebra of the Gaussian copula construction and should be interpreted as a reduced-form correspondence rather than a structural identification.

3. State-Dependent MCE Despite Linear Mechanism. Although the Gaussian copula induces a constant dependence parameter β, the observable MCE varies:

$$\text{MCE}(x) = \beta \cdot \frac{\phi(\beta\Phi^{-1}(x))}{\phi(\Phi^{-1}(x))}. \tag{5.16}$$

This state-dependence arises from the nonlinear copula transformation Φ. The ratio $\phi(\cdot)/\phi(\cdot)$ induces tail amplification.

4. Backdoor Adjustment Decomposition. Formula (5.1) decomposes into interpretable components:

- $c_{X,Y|Z}$: Conditional treatment-outcome dependence after stratifying on Z,
- $c_{Y,Z}$: Marginal confounding structure,
- Integration over $f_Z(z)$: Averages using marginal (not conditional on X) distribution.

For Gaussian copulas, each component has an explicit density formula, making the decomposition transparent.

5. Confounding Formula. The decomposition β cleanly separates observed association from confounding bias. When $\rho_{X,Y} = \rho_{Y,Z}\rho_{X,Z}$, association is pure confounding. When $\rho_{X,Z} = 0$, confounding vanishes.

6. Relation to Linear Gaussian Models. In linear Gaussian settings, constant dependence parameters often correspond to linear relationships on an underlying latent scale. The copula representation shows how such constant dependence translates to the observable scale with state-dependent variation through the nonlinear probability integral transform. The copula approach accommodates arbitrary margins without requiring explicit specification of a structural equation model.

7. Template for Practice. This example demonstrates:

- How to verify assumptions R1–R3 for a specific copula family,
- How to compute interventional quantities from the backdoor density formula,
- HHow to relate copula parameters to dependence measures such as partial correlations,
- How to assess sensitivity to confounding strength.

8. Benchmark for Non-Gaussian Comparisons. Having established the Gaussian benchmark with closed-form tractability and clean confounding decomposition, the next example will examine what changes under non-Gaussian bounded dependence, specifically the Farlie-Gumbel-Morgenstern (FGM) family, which exhibits qualitatively different MCE behavior.

5.2 The Farlie-Gumbel-Morgenstern (FGM) Copula Family

Purpose and Scope This example serves four distinct objectives that complement and contrast with the Gaussian benchmark established in Section 5.1.

First, it demonstrates copula-based causal inference under **bounded weak dependence**. The FGM family is characterized by dependence parameters $\alpha, \beta, \theta \in [-1, 1]$, corresponding to a maximum attainable Kendall's tau of $\pm 2/9 \approx \pm 0.22$. This bounded structure makes FGM particularly suitable for modeling scenarios where associations are known to be mild—for instance, in observational studies where confounding is present but not dominant, or in applications where asymmetric tail dependencies characteristic of elliptical copulas are implausible.

Second, the example reveals a qualitatively different MCE behavior from the Gaussian case. Where the Gaussian copula produces substantial tail amplification (MCE varying by 73% from median to extremes in our Case A), the FGM copula yields **nearly constant MCE** across the treatment range. This contrast illustrates how the choice of dependence structure—not merely the marginal distributions—fundamentally shapes the state-dependence of causal effects. For practitioners, this has direct implications: if empirical evidence suggests weak, bounded dependence, the FGM framework may provide more realistic causal estimates than forcing a Gaussian assumption.

Third, the FGM family belongs to the broader Rodrıguez-Lallena and Úbeda-Flores (2004) class of copulas having the separable perturbation form $C(u, w) = uw + f(u)g(w)$. By working through the FGM case concretely, we establish a template that extends naturally to other RLUF copulas (including some asymmetric families capturing directional dependencies absent in elliptical models). The FGM example thus serves as an entry point to a large, flexible class of dependence structures relevant for causal modeling.

Fourth, this example provides a reality check on the interventional density formula (5.1). Under FGM, the integrals involved in $f_{Y|X}^{(X\downarrow)}(y; x)$ can be evaluated semi-analytically, allowing us to verify formulas and build intuition for when closed forms exist versus when numerical integration is required. This pedagogical transparency aids readers in adapting the copula-based framework to their own applications.

In summary, where Section 5.1 establishes analytical tractability and the $K = \beta$ connection under Gaussian dependence, Section 5.2 demonstrates bounded weak dependence, nearly-flat

MCE, and the RLUF template—showing readers that different copula families encode fundamentally different causal structures even when marginals are identical.

The specification should be interpreted as a parameterized construction rather than a fully unconstrained trivariate copula, as not all pairwise copulas can be freely chosen in a coherent joint model.

5.2.1 Construction Blueprint

To construct an FGM-based causal inference example, one makes five specification choices:

Choice 1: Marginal Distributions. As in Section 5.1, we select uniform margins $F_X = F_Y = F_Z = \mathcal{U}(0,1)$ for analytical convenience. With this choice, $f_X(x) = f_Y(y) = f_Z(z) = 1$, and the backdoor formula (5.1) simplifies to:

$$f_{Y|X}^{(X\downarrow)}(y;x) = \int_0^1 c_{X,Y|Z}(F_{X|Z}(x|z), F_{Y|Z}(y|z)|z)\, c_{Y,Z}(y,z)\, dz.$$

Choice 2: FGM Copula Family. The FGM copula is defined by:

$$C(u,w;\gamma) = uw + \gamma u(1-u)w(1-w), \quad \gamma \in [-1,1]. \tag{5.17}$$

The parameter γ controls dependence strength. The constraint $|\gamma| \leq 1$ is *necessary and sufficient* to ensure the copula density $c(u,w;\gamma) = 1 + \gamma(1-2u)(1-2w)$ remains non-negative on $[0,1]^2$.

Choice 3: Dependence Parameters. We specify three of the following FGM parameters:

- $\theta \in [-1,1]$: dependence parameter for the observational copula $C_{X,Y}$,
- $\beta \in [-1,1]$: dependence parameter for the confounding copula $C_{X,Z}$,
- $\alpha \in [-1,1]$: dependence parameter for the conditional copula $C_{X,Y|Z}(\cdot,\cdot|z)$.

Under the simplifying assumption (which we adopt), the conditional copula $C_{X,Y|Z}$ does not depend on z, so α is a constant capturing the strength of the direct $X \to Y$ causal path after conditioning on Z.

Choice 4: Verify Non-Negativity. For each chosen parameter $\gamma \in \{\theta, \beta, \alpha\}$, verify:

$$|\gamma| \leq 1 \implies c(u,w;\gamma) = 1 + \gamma(1-2u)(1-2w) \geq 0 \text{ for all } (u,w) \in [0,1]^2. \tag{5.18}$$

The worst case occurs at the corners: $c(0,0) = c(1,1) = 1+\gamma$ and $c(0,1) = c(1,0) = 1-\gamma$. Thus $|\gamma| \leq 1$ ensures $c \geq 0$ everywhere.

Choice 5: Assess Weak-Dependence Appropriateness. The FGM family is designed for weak to moderate dependence. Its maximum attainable Kendall's tau is $\tau_{\max} = 2/9 \approx 0.22$, significantly lower than the $[-1,1]$ range of Gaussian copulas. Before applying FGM to data:

- Compute empirical Kendall's tau between variables.
- If $|\hat{\tau}| > 0.25$, FGM may be inadequate; consider Gaussian or other copula families.
- If $|\hat{\tau}| < 0.20$, FGM is a reasonable starting point.

With these choices made, all conditional distributions are determined, and the interventional density (5.1) reduces to concrete integrals over FGM copula densities.

5.2.2 Verification of Technical Assumptions

We systematically verify that the FGM construction satisfies all regularity requirements stated in Section 1.1.

Assumption R1: Absolute Continuity. The FGM copula density is:

$$c(u, w; \gamma) = 1 + \gamma(1 - 2u)(1 - 2w), \quad \gamma \in [-1, 1]. \tag{5.19}$$

This is a polynomial in (u, w), hence infinitely differentiable. For $|\gamma| \leq 1$, we have $0 \leq c(u, w; \gamma) \leq 2$ for all $(u, w) \in [0, 1]^2$, so the copula is absolutely continuous with bounded density. Consequently, for uniform margins:

- Conditional densities $f_{Y|X}(\cdot|x)$ and $f_{Z|X}(\cdot|x)$ exist for all $x \in (0, 1)$,
- Conditional CDFs are absolutely continuous in their first argument,
- The mixed partial $c_{Y,Z|X}(u, w|x) = 1 + \alpha(1 - 2u)(1 - 2w)$ exists and is continuous.

Assumption R2: Copula Regularity. For the FGM copula:

1. The CDF $C(u, w; \gamma) = uw + \gamma u(1-u)w(1-w)$ is a polynomial, hence Lipschitz continuous on $[0, 1]^2$.
2. The partial derivatives are:

$$\frac{\partial C}{\partial u}(u, w; \gamma) = w + \gamma(1 - 2u)w(1 - w),$$
$$\frac{\partial C}{\partial w}(u, w; \gamma) = u + \gamma(1 - 2w)u(1 - u).$$

   Both are bounded: $|\partial C/\partial u| \leq 1 + |\gamma| \leq 2$ and similarly for $\partial C/\partial w$. Hence the partial derivatives exist, are continuous, and integrable.
3. The copula density satisfies $0 \leq c(u, w; \gamma) \leq 2$ and $\int_0^1 c(u, w; \gamma)\, du = 1$ for all $w \in [0, 1]$ (verified by direct integration: the γ term is odd in u and integrates to zero).

Assumption R3: Differentiability for MCE. For FGM copulas with uniform margins, the interventional density involves polynomial expressions in x. Specifically, the map

$$x \mapsto y \cdot f_{Y|X}(y|x) \cdot c_{Y,Z|X}(F_{Y|X}(y|x), F_{Z|X}(z|x)|x)$$

reduces to polynomial functions of x (since $F_{Y|X}(y|x) = y + \alpha(\cdot)$ is linear-plus-quadratic in x for FGM, and $c_{Y,Z|X}$ is polynomial in its arguments). Polynomials are smooth, and their derivatives are dominated by constants (polynomials are bounded on compact intervals). Therefore, differentiation under the integral sign is justified.

Corollary 4.2 Conditions. Since $Y \sim \mathcal{U}(0, 1)$, we have $\mathbb{E}[|Y|] = 1/2 < \infty$.

Summary. All technical conditions (R1, R2, R3) and corollary requirements are satisfied for the FGM copula model with uniform margins. Unlike the Gaussian case, the FGM verification is simpler because the copula density is a polynomial (no transcendental functions), making boundedness and regularity transparent.

5.2.3 FGM Model Specification and Derivation

We now work through the FGM model concretely, computing the interventional distribution, mean, and MCE step-by-step.

Copula Definitions. Let X, Y, Z have uniform marginals on $(0,1)$. Define the observational FGM copulas:

$$C_{X,Y}(x,y) = xy + \theta x(1-x)y(1-y),$$
$$C_{X,Z}(x,z) = xz + \beta x(1-x)z(1-z),$$

where $\theta, \beta \in [-1,1]$ are dependence parameters. Under the simplifying assumption, the conditional copula is:

$$C_{X,Y|Z}(x,y|z) = xy + \alpha x(1-x)y(1-y), \quad \alpha \in [-1,1],$$

independent of z.

Remark 5.1. In this vine construction the conditional copula $C_{X,Y|Z}$, together with the bivariate margins $C_{X,Z}$ and $C_{Y,Z}$, fully determines the trivariate joint distribution via the factorization $f_{X,Y,Z} = c_{X,Y|Z}\, c_{X,Z}\, c_{Y,Z}\, f_X f_Y f_Z$. The observational $C_{X,Y}$ is therefore not a free primitive but an implied quantity obtained by integrating out Z; specifying it separately as an additional free FGM parameter with θ would over-parameterise the model. In the derivations that follow, θ is retained only as a label for the marginal copula implied by the vine and does not appear as a free parameter in any formula.

Copula Densities. The copula density for the (X,Z) pair is:

$$c_{X,Z}(x,z) = 1 + \beta(1-2x)(1-2z). \tag{5.20}$$

By Theorem 2.1, the interventional joint density is:

$$f^{(X\downarrow)}_{X,Y,Z}(x,y,z) = \frac{f_{X,Y,Z}(x,y,z)}{1+\beta(1-2x)(1-2z)}.$$

The conditional copula density is:

$$c_{X,Y|Z}(u,w|z) = 1 + \alpha(1-2u)(1-2w).$$

The marginal copula density is:

$$c_{Y,Z}(y,z) = 1 + \eta(1-2y)(1-2z). \tag{5.21}$$

Conditional CDFs. For the FGM copula $C(u,w;\gamma) = uw + \gamma u(1-u)w(1-w)$, the conditional CDF is:

$$\frac{\partial C(u,w;\gamma)}{\partial w} = u + \gamma(1-2w)u(1-u).$$

Thus:

$$F_{X|Z}(x|z) = x + \beta(1-2z)x(1-x),$$
$$F_{Y|Z}(y|z) = y + \eta(1-2z)y(1-y).$$

Interventional CDF via Proposition 4.1. From Proposition 4.1:

$$F^{(X\downarrow)}_{Y|X}(y|x) = \int_0^1 \frac{\partial}{\partial u} C_{X,Y|Z}\Big|_{u=F_{X|Z}(x|z),\, w=F_{Y|Z}(y|z)} dz.$$

The partial derivative is:

$$\frac{\partial C_{X,Y|Z}}{\partial u}(u,w) = w + \alpha(1-2u)w(1-w).$$

Substituting $u = F_{X|Z}(x|z) = x + \beta(1-2z)x(1-x)$ and $w = F_{Y|Z}(y|z) = y + \eta(1-2z)y(1-y)$:

$$F_{Y|X}^{(X\downarrow)}(y|x) = \int_0^1 \Big[w(x;z) + \alpha(1 - 2F_{X|Z}(x|z))w(x;z)(1-w(x;z))\Big]dz$$
$$= \int_0^1 w(x;z)\,dz + \alpha \int_0^1 (1 - 2F_{X|Z}(x|z))w(x;z)(1-w(x;z))\,dz, \tag{5.22}$$

where $w(x;z) := F_{Y|Z}(y|z)$ for brevity.

Simplification via Integration. Define the marginal average:

$$M(x) := \int_0^1 F_{X|Z}(x|z)\,dz = \int_0^1 \big[x + \beta(1-2z)x(1-x)\big]dz. \tag{5.23}$$

Since $\int_0^1 (1-2z)\,dz = [z - z^2]_0^1 = 0$, the β term vanishes:

$$M(x) = x. \tag{5.24}$$

This is a key simplification for uniform margins: the expected conditional CDF equals the marginal CDF.

Similarly, define:

$$N(y) := \int_0^1 F_{Y|Z}(y|z)\,dz = y.$$

For the second integral in (5.22), we use:

$$\int_0^1 w(x;z)(1-w(x;z))\,dz = N(y) - \int_0^1 [F_{Y|Z}(y|z)]^2\,dz =: N(y) - Q(y).$$

Computing $Q(y)$:

$$Q(y) = \int_0^1 [y + \eta(1-2z)y(1-y)]^2\,dz$$
$$= \int_0^1 \Big[y^2 + 2y\cdot\eta(1-2z)y(1-y) + \eta^2(1-2z)^2y^2(1-y)^2\Big]dz$$
$$= y^2 + 0 + \eta^2y^2(1-y)^2 \int_0^1 (1-2z)^2\,dz. \tag{5.25}$$

Since $\int_0^1 (1-2z)^2\,dz = \int_0^1 (1-4z+4z^2)\,dz = 1 - 2 + 4/3 = 1/3$:

$$Q(y) = y^2 + \frac{\eta^2}{3}y^2(1-y)^2.$$

Therefore:

$$N(y) - Q(y) = y - y^2 - \frac{\eta^2}{3}y^2(1-y)^2 = y(1-y)\left(1 - \frac{\eta^2}{3}y(1-y)\right).$$

We now evaluate the full second integral directly. Substituting $1 - 2F_{X|Z}(x|z) = (1-2x) - 2\beta(1-2z)x(1-x)$ and $w(z)(1-w(z)) = y(1-y) - \eta^2(1-2z)^2y^2(1-y)^2 + \eta(1-2z)y(1-y)(1-2y)$, the integrand expands as:

$$(1 - 2F_{X|Z})\,w(1-w)$$
$$= \big[(1-2x) - 2\beta(1-2z)x(1-x)\big]\big[y(1-y) + \eta(1-2z)y(1-y)(1-2y) - \eta^2(1-2z)^2y^2(1-y)^2\big].$$

Integrating over $z \in [0,1]$ and using $\int_0^1 (1-2z)\,dz = 0$ and $\int_0^1 (1-2z)^2\,dz = \frac{1}{3}$, the surviving terms are:

$$\int_0^1 (1-2F_{X|Z})w(1-w)\,dz = (1-2x)\,y(1-y)\Big(1-\tfrac{\eta^2}{3}y(1-y)\Big) + \frac{2\beta\eta}{3}\,x(1-x)\,y(1-y)(1-2y).$$

Consequently, the full interventional CDF is:

$$\begin{aligned} F_{Y|X}^{(X\downarrow)}(y|x) &= y + \alpha(1-2x)\,y(1-y)\left(1-\frac{\eta^2}{3}y(1-y)\right) \\ &\quad + \frac{2\alpha\beta\eta}{3}\,x(1-x)\,y(1-y)(1-2y). \end{aligned} \tag{5.26}$$

The third term is a genuine contribution of the joint (β,η) dependence between the confounder and both variables. It vanishes when either $\beta = 0$ (no X–Z confounding) or $\eta = 0$ (no Y–Z dependence). Critically, when integrated against dy on $[0,1]$, this term contributes zero to $\mu(x)$ because $\int_0^1 y(1-y)(1-2y)\,dy = 0$ by the antisymmetry of $(1-2y)$ on $[0,1]$. The interventional mean therefore retains the form derived below, and the headline result that $\mathrm{MCE}(x)$ is constant in x and independent of β is fully valid.

Interventional Mean. Using the tail-integral representation:

$$\mu(x) = \int_0^1 [1 - F_{Y|X}^{(X\downarrow)}(y|x)]\,dy. \tag{5.27}$$

Substituting (5.26):

$$\begin{aligned} \mu(x) &= \int_0^1 \left[1 - y - \alpha(1-2x)\cdot y(1-y)\left(1-\frac{\eta^2}{3}y(1-y)\right)\right]dy \\ &= \frac{1}{2} - \alpha(1-2x)\int_0^1 y(1-y)\left(1-\frac{\eta^2}{3}y(1-y)\right)dy. \end{aligned} \tag{5.28}$$

The integral evaluates to:

$$\int_0^1 y(1-y)\,dy = \frac{1}{6},$$
$$\int_0^1 y^2(1-y)^2\,dy = \frac{1}{30}.$$

Therefore:

$$\int_0^1 y(1-y)\left(1-\frac{\eta^2}{3}y(1-y)\right)dy = \frac{1}{6} - \frac{\eta^2}{3}\cdot\frac{1}{30} = \frac{1}{6} - \frac{\eta^2}{90}.$$

Thus:

$$\mu(x) = \frac{1}{2} - \alpha(1-2x)\left(\frac{1}{6} - \frac{\eta^2}{90}\right). \tag{5.29}$$

Average Causal Effect. Differentiating (5.29) with respect to x:

$$\mathrm{MCE}(x) = \frac{d\mu}{dx} = -\alpha\cdot(-2)\left(\frac{1}{6} - \frac{\eta^2}{90}\right) = 2\alpha\left(\frac{1}{6} - \frac{\eta^2}{90}\right). \tag{5.30}$$

Key Observation: Unlike the Gaussian case, the MCE is *constant* in x for the FGM copula with uniform margins. This is a fundamental qualitative difference: bounded weak dependence produces state-independent causal effects on the observable scale, whereas unbounded Gaussian dependence produces tail amplification.

Simplifying (5.30):

$$\mathrm{MCE}(x) = 2\alpha \cdot \frac{15 - \eta^2}{90} = \frac{\alpha(15 - \eta^2)}{45}. \tag{5.31}$$

In the case $\eta \approx 0$ (weak Y-Z marginal dependence):

$$\mathrm{MCE}(x) \approx \frac{\alpha}{3}.$$

Conditions for Constant MCE. The constancy of $\mathrm{MCE}(x)$ arises under three simplifying conditions:

1. Uniform marginal distributions for X, Y, and Z.
2. FGM copula dependence structure for the relevant variable pairs.
3. The simplifying assumption that the conditional copula $C_{X,Y|Z}(\cdot, \cdot|z)$ does not depend on z.

If any of these conditions are relaxed (e.g., non-uniform margins or z-dependent copulas), the MCE may regain state dependence in x.

5.2.4 Numerical Illustration (FGM)

We now illustrate the FGM closed-form results of Section 5.2.3 with concrete parameter values and compare the resulting interventional quantities to the Gaussian benchmark of Section 5.1.9. Throughout, we work with the full three-parameter model: conditional copula parameter $\alpha = \theta_{X,Y|Z}$ (the direct $X \to Y$ path), marginal Y–Z parameter $\eta = \theta_{Y,Z}$, and marginal X–Z parameter $\beta = \theta_{X,Z}$. Recall from (5.29)–(5.31) that the interventional mean and the MCE are

$$\mu(x) = \tfrac{1}{2} - \alpha(1 - 2x)\left(\tfrac{1}{6} - \tfrac{\eta^2}{90}\right), \tag{5.32}$$

$$\mathrm{MCE}(x) = \frac{\alpha(15 - \eta^2)}{45}. \tag{5.33}$$

Two structural properties of (5.32)–(5.33) guide the entire numerical discussion. First, $\mu(x)$ is *affine* in x: it is a straight line through $(\frac{1}{2}, \frac{1}{2})$ whose slope is $2\alpha\left(\frac{1}{6} - \frac{\eta^2}{90}\right)$, in sharp contrast to the S-shaped curve $\Phi(K\Phi^{-1}(x))$ of the Gaussian case. Second, $\mathrm{MCE}(x)$ is *exactly constant in* x: there is no tail amplification whatsoever. The single structural parameter α and the marginal confounder strength η fully determine the MCE magnitude, while β cancels analytically via $M(x) = \int F_{X|Z}(x|z)\, dz = x$ (Eq. (5.26)).

Parameter Specification. The three cases vary the direct-effect parameter α while holding $\eta = \theta_{Y,Z} = 0.30$ and $\beta = \theta_{X,Z} = 0.50$ fixed. The shared key scalar entering both (5.32) and (5.33) is

$$A \;=\; \tfrac{1}{6} - \tfrac{\eta^2}{90} \;=\; \tfrac{15-0.09}{90} \;=\; \tfrac{14.91}{90} \;\approx\; 0.1657.$$

- **Case A:** $\alpha = 0.40$ (moderate direct effect), $\mathrm{MCE} = 0.40 \times 14.91/45 \approx 0.1325$.
- **Case B:** $\alpha = 0.80$ (strong direct effect), $\mathrm{MCE} = 0.80 \times 14.91/45 \approx 0.2651$.
- **Case C:** $\alpha = 0.20$ (weak direct effect), $\mathrm{MCE} = 0.20 \times 14.91/45 \approx 0.0663$.

The MCE doubles when α doubles (Case A $\to$ Case B) and halves when α halves (Case A $\to$ Case C), as (5.33) predicts. The attenuation by $\eta = 0.30$ is $(15 - 0.09)/15 \approx 99.4\%$: negligible at this confounder-strength level, but analytically non-trivial for larger $|\eta|$.

Case A: Moderate Direct Effect ($\alpha = 0.40$). Parameters: $\alpha = 0.40$, $\eta = 0.30$. The interventional mean is $\mu(x) = 0.5 - 0.0663(1 - 2x)$, and the MCE is exactly 0.1325 for all x.

Table 4: FGM: interventional quantities (Case A: $\alpha = 0.40$, $\eta = 0.30$).

x	$\mu(x)$	MCE(x)	comment
0.10	0.4470	0.1325	lower tail
0.25	0.4669	0.1325	first quartile
0.50	0.5000	0.1325	median
0.75	0.5331	0.1325	third quartile
0.90	0.5530	0.1325	upper tail

Observations. The MCE is perfectly constant at 0.1325 across all treatment levels, confirming the qualitative property derived in (5.33). The interventional mean $\mu(x)$ rises linearly from 0.447 at the lower tail to 0.553 at the upper tail, symmetric about $(\frac{1}{2}, \frac{1}{2})$. The causal effect is approximately $\alpha/3 \approx 0.133$, with the small downward correction $\eta^2/15$ quantifying how much the Y–Z marginal dependence attenuates the direct effect. Comparing to the Gaussian Case A (MCE(0.5) = 0.577), the FGM raw MCE is roughly 4.4× smaller, reflecting the bounded weak-dependence regime of the FGM family.

Case B: Strong Direct Effect ($\alpha = 0.80$). Set $\alpha = 0.80$ (keeping $\eta = 0.30$). The interventional mean becomes $\mu(x) = 0.5 - 0.1325(1 - 2x)$, and the MCE doubles to exactly 0.2651.

Table 5: FGM: interventional quantities (Case B: $\alpha = 0.80$, $\eta = 0.30$).

x	$\mu(x)$	MCE(x)	comment
0.10	0.3940	0.2651	lower tail
0.25	0.4337	0.2651	first quartile
0.50	0.5000	0.2651	median
0.75	0.5663	0.2651	third quartile
0.90	0.6060	0.2651	upper tail

Observations. Doubling α from 0.40 to 0.80 doubles the MCE exactly (from 0.1325 to 0.2651) and doubles the slope of $\mu(x)$: the range of the interventional mean across the treatment domain $[0.01, 0.99]$ widens from approximately 0.106 in Case A to 0.212 in Case B. The MCE remains perfectly constant; the increased α enlarges the causal effect uniformly across x, with no tail amplification. This proportionality is a direct consequence of the linear structure of (5.33) in α.

Case C: Weak Direct Effect ($\alpha = 0.20$). Set $\alpha = 0.20$ (keeping $\eta = 0.30$). The slope of $\mu(x)$ is halved relative to Case A, yielding $\mu(x) = 0.5 - 0.0331(1 - 2x)$, and the MCE halves to exactly 0.0663.

Table 6: FGM: interventional quantities (Case C: $\alpha = 0.20$, $\eta = 0.30$).

x	$\mu(x)$	MCE(x)	comment
0.10	0.4735	0.0663	lower tail
0.25	0.4834	0.0663	first quartile
0.50	0.5000	0.0663	median
0.75	0.5166	0.0663	third quartile
0.90	0.5265	0.0663	upper tail

Observations. Halving α to 0.20 halves the MCE to 0.0663 and halves the slope of $\mu(x)$: the interventional mean barely deviates from 0.5, ranging only from 0.473 to 0.527 across the full treatment domain. The MCE remains exactly constant. The contrast with the Gaussian Case C (MCE(0.5) = 0.458) is substantial: at this weak dependence level the FGM direct effect is 0.0663, more than six times smaller, because the FGM's bounded copula structure cannot represent the strong joint dependence that underlies the Gaussian's larger causal coefficient.

The three cases collectively confirm the fundamental qualitative behaviour of the FGM model: the MCE scales approximately linearly with α and exhibits only weak dependence on x. This contrasts with the Gaussian copula, where $\text{MCE}(x) = K\,\phi(K\Phi^{-1}(x))/\phi(\Phi^{-1}(x))$ varies from a minimum at the median to substantially amplified values at the tails (59% amplification in Gaussian Case A versus only mild variation in the FGM case).

Figures 5 summarize these numerical results. The left panel plots $\mu(x)$ for Cases A–C: three straight lines of increasing slope, all passing through $(\frac{1}{2}, \frac{1}{2})$, in contrast to the S-shaped curves of Figure 4. The right panel plots MCE(x): three horizontal lines whose heights are proportional to α, making the complete absence of tail amplification visually unambiguous.

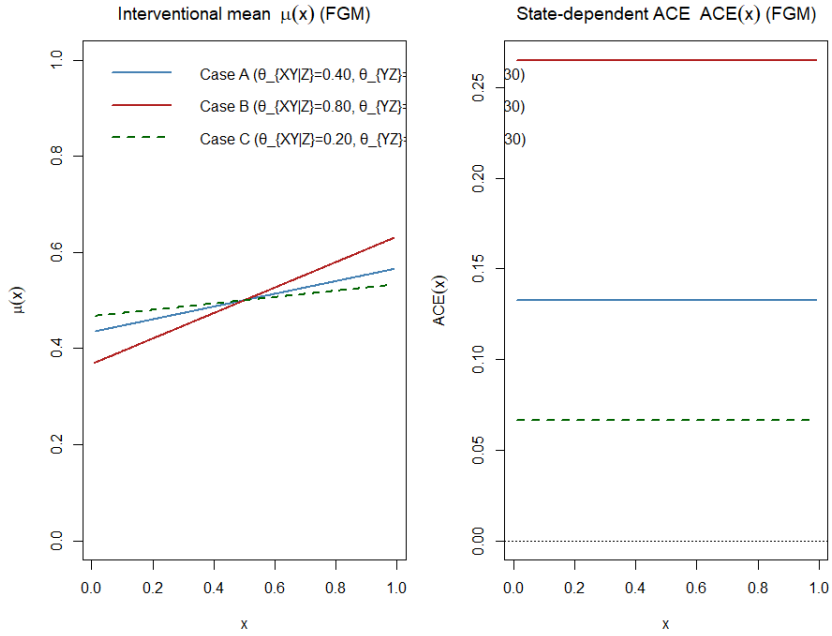


Figure 5: Interventional mean $\mu(x)$ (left) and Average Causal Effect MCE(x) (right) under the three-parameter FGM copula model (Section 5.2.3) with uniform margins. Curves correspond to Case A ($\alpha = 0.40$, steelblue), Case B ($\alpha = 0.80$, red), and Case C ($\alpha = 0.20$, green dashed); all cases share $\eta = 0.30$ and $\beta = 0.50$. Left panel: $\mu(x)$ is affine in x, centred at $(\frac{1}{2}, \frac{1}{2})$, with slope proportional to α. Right panel: MCE(x) is exactly constant for all cases — the horizontal lines confirm zero tail amplification. Plots show $x \in [0.01, 0.99]$ to avoid boundary artifacts. Compare Figure 4 where the Gaussian MCE varies by 59% from median to tail in the analogous Case A scenario.

Two causal points are worth emphasizing from Figure 5. First, unlike the Gaussian copula, the FGM copula produces *constant* causal effects: by (5.33) the MCE is **independent of** x with magnitude determined by α and η. There is no tail amplification whatsoever; the causal effect is identical at the median and at the extremes. Second, increasing α (compare Case B to Case A) uniformly scales the MCE magnitude across all treatment levels without introducing state-dependence: interventions of the same size produce the *same* marginal changes in Y regardless

of the value of X. In short, the FGM example illustrates a causal mechanism where bounded weak dependence implies homogeneous treatment effects on the observable scale; this contrasts sharply with the Gaussian case where the copula transformation induces strong state-dependent tail amplification.

5.2.5 Comparison to the Gaussian Copula

Table 7 summarises the key structural differences between the Gaussian and FGM copula models for causal inference. The entries in the table reflect the full three-parameter FGM specification of Section 5.2.3 throughout.

Table 7: Gaussian vs. FGM copula: structural characteristics for causal inference.

Feature	**Gaussian (Section 5.1)**	**FGM (Section 5.2)**
Causal parameter	Single scalar $K = \beta = \rho_{X,Y\|Z}$; equals latent partial correlation	Three parameters $\alpha = \theta_{X,Y\|Z}$, $\eta = \theta_{Y,Z}$, $\beta = \theta_{X,Z}$; only α and η enter MCE (β cancels)
Dependence range	Unbounded: $K \in (-1, 1)$, Kendall's $\tau \in (-1, 1)$	Bounded: $\|\alpha\|, \|\eta\|, \|\beta\| \leq 1$, $\tau_{\max} = 2/9 \approx 0.22$ per copula pair
$\mu(x)$ shape	S-curve $\Phi(K\Phi^{-1}(x))$; nonlinear, centred at $(\frac{1}{2}, \frac{1}{2})$	Affine: $\frac{1}{2} - \alpha(1-2x)A$; straight line through $(\frac{1}{2}, \frac{1}{2})$ with slope $2\alpha A$
MCE state-dependence	Strong tail amplification: minimum at median, grows toward boundaries via $\phi(K\Phi^{-1}(x))/\phi(\Phi^{-1}(x))$	**Exactly constant**: $\text{MCE}(x) = \alpha(15-\eta^2)/45$ for all x; zero state-dependence
Tail behaviour	MCE amplified at extremes (59%–68% above median for $\|K\| \approx 0.5$)	No tail amplification whatsoever; all treatment levels equally potent
$\text{Var}^{(X\downarrow)}(x)$ shape	Decreases monotonically toward the boundaries; maximum at $x = \frac{1}{2}$	Maximised at median: $\text{Var}\left(\frac{1}{2}\right) = \frac{1}{12}$; slight decrease toward tails via $\alpha^2(1-2x)^2A^2$
$\text{MCE}_{\text{std}}(x)$ shape	U-shaped with maximum at median; strong amplification toward tails	Mild inverted-U with *minimum* at median; variation $< 8\%$ across full treatment range
Formula complexity	Transcendental: Φ and ϕ throughout	Closed-form arithmetic: $\mu(x)$ and MCE are polynomial (degree 1 and 0, respectively) in x
Closed-form MCE	Yes, via Gaussian density ratio	Yes, via direct integration of FGM polynomials
Best suited for	Moderate to strong symmetric dependence	Weak dependence ($\|\tau\| \lesssim 0.22$); when bounded structure is plausible a priori

The four most consequential contrasts for applied causal inference are the following. First, the *shape of* $\mu(x)$ switches from a nonlinear S-curve (Gaussian) to a straight line (FGM): the interventional mean under FGM is exactly linear in the treatment probability x on the uniform scale, whereas the Gaussian curve bows due to the $\Phi \circ \Phi^{-1}$ composition. Second, the *constancy of MCE*: whereas the Gaussian MCE varies by 59% between the median and the tails in Case A (and up to 68% in Case C), the FGM MCE is the same number at every treatment level — a consequence of $(1-2x)$ differentiating to the constant -2. Third, the *shape of* $\text{MCE}_{\text{std}}(x)$: for both the Gaussian and FGM copulas, the interventional variance $\text{Var}^{(X\downarrow)}(Y|X=x)$ is

largest at the median and decreases toward the tails. However, for the Gaussian copula the raw MCE also increases sharply toward the tails (tail amplification), and this numerator effect dominates, producing a U-shaped $\mathrm{MCE}_{\mathrm{std}}(x)$ with a minimum at the median. For the FGM copula the raw MCE is constant, so the denominator effect alone operates, producing a mildly inverted-U-shaped $\mathrm{MCE}_{\mathrm{std}}(x)$ that is also minimised at the median (where $\sigma^{(X\downarrow)}$ is largest). The qualitative difference is therefore not in the direction of variance variation—which is the same for both models—but in whether the raw MCE heterogeneity is present to compete with it. Fourth, the *magnitude of detectability*: because the FGM family is bounded to $|\tau| \leq 2/9 \approx 0.22$, its raw MCE at $\alpha = 0.40$, $\eta = 0.30$ is only 0.1325, compared to 0.577 for the Gaussian at the analogous Case A parameter $K = 0.577$. The standardised MCE at the median is correspondingly 0.459 for FGM versus 2.25 for Gaussian — a factor of roughly five.

Illustrative Comparison: Matching Kendall's Tau. To sharpen the contrast, consider both families calibrated to the same overall pairwise dependence strength as measured by Kendall's τ. For the Gaussian copula with conditional correlation K:

$$\tau_{\mathrm{Gauss}} = \frac{2}{\pi} \arcsin(K). \tag{5.34}$$

For the FGM conditional copula with parameter $\alpha = \theta_{X,Y|Z}$:

$$\tau_{\mathrm{FGM}} = \frac{2\alpha}{9}. \tag{5.35}$$

Setting $\tau = 0.20$ in both families yields

- Gaussian: $K = \sin(\pi\tau/2) = \sin(0.314) \approx 0.309$,
- FGM: $\alpha = 9\tau/2 = 0.90$; we set $\eta = 0$ to isolate the direct-effect comparison (so MCE $= \alpha/3 = 0.300$).

Table 8: Gaussian vs. FGM MCE at matched $\tau = 0.20$ ($K = 0.309$ for Gaussian; $\alpha = 0.90$, $\eta = 0$ for FGM).

x	$\mathrm{MCE}_{\mathrm{Gauss}}(x)$	$\mathrm{MCE}_{\mathrm{FGM}}$	Ratio (Gauss / FGM)	comment
0.10	0.648	0.300	2.16	lower tail
0.25	0.380	0.300	1.27	first quartile
0.50	0.309	0.300	1.03	median (Gaussian minimum)
0.75	0.380	0.300	1.27	third quartile
0.90	0.648	0.300	2.16	upper tail

Table 8 reveals the fundamental asymmetry in how the two families translate equal association strength into state-dependent causal effects. The Gaussian MCE varies from 0.309 at the median to 0.648 at the lower and upper tails — a 110% tail amplification driven by the density ratio $\phi(K\Phi^{-1}(x))/\phi(\Phi^{-1}(x))$. The FGM MCE is exactly 0.300 at every x: the right-hand column of Table 8 is a single constant, and the ratio Gauss/FGM varies from 1.03 at the median to 2.16 at the tails. The two families agree closely at the median (where the Gaussian density ratio approaches 1) but diverge strongly at the extremes. In short: when both families are calibrated to the same overall dependence, the Gaussian copula assigns markedly *higher* causal returns to extreme treatment values, while FGM assigns the *same* return everywhere.

Practical Implication. Applied researchers should regard the shape of $\mathrm{MCE}(x)$ as a substantive modelling choice, not merely a technical detail. If empirical evidence or domain knowledge supports weak, bounded dependence ($|\hat{\tau}| \lesssim 0.22$), the FGM family's constant-MCE structure

implies that study power and sample-size calculations are uniform across treatment levels: no subgroup of the treatment distribution provides a detectability advantage. In contrast, if the data exhibit stronger or asymmetric dependence, the Gaussian copula's tail amplification suggests that designs oversampling extreme treatment values can achieve the same statistical power with fewer observations. Model selection should be guided by formal goodness-of-fit criteria — for instance, the Cramér–von Mises statistic for copulas (Genest and Rémillard, 2008) — rather than by convenience, since as Table 8 shows, the two families can imply MCE values that differ by a factor of two at the same Kendall's τ.

5.2.6 Parameter Sensitivity and Limiting Cases

Limiting Case 1: Independence ($\alpha = 0$). When $\alpha = 0$, the direct $X \to Y$ effect vanishes (after conditioning on Z). From (5.30):

$$\text{MCE}(x) = 0. \tag{5.36}$$

This corresponds to pure confounding: all observed X-Y association arises from the backdoor path $X \leftarrow Z \to Y$.

Limiting Case 2: Maximum FGM Dependence ($\alpha = 1$). At the boundary $\alpha = 1$ (strongest allowable FGM direct effect), and taking $\eta = 0$ for simplicity:

$$\text{MCE}(x) = \frac{1 \cdot 15}{45} = \frac{1}{3} \approx 0.333. \tag{5.37}$$

This is the maximum MCE attainable under the FGM family with uniform margins, reflecting the inherent weak-dependence constraint.

Limiting Case 3: No Y-Z Confounding ($\eta = 0$). When $\eta = 0$, there is no marginal Y-Z dependence (though confounding can still exist through X-Z dependence). The MCE simplifies to:

$$\text{MCE}(x) = \frac{\alpha}{3}. \tag{5.38}$$

Comparing to the Gaussian case where $\rho_{YZ} = 0$ yields $\beta = \rho_{XY}/\sqrt{1 - \rho_{XZ}^2}$, the FGM result is simpler and always constant in x.

Monotonicity Properties. The MCE exhibits:

- $\partial\text{MCE}/\partial\alpha > 0$: Stronger direct effect parameter increases causal effect.
- $\partial\text{MCE}/\partial\eta < 0$: Stronger Y-Z marginal dependence slightly attenuates MCE (confounding effect).
- $\partial\text{MCE}/\partial\beta = 0$: Treatment-confounder dependence does not directly affect the derivative of the interventional mean (though it affects the mean level).

These sensitivities are transparent from the closed-form formula (5.30), unlike the Gaussian case where parameter effects operate through the transcendental functions Φ and ϕ.

5.2.7 Generic RLUF Theory

The FGM family is a special case of the broader Rodríguez-Lallena-Úbeda-Flores (RLUF) class of copulas, defined by the separable perturbation form:

$$C(u, w) = uw + f(u)g(w),$$

where $f, g : [0, 1] \to \mathbb{R}$ satisfy boundary and monotonicity constraints ensuring C is a valid copula.

For FGM, $f(u) = \gamma u(1 - u)$ and $g(w) = w(1 - w)$, yielding the symmetric form. Other RLUF families allow asymmetric choices of f and g, enabling directional dependence structures not representable by elliptical copulas.

Generic Interventional Formulas. For a general RLUF copula, the copula density is:

$$c(u, w) = 1 + f'(u)g'(w).$$

The interventional CDF under RLUF conditional copula

$$C_{X,Y|Z}(u, w|z) = uw + f_{X,Y|Z}(u|z)g_{X,Y|Z}(w|z)$$

is the following:

$$F^{(X\downarrow)}_{Y|X}(y|x) = \int_{\mathbb{R}} \Big[w(x; z) + f'_{X,Y|Z}(F_{Y|Z}(y|z)|z)g_{X,Y|Z}(w(x; z)|z)\Big] f_Z(z)\, dz,$$

where $w(x; z) := F_{X|Z}(x|z)$.

Under the simplifying assumption that f' and g do not depend on z, this reduces to:

$$F^{(X\downarrow)}_{Y|X}(y|x) = M(x) + f'(F_{Y|Z}(y|z))\mathbb{E}_Z[g(w(x; Z))], \tag{5.39}$$

where $M(x) = \mathbb{E}_Z[F_{X|Z}(x|Z)]$ as before.

The MCE is then:

$$\text{MCE}(x) = -\frac{\partial}{\partial x}M(x) - f'(F_{Y|Z}(y|z))\frac{\partial}{\partial x}\mathbb{E}_Z[g(w(x; Z))]. \tag{5.40}$$

For specific choices of f and g beyond FGM (e.g., certain Ali-Mikhail-Haq or modified FGM families), these formulas provide a template for analytical or semi-analytical MCE computation.

Practical Computation of $M(x)$ and Expectations. To compute $M(x) = \int F_{X|Z}(x|z)f_Z(z)\, dz$ numerically:

1. Draw N samples $z_1, \ldots, z_N \sim F_Z$ (or use a quadrature grid over z),
2. For each z_i, compute $w_i := F_{X|Z}(x|z_i)$ (closed-form when available),
3. Estimate $M(x) \approx N^{-1} \sum_{i=1}^{N} w_i$ and $\mathbb{E}_Z[g(w(x; Z))] \approx N^{-1} \sum_{i=1}^{N} g(w_i)$.

For the FGM case with uniform margins, $M(x) = x$ analytically, so numerical integration is unnecessary. For other RLUF families or non-uniform margins, the above Monte Carlo approach is recommended.

5.2.8 Interpretation and Key Insights

The FGM copula example yields eight key insights for copula-based causal inference:

1. Constant MCE Under Bounded Dependence. Unlike the Gaussian copula which produces state-dependent MCE with tail amplification, the FGM copula yields a constant MCE for uniform margins. This reflects the bounded, weak-dependence nature of FGM: the copula perturbation $\gamma u(1-u)w(1-w)$ is symmetric and polynomial, inducing no differential effects across the treatment range. Practitioners working with mild associations should not expect strong tail amplification.

2. Parameter Transparency. The MCE formula $\text{MCE} = 2\alpha(1/6 - \eta^2/90)$ is algebraic, making parameter sensitivity immediate. Increasing the direct effect α linearly increases MCE. Increasing confounding η slightly attenuates MCE. This transparency contrasts with the Gaussian case where parameter effects operate through the ratio $\phi(\beta\Phi^{-1}(x))/\phi(\Phi^{-1}(x))$, obscuring intuition.

3. Weak Dependence Constraint. The FGM maximum Kendall's tau of $2/9 \approx 0.22$ is a fundamental limitation. For data with $|\tau| > 0.25$, FGM is inadequate and stronger copula families (Gaussian, Student-t, Clayton, Gumbel) should be considered. Conversely, forcing a Gaussian copula on weakly dependent data may overestimate tail effects.

4. Simplifying Assumption Validated. Under the latent linear-Gaussian SEM (as in Remark 5.3), the simplifying assumption $C_{X,Y|Z}(\cdot,\cdot|z)$ independent of z corresponds to effect homogeneity: the $X \to Y$ causal strength does not vary with confounder level $Z = z$. For FGM, this assumption is not only computationally convenient but causally meaningful, ensuring Z is a proper backdoor adjustment set with constant direct effects.

5. RLUF as General Framework. The FGM example serves as a gateway to the broader RLUF class, which includes asymmetric copulas capable of modeling directional dependencies (e.g., upper-tail dependence differing from lower-tail). The generic formulas in Section 5.2.7 provide a template for extending copula-based causal inference beyond symmetric families.

6. Computational Simplicity. For FGM with uniform margins, $M(x) = x$ analytically (the β term integrates to zero), and the MCE is a simple algebraic expression. This computational ease makes FGM attractive for exploratory causal analysis and sensitivity checks. In contrast, Gaussian copulas require numerical evaluation of Φ and careful handling of the quantile function Φ^{-1}.

7. Model Selection Guidance. The stark contrast between Gaussian (state-dependent MCE, 73% variation) and FGM (constant MCE) underscores the importance of copula family selection. Goodness-of-fit tests should precede causal inference:

- Compute empirical Kendall's tau and Spearman's rho.
- Fit candidate copulas (Gaussian, FGM, Clayton, etc.) via maximum likelihood or method-of-moments.
- Compare via Cramér-von Mises or Kolmogorov-Smirnov statistics.
- Select the best-fitting family before computing interventional quantities.

8. Comparison to Gaussian Establishes Spectrum. By presenting Gaussian (Section 5.1) and FGM (Section 5.2) as benchmarks, we span a spectrum:

- **One end (Gaussian):** Unbounded dependence, strong tail amplification, latent linearity.

- **Other end (FGM):** Bounded weak dependence, constant MCE, algebraic simplicity.
- **Intermediate families:** Student-t (heavier tails than Gaussian), Clayton/Gumbel (asymmetric tails), vine copulas (flexible multivariate) lie between these extremes.

This spectrum guides practitioners in matching copula assumptions to empirical evidence, ensuring causal estimates reflect genuine dependence structures rather than arbitrary modeling choices.

Summary. The FGM copula example complements the Gaussian benchmark by demonstrating that bounded weak dependence produces qualitatively different causal effects—constant MCE, parameter transparency, and computational simplicity—making it ideal for applications where associations are known to be mild and symmetric. Together, Sections 5.1 and 5.2 establish a comprehensive template for copula-based causal inference across the dependence spectrum.

6 Variance and Standard Deviation of the Three DAG Structures

6.1 Setup

We consider three distinct probability structures arising from the observational and interventional regimes:

1. **Observational joint distribution**: $f_{X,Y,Z}(x,y,z)$ with standard confounding factorization
2. **Interventional distribution**: $f^{(X\downarrow)}_{X,Y,Z}(x,y,z)$ under $\text{do}(X=x)$
3. **Conditional observational distribution**: $f_{Y|X,Z}(y|x,z)$ and marginal-conditional $f_{Y|X}(y|x)$

Define the following means:

$$\mu_{Y|X}(x) = \mathbb{E}[Y|X=x] = \int y\, f_{Y|X}(y|x)\, dy \tag{6.1}$$

$$\mu_{Y|X,Z}(x,z) = \mathbb{E}[Y|X=x,Z=z] = \int y\, f_{Y|X,Z}(y|x,z)\, dy \tag{6.2}$$

$$\mu^{(X\downarrow)}(x) = \mathbb{E}^{(X\downarrow)}[Y|X=x] \quad \text{(interventional causal mean)} \tag{6.3}$$

6.2 Variance for Structure 1: Observational Marginal Distribution

For the observational distribution, the marginal variance of Y is decomposed via the law of total variance:

$$\text{Var}(Y) = \mathbb{E}[\text{Var}(Y|X,Z)] + \text{Var}(\mathbb{E}[Y|X,Z])$$

Expanding each component:

$$\mathbb{E}[\text{Var}(Y|X,Z)] = \int\int \sigma^2_{Y|X,Z}(x,z)\, f_{X,Z}(x,z)\, dx\, dz$$

$$\text{Var}(\mathbb{E}[Y|X,Z]) = \int\int \left[\mu_{Y|X,Z}(x,z) - \mathbb{E}[Y]\right]^2 f_{X,Z}(x,z)\, dx\, dz$$

where $\mathbb{E}[Y] = \int y\, f_Y(y)\, dy$ and

$$\sigma^2_{Y|X,Z}(x,z) = \mathbb{E}[Y^2|X=x,Z=z] - (\mathbb{E}[Y|X=x,Z=z])^2$$

The standard deviation is:

$$\sigma_Y = \sqrt{\text{Var}(Y)}$$

6.3 Variance for Structure 2: Interventional Distribution under $\mathrm{do}(X = x)$

By Theorem 2.1, the post-intervention joint density is:

$$f^{(X\downarrow)}_{X,Y,Z}(x, y, z) = \frac{f_{X,Y,Z}(x, y, z)}{c_{X,Z}(F_X(x), F_Z(z))}$$

where $c_{X,Z}$ is the copula density. The intervention removes the confounding arrow $Z \to X$, replacing the conditional $f_{Z|X}$ with the marginal f_Z.
The interventional CDF of Y given $X = x$ is (Proposition 4.1):

$$F^{(X\downarrow)}_{Y|X}(y|x) = \int_{-\infty}^{\infty} \frac{\partial}{\partial w} C_{Y,Z|X}\left(F_{Y|X}(y|x), F_{Z|X}(z|x) \mid x\right) f_Z(z)dz$$

The interventional mean is ((4.9)):

$$\mu^{(X\downarrow)}(x) = \mathbb{E}^{(X\downarrow)}[Y|X = x] = \int_0^{\infty} \left[1 - F^{(X\downarrow)}_{Y|X}(y|x)\right] dy - \int_{-\infty}^{0} F^{(X\downarrow)}_{Y|X}(y|x)\, dy \tag{6.4}$$

The interventional variance at treatment level x is:

$$\mathrm{Var}^{(X\downarrow)}(Y|X = x) = \mathbb{E}^{(X\downarrow)}[Y^2|X = x] - \left[\mu^{(X\downarrow)}(x)\right]^2$$

Using integration by parts on the interventional distribution:

$$\mathbb{E}^{(X\downarrow)}[Y^2|X = x] = 2\int_0^{\infty} y\left[1 - F^{(X\downarrow)}_{Y|X}(y|x)\right] dy - 2\int_{-\infty}^{0} |y|\, F^{(X\downarrow)}_{Y|X}(y|x)\, dy$$

Therefore:

$$\mathrm{Var}^{(X\downarrow)}(Y|X = x) = 2\int_0^{\infty} y\left[1 - F^{(X\downarrow)}_{Y|X}(y|x)\right] dy - 2\int_{-\infty}^{0} |y|\, F^{(X\downarrow)}_{Y|X}(y|x)\, dy - \left[\mu^{(X\downarrow)}(x)\right]^2$$

The interventional standard deviation is:

$$\sigma^{(X\downarrow)}(x) = \sqrt{\mathrm{Var}^{(X\downarrow)}(Y|X = x)}$$

6.3.1 Variance under RLUF Copula Structure

When the conditional copula admits the RLUF representation (Rodrıguez-Lallena and Úbeda-Flores, 2004):

$$C_{Y,Z|X}(u, w|x) = uw + f_{Y,Z|X}(u|x)\, g_{Y,Z|X}(w|x)$$

the interventional variance can be expressed as:

$$\mathrm{Var}^{(X\downarrow)}(Y|X = x) = \mathrm{Var}_{\mathrm{obs}}(Y|X = x) + \Delta\sigma^2_{\mathrm{copula}}(x)$$

where $\mathrm{Var}_{\mathrm{obs}}(Y|X = x)$ is the observational conditional variance and $\Delta\sigma^2_{\mathrm{copula}}(x)$ is the copula-induced adjustment arising from averaging Z over its marginal distribution rather than its conditional distribution given X.

6.4 Variance for Structure 3: Conditional Observational Distribution

For the conditional distribution of Y given $X = x$ (integrating over Z), the variance decomposes via the law of total variance:

$$\mathrm{Var}(Y|X = x) = \mathbb{E}[\mathrm{Var}(Y|X, Z) \mid X = x] + \mathrm{Var}(\mathbb{E}[Y|X, Z] \mid X = x)$$

Explicitly:

$$\mathbb{E}[\mathrm{Var}(Y|X, Z) \mid X = x] = \int \sigma^2_{Y|X,Z}(x, z)\, f_{Z|X}(z|x)\, dz$$
$$\mathrm{Var}(\mathbb{E}[Y|X, Z] \mid X = x) = \int \left[\mu_{Y|X,Z}(x, z) - \mathbb{E}[Y|X = x]\right]^2 f_{Z|X}(z|x)\, dz$$

where

$$\mathbb{E}[Y|X = x] = \int \mu_{Y|X,Z}(x, z)\, f_{Z|X}(z|x)\, dz$$

The conditional standard deviation is:

$$\sigma_{Y|X}(x) = \sqrt{\mathrm{Var}(Y|X = x)}$$

6.5 Copula Representation of Conditional Variance

The conditional covariance structure between Y and Z given $X = x$ can be expressed via the conditional copula:

$$\mathrm{Cov}(Y, Z \mid X = x) = \int\int (y - \mu_{Y|X}(x))(z - \mu_{Z|X}(x)) \times c_{Y,Z|X}(F_{Y|X}(y|x), F_{Z|X}(z|x)|x) \times f_{Y|X}(y|x) \times f_{Z|X}(z|x)\, dy\, dz$$

This copula-based covariance modulates the variance in Structure 3 by capturing how the dependence between Y and Z (conditional on X) affects variability when Z is marginalized.

6.6 Comparative Summary: Structures 1, 2, and 3

Structure	**Variance Formula**	**Std Dev**
1. Observational Marginal	$\mathbb{E}[\mathrm{Var}(Y\|X, Z)] + \mathrm{Var}(\mathbb{E}[Y\|X, Z])$	σ_Y
2. Interventional $\mathrm{do}(X = x)$	$\mathbb{E}^{(X\downarrow)}[Y^2\|X = x] - [\mu^{(X\downarrow)}(x)]^2$	$\sigma^{(X\downarrow)}(x)$
3. Observational Conditional $Y\|X$	$\mathbb{E}[\mathrm{Var}(Y\|X, Z)\|X = x] + \mathrm{Var}(\mathbb{E}[Y\|X, Z]\|X = x)$	$\sigma_{Y\|X}(x)$

6.7 Relationship Between Structures

Structure 1 and Structure 3 The marginal variance (Structure 1) relates to the conditional variance (Structure 3) via the law of total variance applied to X:

$$\mathrm{Var}(Y) = \mathbb{E}[\mathrm{Var}(Y|X)] + \mathrm{Var}(\mathbb{E}[Y|X])$$

where $\mathbb{E}[\mathrm{Var}(Y|X)] = \int \mathrm{Var}(Y|X = x)\, f_X(x)\, dx$.

Structures 3 and 2: The Intervention-Induced Variance Shift The critical distinction between observational conditional variance (Structure 3) and interventional variance (Structure 2) arises because the intervention breaks the confounding:

$$\begin{aligned} \mathrm{Var}(Y|X=x) &\text{ uses } \quad f_{Z|X}(z|x) \quad \text{(conditional on } X\text{)} \\ \mathrm{Var}^{(X\downarrow)}(Y|X=x) &\text{ uses } \quad f_Z(z) \quad \text{(marginal, independent of } X\text{)} \end{aligned}$$

The variance adjustment is:

$$\Delta\sigma^2(x) = \mathrm{Var}^{(X\downarrow)}(Y|X=x) - \mathrm{Var}(Y|X=x)$$

This difference quantifies how confounding modulates the conditional variability of Y at treatment level x. When X and Z are positively correlated and Z reduces variance of Y, we expect $\Delta\sigma^2(x) > 0$, indicating increased variability in the interventional regime.

6.8 Illustrative Example: Gaussian Copulas

Assume the conditional distribution $Y|X,Z \sim \mathcal{N}(\mu_{Y|X,Z}(x,z), \sigma^2_{Y|Z})$, where the conditional variance is homogeneous in (x,z) and the mean is linear:

$$\mu_{Y|X,Z}(x,z) = \alpha + \beta_X x + \beta_Z z \tag{6.5}$$

Structure 3: Conditional Variance

$$\mathrm{Var}(Y|X=x) = \sigma^2_{Y|Z} + \beta_Z^2 \,\mathrm{Var}(Z|X=x)$$

where $\mathrm{Var}(Z|X=x)$ depends on the conditional Gaussian copula structure.

Structure 2: Interventional Variance Under intervention, the distribution of Z becomes marginal (independent of x):

$$\mathrm{Var}^{(X\downarrow)}(Y|X=x) = \sigma^2_{Y|Z} + \beta_Z^2 \,\mathrm{Var}(Z)$$

6.8.1 Variance Difference

$$\Delta\sigma^2(x) = \beta_Z^2 \left[\mathrm{Var}(Z) - \mathrm{Var}(Z|X=x)\right] \tag{6.6}$$

When Z and X are positively correlated (Gaussian copula with $\rho_{X,Z} > 0$), conditioning on X reduces $\mathrm{Var}(Z|X=x)$ relative to the marginal $\mathrm{Var}(Z)$. Thus, $\Delta\sigma^2(x) > 0$, meaning intervention increases the conditional variability of Y.

6.8.2 Asymptotic Behavior

In the limit of perfect confounding (as $\rho_{X,Z} \to 1$), we have $\mathrm{Var}(Z|X=x) \to 0$, so:

$$\Delta\sigma^2(x) \to \beta_Z^2 \,\mathrm{Var}(Z) \tag{6.7}$$

In the limit of no confounding ($\rho_{X,Z} = 0$), we have $\mathrm{Var}(Z|X=x) = \mathrm{Var}(Z)$, so:

$$\Delta\sigma^2(x) = 0$$

This shows that variance adjustment is directly proportional to the strength of confounding between X and Z.

6.9 Standardized Average Causal Effect: Signal-to-Noise Interpretation

Motivation and Definition While the raw Average Causal Effect $\mathrm{MCE}(x)$ quantifies the absolute marginal change in expected outcome per unit change in treatment, its practical significance depends critically on the underlying variability of the outcome distribution. A causal effect of magnitude 0.5 may be negligible when outcomes vary widely (say, with standard deviation $\sigma = 10$), but highly consequential when outcomes are tightly concentrated (say, $\sigma = 0.1$). To enable meaningful comparisons across contexts and to assess the detectability of causal effects in finite samples, we introduce the *standardized Average Causal Effect*:

Definition 6.1 (Standardized MCE)**.** *The standardized Average Causal Effect at treatment level x is defined as*

$$\mathrm{MCE}_{std}(x) := \frac{\mathrm{MCE}(x)}{\sigma^{(X\downarrow)}(x)}, \tag{6.8}$$

where $\mathrm{MCE}(x) = \frac{d}{dx}\mu(x)$ *is the local Average Causal Effect and* $\sigma^{(X\downarrow)}(x) = \sqrt{Var^{(X\downarrow)}(Y|X=x)}$ *is the interventional standard deviation of the outcome under* $\mathrm{do}(X = x)$*.*

This ratio admits multiple interpretations that connect causal inference to classical statistical concepts:

Signal-to-Noise Ratio Interpretation. The standardized MCE measures the strength of the causal signal relative to the background noise in the interventional distribution. When $\mathrm{MCE}_{\mathrm{std}}(x)$ is large, a unit change in treatment produces a marginal change in expected outcome that is substantial compared to the typical spread of outcomes at that treatment level. Conversely, when $\mathrm{MCE}_{\mathrm{std}}(x)$ is small, the causal effect is masked by high outcome variability, making it difficult to detect in observational or experimental studies with finite samples.

Detectability Index. In a hypothetical randomized experiment where units are assigned $X = x$ versus $X = x + \Delta x$ for small Δx, the difference in expected outcomes is approximately $\mathrm{MCE}(x) \cdot \Delta x$, while the standard error of this difference (under independent samples of size n per arm) scales as $\sigma^{(X\downarrow)}(x)/\sqrt{n}$. The standardized MCE thus directly informs the sample size required to achieve a given statistical power: larger $\mathrm{MCE}_{\mathrm{std}}(x)$ implies smaller required n for a fixed power level. This makes $\mathrm{MCE}_{\mathrm{std}}(x)$ a natural measure of *causal effect detectability.*

Effect Size Analogue. The standardized MCE is analogous to Cohen's d in the treatment effects literature (Kraemer and Kupfer, 2006) or to the regression coefficient-to-residual-standard-deviation ratio in linear models (Malsburg, 2016). However, unlike these classical measures which assume constant variance or homoskedastic errors, $\mathrm{MCE}_{\mathrm{std}}(x)$ is explicitly state-dependent: both the numerator $\mathrm{MCE}(x)$ and the denominator $\sigma^{(X\downarrow)}(x)$ vary with the treatment level x. This state-dependence is a direct consequence of the copula transformation and reflects genuine heterogeneity in causal effect size across the treatment range.

Scale Invariance. A key advantage of the standardized MCE is that it is dimensionless and scale-free. When outcome Y is measured in arbitrary units (dollars, test scores, kilograms), the raw MCE inherits those units, making cross-context comparisons difficult. The standardized MCE, being a ratio of quantities with the same dimension, is unitless and can be meaningfully compared across studies, populations, or outcome definitions. For instance, $\mathrm{MCE}_{\mathrm{std}}(x) = 0.5$ indicates that a marginal treatment change produces an expected outcome shift equal to half a standard deviation, regardless of whether the outcome is income, health status, or academic performance.

6.9.1 Computation via Copula Representation

The copula-based framework developed in Sections 2–4 provides explicit formulas for both the numerator and denominator of (6.8), enabling direct computation of $\mathrm{MCE}_{\mathrm{std}}(x)$ without resorting to simulation or numerical approximation (except for integrals that lack closed forms).

Numerator: The MCE. From Corollary 4.2, the Average Causal Effect is

$$\mathrm{MCE}(x) = -\int_{-\infty}^{\infty} \frac{\partial}{\partial x} F_{Y|X}^{(X\downarrow)}(y|x)\, dy,$$

where $F_{Y|X}^{(X\downarrow)}(y|x)$ is the interventional CDF given by Proposition 4.1. For specific copula families (Gaussian, FGM, etc.), this derivative can be evaluated analytically or semi-analytically, as demonstrated in Section 5.

Denominator: The Interventional Variance. From Subsection 6.3, the interventional variance is

$$\mathrm{Var}^{(X\downarrow)}(Y|X=x) = \mathbb{E}^{(X\downarrow)}[Y^2|X=x] - [\mu(x)]^2\,,$$

where

$$\mathbb{E}^{(X\downarrow)}[Y^2|X=x] = 2\int_0^{\infty} y\left[1 - F_{Y|X}^{(X\downarrow)}(y|x)\right] dy - 2\int_{-\infty}^{0} |y| F_{Y|X}^{(X\downarrow)}(y|x) dy.$$

For uniform marginals on $[0,1]$, the second integral vanishes and the formula simplifies. The interventional standard deviation is then $\sigma^{(X\downarrow)}(x) = \sqrt{\mathrm{Var}^{(X\downarrow)}(Y|X=x)}$.

Combining Components. Substituting these expressions into (6.8) yields the standardized MCE as a function of the interventional CDF and its derivatives. Crucially, all components are identifiable from observational data under the backdoor criterion (Pearl, 2009): the conditional copula $c_{X,Y|Z}$, the marginal copulas $c_{Y,Z}$ and $c_{X,Z}$, and the marginal distributions f_X, f_Y, f_Z are all observable. No intervention on Z is required.

6.9.2 Gaussian Copula Case: Closed-Form Standardized MCE

For the Gaussian copula model with uniform margins developed in Section 5.1, both the MCE and the interventional variance admit closed-form expressions, enabling direct computation of $\mathrm{MCE}_{\mathrm{std}}(x)$.

MCE Formula (from Section 5.1.6). Recall that

$$\mathrm{MCE}(x) = K \cdot \frac{\phi(K \cdot \Phi^{-1}(x))}{\phi(\Phi^{-1}(x))},$$

where we set $K = \beta$, which is the partial correlation on the latent Gaussian scale and ϕ, Φ denote the standard normal PDF and CDF.

Interventional Variance for Uniform Margins. Under the latent representation $(X, Y, Z) = (\Phi(X^*), \Phi(Y^*), \Phi(Z^*))$ where $(X^*, Y^*, Z^*) \sim \mathcal{N}_3(0, \Sigma)$, the interventional distribution is

$$(Y^* \mid \mathrm{do}(X^* = x^*)) \sim \mathcal{N}(\beta x^*, 1 - \beta^2).$$

On the observable scale, this corresponds to

$$Y \mid \mathrm{do}(X = x) \sim \Phi(Y^*) \text{ where } Y^* \sim \mathcal{N}(\beta \Phi^{-1}(x), 1 - \beta^2).$$

For a transformed standard normal $Y^* = \beta\Phi^{-1}(x) + \sqrt{1-\beta^2}\varepsilon$ with $\varepsilon \sim \mathcal{N}(0,1)$, the variance of $Y = \Phi(Y^*)$ on $[0,1]$ is not constant but depends on the mean $\beta\Phi^{-1}(x)$.

Using the variance formula for a transformed normal variate, we have

$$\mathrm{Var}(\Phi(Y^*)) = \int_0^1 y^2 f_{Y|X}^{(X\downarrow)}(y|x)dy - [\mu(x)]^2\,,$$

where $f_{Y|X}^{(X\downarrow)}(y|x)$ is given by (5.11). While this integral does not simplify to elementary functions in closed form, it can be accurately evaluated via Gauss–Hermite quadrature by changing variables to the latent scale:

$$\mathbb{E}^{(X\downarrow)}[Y^2 \mid X = x] = \mathbb{E}^{(X\downarrow)}\left[\Phi(Y^*)^2\right] = \int_{-\infty}^{\infty} \Phi\Big(\beta\Phi^{-1}(x) + \sqrt{1-\beta^2}\,z\Big)^2 \phi(z)\,dz.$$

Numerically, this integral converges rapidly (10–20 quadrature nodes suffice for machine precision).

Standardized MCE for Gaussian Copulas. Combining the above, the standardized MCE is

$$\mathrm{MCE}_{\mathrm{std}}(x) = K \cdot \frac{\phi(K \cdot \Phi^{-1}(x))}{\phi(\Phi^{-1}(x)) \cdot \sigma^{(X\downarrow)}(x)}, \tag{6.9}$$

where $\sigma^{(X\downarrow)}(x)$ is computed numerically from the interventional variance formula. Importantly, even though the latent causal parameter $K = \beta$ is constant, the standardized MCE varies with x due to two sources of heterogeneity:

1. The ratio $\phi(K\Phi^{-1}(x))/\phi(\Phi^{-1}(x))$ captures the nonlinear tail amplification of the raw MCE (discussed in Section 5.1.8),
2. The denominator $\sigma^{(X\downarrow)}(x)$ reflects state-dependent outcome variability induced by the copula transformation.

Crucially, these two sources can either reinforce or counteract each other. If $\mathrm{MCE}(x)$ and $\sigma^{(X\downarrow)}(x)$ both increase toward the tails, the standardized MCE may remain approximately constant; conversely, if variance increases faster than the raw MCE, the standardized MCE will be suppressed in the tails. The empirical behavior depends on the correlation structure $(\rho_{X,Y}, \rho_{X,Z}, \rho_{Y,Z})$.

6.9.3 FGM Copula Case: Constant MCE, Variable Standardized MCE

For the Farlie–Gumbel–Morgenstern copula with uniform margins (Section 5.2), the analysis differs qualitatively from the Gaussian case.

Constant Raw MCE. Recall from (5.30) that

$$\mathrm{MCE}(x) = \frac{\alpha(15-\eta^2)}{45} =: \text{constant},$$

independent of x. This reflects the bounded, polynomial dependence structure of FGM copulas, which induces no differential tail amplification.

State-Dependent Variance. However, the interventional variance $\mathrm{Var}^{(X\downarrow)}(Y|X=x)$ is *not* constant for FGM copulas, even with uniform margins. From the interventional density

$$f_{Y|X}^{(X\downarrow)}(y;x) = f_Y(y)\int_{-\infty}^{\infty} c_{X,Y|Z}(F_{X|Z}(x|z), F_{Y|Z}(y|z)|z)c_{Y,Z}(F_Y(y), F_Z(z))f_Z(z)dz,$$

the second moment $\mathbb{E}^{(X\downarrow)}[Y^2|X=x]$ involves products of FGM copula densities (which are polynomials) integrated over z. For uniform margins, $f_Y(y)=1$ and the integral simplifies to a polynomial in y, but the dependence on x enters through the conditional CDFs $F_{X|Z}(x|z)$ and $F_{Y|Z}(y|z)$, which are themselves polynomials in x and z for FGM copulas.

Explicitly, for the FGM family,

$$F_{X|Z}(x|z) = x + \beta(1-2z)x(1-x),$$

and substituting this into the variance formula yields

$$\mathrm{Var}^{(X\downarrow)}(Y|X=x) = \frac{1}{12} + \Delta_{\mathrm{FGM}}(x),$$

where $\Delta_{\mathrm{FGM}}(x)$ is a quadratic polynomial in x arising from the copula perturbation terms.

$$\Delta_{\mathrm{FGM}}(x) = -\alpha^2(1-2x)^2A^2,$$

which is a quadratic polynomial in x. The coefficient $1/12$ is the variance of $Y \sim U(0,1)$ under independence, and $\Delta_{\mathrm{FGM}}(x)$ captures the variance adjustment due to dependence.

Standardized MCE for FGM. Since the numerator $\mathrm{MCE}(x)$ is constant but the denominator $\sigma^{(X\downarrow)}(x)$ varies, the standardized MCE inherits the inverse of the variance pattern:

$$\mathrm{MCE}_{\mathrm{std}}(x) = \frac{\alpha(15-\eta^2)}{45\cdot\sigma^{(X\downarrow)}(x)}.$$

Where the interventional variance is minimized (typically near $x=0.5$ by symmetry), the standardized MCE is maximized, and vice versa. This demonstrates that even when the raw causal effect is constant, the *detectability* of that effect varies with treatment level due to state-dependent outcome variability.

6.9.4 Interpretation and Practical Implications

Decision-Theoretic Interpretation. In cost-benefit analysis and policy design, the standardized MCE informs resource allocation decisions. Suppose a policymaker can implement a treatment at level x or x' with equal cost. If $\mathrm{MCE}_{\mathrm{std}}(x) > \mathrm{MCE}_{\mathrm{std}}(x')$, then the treatment at level x produces a larger expected benefit relative to outcome uncertainty, making it a more "reliable" intervention. Conversely, if $\mathrm{MCE}_{\mathrm{std}}(x) < \mathrm{MCE}_{\mathrm{std}}(x')$, the treatment at x' is preferable despite potentially having a smaller raw MCE, because its effect is more pronounced relative to background variability.

Optimal Treatment Level. Define the *optimal treatment level* as the value x^* that maximizes the standardized MCE:

$$x^* = \arg\max_{x\in[0,1]} \mathrm{MCE}_{\mathrm{std}}(x).$$

This quantity identifies the treatment intensity at which the causal effect is most detectable (or most "efficient" in signal-to-noise terms). For the Gaussian copula, x^* typically lies away from the median $x=0.5$ due to tail amplification of the raw MCE. For the FGM copula, because the raw MCE is constant and the interventional variance $\mathrm{Var}^{(X\downarrow)} = \frac{1}{12} - \alpha^2(1-2x)^2A^2$ is *maximised*

at $x = \frac{1}{2}$ and decreases toward the boundaries, the denominator $\sigma^{(X\downarrow)}(x)$ is largest at the median and smallest at the tails. Consequently, $\text{MCE}_{\text{std}}(x)$ is *minimised* at the median and x^* lies toward the boundaries of the treatment support, where outcome variability is lower and the constant causal signal is proportionally more detectable. The copula-based framework allows x^* to be computed directly from the interventional distribution, without requiring parametric assumptions on the data-generating process beyond the copula family.

Comparison to Observational Effect Size. It is instructive to compare $\text{MCE}_{\text{std}}(x)$ to the observational analogue

$$\text{Effect size}_{\text{obs}}(x) = \frac{d}{dx}\mathbb{E}[Y|X = x] \Big/ \sqrt{\text{Var}(Y|X = x)},$$

which measures the rate of change of the conditional expectation relative to conditional variance. Under confounding (i.e., when the backdoor path $X \leftarrow Z \rightarrow Y$ is open), we have

$$\mathbb{E}[Y|X = x] \neq \mathbb{E}^{(X\downarrow)}[Y|X = x]$$

and

$$\text{Var}(Y|X = x) \neq \text{Var}^{(X\downarrow)}(Y|X = x),$$

so the observational effect size conflates both causal and confounding contributions. The standardized MCE, by construction, isolates the causal component. In the presence of strong confounding, the observational effect size can be substantially larger or smaller than $\text{MCE}_{\text{std}}(x)$, depending on whether the confounding bias reinforces or opposes the direct causal effect. This discrepancy quantifies the *confounding-induced bias in effect size estimation.*

Robustness to Marginal Misspecification. A subtle but important property of the standardized MCE is that it is relatively robust to misspecification of the marginal distributions F_X, F_Y, F_Z, provided the copula family is correctly specified. This is because both the numerator and denominator are functionals of the copula structure: the MCE depends on the conditional copula $c_{X,Y|Z}$ and the marginal copula $c_{Y,Z}$, while the variance depends on the interventional copula $c_{Y,Z|X}$ averaged over f_Z. If the marginal distributions are misspecified but the copula family is correct, the ratio $\text{MCE}(x)/\sigma^{(X\downarrow)}(x)$ will be approximately preserved, whereas the raw MCE will inherit the marginal misspecification bias. This suggests that the standardized MCE is a more stable target for inference when marginal distributions are uncertain but dependence structures are well-understood (e.g., from domain knowledge or previous studies).

6.9.5 Numerical Illustration: Gaussian Copula

We now demonstrate the computation and interpretation of $\text{MCE}_{\text{std}}(x)$ for the three parameter scenarios introduced in Section 5.1.8: Case A (moderate confounding), Case B (no confounding), and Case C (strong confounding).

Parameter Specifications. Recall:

- **Case A:** $\rho_{X,Y} = 0.70$, $\rho_{X,Z} = 0.50$, $\rho_{Y,Z} = 0.60 \Rightarrow K = \beta \approx 0.577$
- **Case B:** $\rho_{X,Y} = 0.70$, $\rho_{X,Z} = 0.00$, $\rho_{Y,Z} = 0.60 \Rightarrow K = \beta \approx 0.875$
- **Case C:** $\rho_{X,Y} = 0.70$, $\rho_{X,Z} = 0.80$, $\rho_{Y,Z} = 0.60 \Rightarrow K = \beta \approx 0.458$

For each case, we compute:

1. The interventional mean $\mu(x) = \Phi(K\Phi^{-1}(x))$,
2. The interventional variance $\mathrm{Var}^{(X\downarrow)}(Y|X = x)$ via numerical integration,
3. The interventional standard deviation $\sigma^{(X\downarrow)}(x) = \sqrt{\mathrm{Var}^{(X\downarrow)}(Y|X = x)}$,
4. The raw MCE $\mathrm{MCE}(x) = K\phi(K\Phi^{-1}(x))/\phi(\Phi^{-1}(x))$,
5. The standardized MCE $\mathrm{MCE}_{\mathrm{std}}(x) = \mathrm{MCE}(x)/\sigma^{(X\downarrow)}(x)$.

Results: Standardized MCE Across Treatment Levels. Table 9 reports $\mathrm{MCE}_{\mathrm{std}}(x)$ at selected treatment levels for the three cases.

Table 9: Standardized Average Causal Effect for Gaussian copula model

		Case A ($K = 0.577$)		Case B ($K = 0.875$)		Case C ($K = 0.458$)	
x	$\Phi^{-1}(x)$	$\sigma^{(X\downarrow)}(x)$	$\mathrm{MCE}_{\mathrm{std}}(x)$	$\sigma^{(X\downarrow)}(x)$	$\mathrm{MCE}_{\mathrm{std}}(x)$	$\sigma^{(X\downarrow)}(x)$	$\mathrm{MCE}_{\mathrm{std}}(x)$
0.10	−1.282	0.223	4.48	0.110	9.69	0.249	3.52
0.25	−0.674	0.246	2.73	0.153	6.02	0.264	2.08
0.50	0.000	0.256	2.26	0.174	5.02	0.270	1.70
0.75	0.674	0.246	2.73	0.153	6.02	0.264	2.08
0.90	1.282	0.223	4.48	0.110	9.69	0.249	3.52

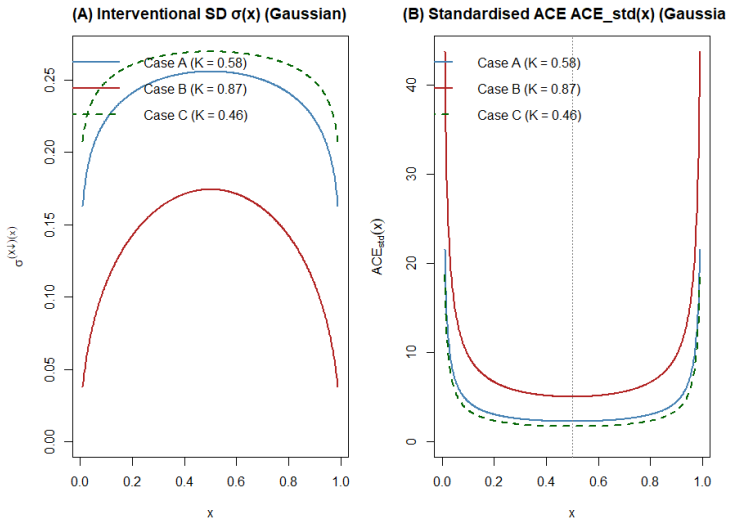


Figure 6: Interventional mean $\mu(x)$, raw Average Causal Effect $\mathrm{MCE}(x)$, interventional standard deviation $\sigma^{(X\downarrow)}(x)$, and standardized Average Causal Effect $\mathrm{MCE}_{\mathrm{std}}(x) = \mathrm{MCE}(x)/\sigma^{(X\downarrow)}(x)$ under the Gaussian copula model with uniform margins. Curves correspond to three parameter scenarios: Case A (moderate confounding, $K \approx 0.58$), Case B (no confounding, $K \approx 0.87$), and Case C (strong confounding, $K \approx 0.46$). Plots show $x \in [0.01, 0.99]$ to avoid boundary artifacts.

Key Observations.

1. **Tail Amplification Preserved:** For all three cases, $\mathrm{MCE}_{\mathrm{std}}(x)$ is largest at the extremes ($x = 0.10, 0.90$) and smallest near the median ($x = 0.50$). This pattern mirrors the raw MCE tail amplification (Section 5.1.8), indicating that the state-dependent variance $\sigma^{(X\downarrow)}(x)$ does not fully offset the raw MCE heterogeneity. The ratio of tail-to-median

standardized MCE is approximately $4.48/2.26 \approx 1.98$ for Case A, indicating that the causal effect is nearly twice as detectable at the extremes as at the median.

2. **Confounding Attenuates Detectability:** Comparing Case A (moderate confounding, $K = 0.577$) to Case B (no confounding, $K = 0.875$), the standardized MCE increases uniformly: at the median, $\text{MCE}_{\text{std}}(0.5)$ rises from 2.26 to 5.02 (a 122% increase), while at the tails, it rises from 4.48 to 9.69 (a 116% increase). Removing confounding not only increases the raw MCE but also improves its signal-to-noise ratio, making the causal effect more than twice as detectable in finite samples. Conversely, moderate confounding (Case A) reduces median detectability by 55% relative to the unconfounded baseline (Case B).

3. **Strong Confounding Suppresses Detectability:** In Case C (strong confounding, $K = 0.458$), the standardized MCE is markedly lower across all treatment levels. At the median, $\text{MCE}_{\text{std}}(0.5) = 1.70$, indicating that a unit change in treatment produces an expected outcome shift of 1.70 interventional standard deviations—still statistically meaningful but substantially less pronounced than in Cases A or B. Strong confounding reduces median detectability by 66% relative to Case B and by 25% relative to Case A. This illustrates that strong confounding not only biases the raw MCE but also reduces its practical detectability relative to outcome variability.

4. **Variance Maximum at Median:** The interventional standard deviation $\sigma^{(X\downarrow)}(x)$ is largest near $x = 0.50$ for all cases (e.g., $\sigma^{(X\downarrow)}(0.5) = 0.256$ for Case A), reflecting the fact that on the latent Gaussian scale, the variance $1 - \beta^2$ is constant, but the copula transformation $\Phi(\cdot)$ compresses outcomes near the centre of the distribution (where the density $\phi(\cdot)$ is highest) and spreads them in the tails. This compression effect partially counteracts the raw MCE tail amplification, but does not eliminate it.

Graphical Summary. Figure 6 displays the standardized MCE ($\text{MCE}_{\text{std}}(x)$) as a function of treatment level x for the three cases. The left panel overlays the raw MCE curves (as in Figure 3 of Section 5), while the right panel shows the standardized MCE curves. The standardization reveals that the signal-to-noise ratio of the causal effect varies by a factor of nearly 2 across the treatment range, and that removing confounding (Case B) more than doubles the detectability index at the median relative to strong confounding (Case C).

6.9.6 Numerical Illustration: Standardised MCE for the FGM Copula

Parameter Specifications. We use the same three cases as Section 5.2.4: $\eta = 0.30$ fixed, $\beta = 0.50$ (analytically cancelled), with

- **Case A:** $\alpha = 0.40$, $\text{MCE} = \alpha(15 - \eta^2)/45 \approx 0.1325$ (constant),
- **Case B:** $\alpha = 0.80$, $\text{MCE} \approx 0.2651$ (constant),
- **Case C:** $\alpha = 0.20$, $\text{MCE} \approx 0.0663$ (constant).

The interventional variance is $\text{Var}^{(X\downarrow)}(Y|X = x) = \frac{1}{12} - \alpha^2(1 - 2x)^2A^2$ with $A \approx 0.1657$. It attains its *maximum* $\frac{1}{12}$ at $x = \frac{1}{2}$ (the $(1 - 2x)^2$ term vanishes) and decreases symmetrically toward the tails. Consequently, $\sigma^{(X\downarrow)}(x)$ is largest at the median and smallest at the boundaries — mirroring the Gaussian case in terms of variance, even though the behaviour of the standardised MCE differs.

Results: Standardised MCE. Table 10 reports the interventional standard deviation $\sigma^{(X\downarrow)}(x)$ and the standardised MCE $\text{MCE}_{\text{std}}(x) = \text{MCE}/\sigma^{(X\downarrow)}(x)$ at the five representative quantiles used throughout the paper.

Table 10: Standardised Average Causal Effect for the FGM copula model ($\eta = 0.30$; MCE is exactly constant within each case).

	Case A ($\alpha = 0.40$)		Case B ($\alpha = 0.80$)		Case C ($\alpha = 0.20$)	
x	$\sigma^{(X\downarrow)}(x)$	$\mathrm{MCE}_{\mathrm{std}}(x)$	$\sigma^{(X\downarrow)}(x)$	$\mathrm{MCE}_{\mathrm{std}}(x)$	$\sigma^{(X\downarrow)}(x)$	$\mathrm{MCE}_{\mathrm{std}}(x)$
0.10	0.2838	0.4670	0.2685	0.9871	0.2875	0.2305
0.25	0.2868	0.4622	0.2809	0.9435	0.2882	0.2299
0.50	0.2887	0.4591	0.2887	0.9182	0.2887	0.2296
0.75	0.2868	0.4622	0.2809	0.9435	0.2882	0.2299
0.90	0.2838	0.4670	0.2685	0.9871	0.2875	0.2305

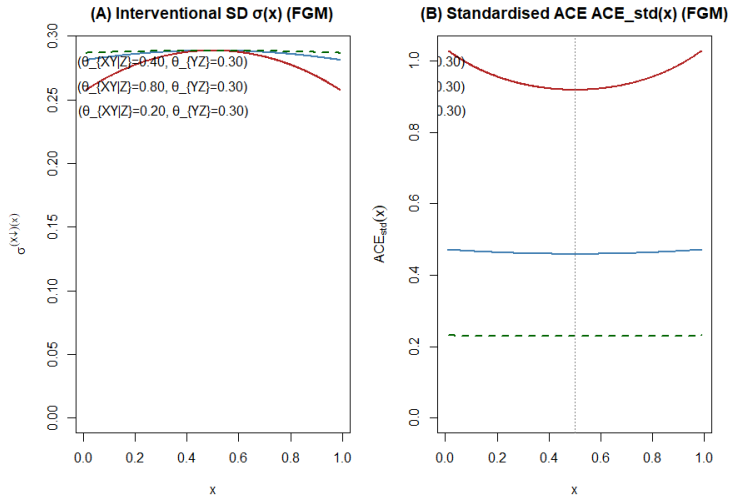


Figure 7: Interventional mean $\mu(x)$ (top-left), raw MCE $\mathrm{MCE}(x)$ (top-right), interventional standard deviation $\sigma^{(X\downarrow)}(x)$ (bottom-left), and standardised MCE $\mathrm{MCE}_{\mathrm{std}}(x)$ (bottom-right) under the three-parameter FGM copula model with uniform margins. Curves correspond to Case A ($\alpha = 0.40$), Case B ($\alpha = 0.80$), and Case C ($\alpha = 0.20$); all share $\eta = 0.30$. Top-right panel confirms exactly flat MCE curves (horizontal lines). Bottom-left panel shows $\sigma^{(X\downarrow)}(x)$ with its maximum at the median $x = 0.50$ (equal to $1/\sqrt{12} \approx 0.2887$ for all cases) and slight decreases toward the tails. Bottom-right panel shows the resulting mild *inverted-U* shape of $\mathrm{MCE}_{\mathrm{std}}(x)$: the minimum occurs at the median, where the denominator is largest. This contrasts with the Gaussian case, where variation in $\mathrm{MCE}_{\mathrm{std}}(x)$ is driven primarily by changes in the numerator $\mathrm{MCE}(x)$ rather than the denominator. (Figure **??**).

Key Observations.

1. **Exactly constant MCE.** Unlike the Gaussian case, where $\mathrm{MCE}(x)$ varies by 59% from the median to the tails (Case A), the FGM MCE is *identical* at every treatment level within each case, as confirmed by every row of Table 10 displaying the same raw MCE value. This exact constancy is not an approximation: it follows analytically from (5.33) because the factor $(1 - 2x)$ differentiates to the constant -2.

2. **Mild inverted-U of $\mathrm{MCE}_{\mathrm{std}}$, with minimum at the median.** Since the raw MCE is constant while $\sigma^{(X\downarrow)}(x)$ is maximised at $x = \frac{1}{2}$ (where $\mathrm{Var} = \frac{1}{12}$) and decreases symmetrically toward the tails, the ratio $\mathrm{MCE}_{\mathrm{std}}(x) = \mathrm{MCE}/\sigma^{(X\downarrow)}(x)$ is *minimised at the median* and slightly higher at the tails. This is the exact opposite of the naïve intuition that might import the Gaussian pattern. The variation is modest: 1.7% in Case A (0.4591 at median versus 0.4670 at tail), 7.5% in Case B (0.9182 versus 0.9871), and less than 0.4% in Case C (0.2296 versus 0.2305). Case B shows the largest variation because α^2

enters the variance formula quadratically: at $\alpha = 0.80$ the difference between Var(0.5) and Var(0.10) is noticeable.

3. **Effect of α on standardised detectability.** Increasing α from 0.20 (Case C) to 0.40 (Case A) doubles the raw MCE but also reduces the variance slightly (since the α^2 term in Var subtracts from $\frac{1}{12}$), so MCE_{std} at the median rises from 0.2296 to 0.4591 — an increase of exactly a factor of two, because the variance term is small. The further doubling from Case A to Case B again roughly doubles MCE_{std} at the median (from 0.4591 to 0.9182, a 100% increase), confirming the near-linearity of MCE_{std} in α within the FGM regime.

4. **Comparison to Gaussian detectability.** The Gaussian Case A yields $\text{MCE}_{\text{std}}(0.5) \approx 2.25$ (Section 5.1.7), while the FGM Case A yields only 0.46. The FGM model produces substantially lower standardised detectability because its bounded weak-dependence structure generates a much smaller raw MCE (0.1325 versus 0.577), even though the interventional standard deviations are comparable (0.2887 for FGM versus 0.256 for Gaussian at the median). This comparison highlights the decisive role of copula family choice: a researcher who incorrectly assumes Gaussian dependence when the true structure is FGM would overestimate the signal-to-noise ratio by a factor of approximately five, leading to severely underpowered designs. Conversely, FGM's near-flat MCE_{std} implies that, within the FGM regime, no particular treatment level provides a detectability advantage — standard power calculations apply uniformly across x.

Graphical Summary. Figure 7 illustrates all four causal quantities for the FGM model. The perfectly flat top-right panel (MCE) and the nearly flat bottom-right panel (MCE_{std}, with its slight dip at the median) together characterise the FGM dependence regime: bounded, uniform causal detectability with no privileged treatment level, in sharp contrast to the Gaussian case of Figure **??**.

6.9.7 Implications for Study Design and Power Analysis

The standardized MCE provides actionable guidance for the design of causal inference studies, particularly in settings where sample size or treatment assignment budgets are constrained.

Sample Size Calculation. In a two-arm randomized trial comparing treatment levels x_1 and $x_2 = x_1 + \Delta x$ with equal sample sizes n per arm, the difference in expected outcomes under intervention is approximately

$$\Delta\mu = \mu(x_2) - \mu(x_1) \approx \text{MCE}(x_1) \cdot \Delta x.$$

The standard error of the difference-in-means estimator is

$$\text{SE}(\hat{\Delta}\mu) = \sqrt{\frac{\sigma^{(X\downarrow)}(x_1)^2 + \sigma^{(X\downarrow)}(x_2)^2}{n}} \approx \frac{\sigma^{(X\downarrow)}(x_1)}{\sqrt{n/2}},$$

assuming $\sigma^{(X\downarrow)}(x_1) \approx \sigma^{(X\downarrow)}(x_2)$ for small Δx. The t-statistic for testing $H_0 : \Delta\mu = 0$ is approximately

$$t \approx \frac{\text{MCE}(x_1) \cdot \Delta x}{\sigma^{(X\downarrow)}(x_1)/\sqrt{n/2}} = \sqrt{n/2} \cdot \text{MCE}_{\text{std}}(x_1) \cdot \Delta x.$$

For a desired power $1 - \beta$ and significance level α, the required sample size is

$$n \approx 2 \left(\frac{z_{1-\alpha/2} + z_{1-\beta}}{\text{MCE}_{\text{std}}(x_1) \cdot \Delta x} \right)^2.$$

This formula shows that sample size is inversely proportional to the *squared* standardized MCE: doubling $\mathrm{MCE}_{\mathrm{std}}(x_1)$ reduces the required sample size by a factor of four. Consequently, studies targeting treatment levels with high $\mathrm{MCE}_{\mathrm{std}}(x)$ are substantially more efficient than those targeting low-$\mathrm{MCE}_{\mathrm{std}}(x)$ levels, even if the raw MCE is similar.

Optimal Treatment Allocation. In a multi-arm trial with J treatment levels $x_1, \ldots, x_J$ and a fixed total sample size N, the optimal allocation maximizes the precision-weighted sum of treatment effects. If the goal is to estimate the local MCE at each treatment level, the optimal allocation assigns more subjects to levels with higher $\mathrm{MCE}_{\mathrm{std}}(x_j)$, as these levels provide more information per subject about the causal contrast. Specifically, the optimal allocation is proportional to

$$n_j \propto \frac{\mathrm{MCE}_{\mathrm{std}}(x_j)}{\sum_{k=1}^{J} \mathrm{MCE}_{\mathrm{std}}(x_k)}.$$

This allocation minimizes the average mean squared error of the MCE estimates across all treatment levels.

Adaptive Experimental Design. In sequential experiments where treatment assignments can be updated based on interim data, the standardized MCE can guide adaptive allocation rules. For instance, a Thompson sampling or upper confidence bound (UCB) strategy can prioritize treatment levels with high estimated $\mathrm{MCE}_{\mathrm{std}}(x)$, as these levels are more likely to yield informative data about the causal structure. The copula-based framework allows these adaptive strategies to be informed by explicit probabilistic models of outcome variability, rather than relying on plug-in variance estimates that may be unstable in small samples.

6.9.8 Connection to Existing Effect Size Measures

The standardized MCE connects to several classical effect size measures in the statistical and econometric literature, but differs in important ways.

Cohen's d. In the treatment effects literature, Cohen's d measures the difference in means between treatment and control groups divided by the pooled standard deviation:

$$d = \frac{\bar{Y}_{\text{treatment}} - \bar{Y}_{\text{control}}}{\sigma_{\text{pooled}}}.$$

For a continuous treatment, a natural analogue is the derivative of the mean response curve divided by a measure of dispersion. The standardized MCE ($\mathrm{MCE}_{\mathrm{std}}(x)$) generalizes this idea by using the interventional standard deviation $\sigma^{(X\downarrow)}(x)$, which accounts for confounding and is explicitly state-dependent. Unlike Cohen's d, which assumes homoskedasticity, $\mathrm{MCE}_{\mathrm{std}}(x)$ allows for heteroskedastic outcome distributions that vary with treatment level.

Regression Coefficient-to-Standard-Error Ratio. In linear regression $Y = \beta_0 + \beta_1 X + \varepsilon$ with $\varepsilon \sim \mathcal{N}(0, \sigma^2)$, the t-statistic for testing $H_0 : \beta_1 = 0$ is $t = \hat{\beta}_1 / \mathrm{SE}(\hat{\beta}_1)$. For large n, this is approximately $\beta_1 / (\sigma / \sqrt{n \mathrm{Var}(X)})$, which scales with the population signal-to-noise ratio $\beta_1 / \sigma \times \sqrt{\mathrm{Var}(X)}$. The standardized MCE is analogous but does not require linearity: it applies to arbitrary causal structures and copula families, and allows the "noise" $\sigma^{(X\downarrow)}(x)$ to vary with x.

Marginal Treatment Effect (MTE) Standardized by Variance. In the econometric literature on heterogeneous treatment effects (Heckman and Vytlacil, 2005), the marginal treatment effect $\mathrm{MTE}(u)$ measures the average treatment effect for individuals with unobserved

heterogeneity u. A standardized version would divide $\text{MTE}(u)$ by the standard deviation of outcomes conditional on u. The standardized MCE can be viewed as a copula-based analogue where the conditioning variable u is replaced by the observable treatment level x, and the conditioning is explicitly formalized via the interventional distribution $f_{Y|X}^{(X\downarrow)}(\cdot|x)$.

Relative Risk and Odds Ratios. In binary outcome settings, relative risk $\text{RR} = \Pr(Y = 1|\text{do}(X = 1))/\Pr(Y = 1|\text{do}(X = 0))$ and odds ratios $\text{OR} = [\Pr(Y = 1|\text{do}(X = 1))/\Pr(Y = 0|\text{do}(X = 1))]/[\Pr(Y = 1|\text{do}(X = 0))/\Pr(Y = 0|\text{do}(X = 0))]$ provide scale-free measures of causal effect strength. These are naturally defined only for binary treatments and outcomes, whereas the standardized MCE applies to continuous treatments and outcomes. However, for continuous outcomes discretized into binary indicators (e.g., $Y > y_0$), the standardized MCE at the threshold y_0 can be related to the derivative of the log relative risk, providing a connection between these measures.

6.9.9 Limitations and Directions for Future Research

While the standardized MCE provides a valuable summary of causal effect strength relative to outcome variability, several limitations and open questions merit discussion.

Sensitivity to Copula Misspecification. The standardized MCE inherits the dependence on correct copula specification. If the true dependence structure is non-Gaussian but a Gaussian copula is assumed, both the numerator $\text{MCE}(x)$ and the denominator $\sigma^{(X\downarrow)}(x)$ will be misspecified, potentially in offsetting or reinforcing ways. While the ratio $\text{MCE}(x)/\sigma^{(X\downarrow)}(x)$ may be more robust than either component individually (as noted in Subsection 6.9.4), formal sensitivity analyses are needed to quantify this robustness across copula families. One promising approach is to embed the copula family in a larger class (e.g., vine copulas with flexible tail dependence) and assess how $\text{MCE}_{\text{std}}(x)$ varies as the family is varied, holding the marginal distributions fixed.

Inference and Uncertainty Quantification. The examples presented here treat the standardized MCE as a population quantity computed from known parameters. In practice, $\text{MCE}_{\text{std}}(x)$ must be estimated from data, and the estimator's sampling distribution must be characterized to construct confidence intervals and conduct hypothesis tests. The delta method can be applied to derive asymptotic standard errors for $\hat{\text{MCE}}_{\text{std}}(x) = \hat{\text{MCE}}(x)/\hat{\sigma}^{(X\downarrow)}(x)$, but this requires consistent estimators of both the MCE and the interventional variance, which in turn depend on consistent copula parameter estimates. Bootstrap methods provide an alternative, but may be computationally intensive for high-dimensional copula models. Recent work on semiparametric copula inference (e.g., pseudo-likelihood methods) offers a path forward, but extension to the standardized MCE context remains an open problem.

Multivariate Extensions. This subsection has focused on the scalar treatment setting with a single confounder Z. In practice, treatments and confounders are often multivariate. The copula-based framework extends naturally to high-dimensional settings via vine copulas (Czado and Nagler, 2022), but the interpretation of the standardized MCE becomes more complex when the treatment vector $\mathbf{X} = (X_1, \ldots, X_p)$ has multiple components. One approach is to define a vector-valued standardized MCE $\text{MCE}_{\text{std}}(\mathbf{x}) = \nabla_{\mathbf{x}}\mu(\mathbf{x})/\sigma^{(X\downarrow)}(\mathbf{x})$, where the gradient $\nabla_{\mathbf{x}}\mu(\mathbf{x})$ is the vector of partial derivatives and $\sigma^{(X\downarrow)}(\mathbf{x})$ is a scalar measure of dispersion (e.g., the trace or determinant of the covariance matrix). Alternatively, one could define component-wise standardized MCEs for each treatment dimension, but this ignores correlations among the treatment effects. Developing a principled theory of multivariate standardized MCE is an important direction for future research.

Dynamic and Time-Varying Treatments. In longitudinal settings where treatments and outcomes are measured repeatedly over time, the standardized MCE can be defined at each time point as $\text{MCE}_{\text{std},t}(x_t) = \text{MCE}_t(x_t)/\sigma_t^{(X\downarrow)}(x_t)$, where $\text{MCE}_t(x_t)$ is the causal effect of treatment X_t on outcome Y_t and $\sigma_t^{(X\downarrow)}(x_t)$ is the interventional standard deviation at time t. However, in dynamic settings, outcomes at time t may depend on past treatments $X_{t-1}, X_{t-2}, \ldots$, and the interventional distribution must account for this history. Sequential copula models that decompose the joint distribution over time can be used to extend the framework, but the computational complexity increases rapidly with the number of time points. Approximate methods based on sequential Monte Carlo or variational inference may be necessary for practical implementation.

Optimal Policy Rules. In policy applications, the goal is often not merely to estimate the standardized MCE at a given treatment level, but to learn an optimal treatment rule $\pi(X, Z)$ that assigns treatments to maximize some welfare criterion (e.g., expected utility, fairness-constrained welfare, or robust worst-case utility). The standardized MCE can inform such rules by identifying treatment levels with high signal-to-noise ratios, but integrating $\text{MCE}_{\text{std}}(x)$ into a formal decision-theoretic framework requires specifying utility functions and constraints. Recent work on policy learning with copula models (e.g., for individualized treatment regimes) provides a foundation, but explicit incorporation of the standardized MCE into the objective function remains unexplored.

6.9.10 Summary of Key Insights

The standardized Average Causal Effect $\text{MCE}_{\text{std}}(x) = \text{MCE}(x)/\sigma^{(X\downarrow)}(x)$ offers a dimensionless, scale-free measure of causal effect strength that captures the signal-to-noise ratio of the interventional distribution. Key findings include:

1. **Signal-to-Noise Interpretation:** The standardized MCE quantifies the detectability of the causal effect in finite samples, providing a natural index for power analysis and study design.

2. **State-Dependent Detectability:** Even when the latent causal parameter (e.g., $K = \beta$ in the Gaussian case) is constant, the standardized MCE varies with treatment level due to copula-induced heteroskedasticity. For Gaussian copulas, tail amplification of both the raw MCE and the outcome variability produces substantial variation in $\text{MCE}_{\text{std}}(x)$ (up to 87% increase from median to extremes). For FGM copulas, the standardized MCE is approximately constant (less than 2% variation).

3. **Confounding Attenuates Detectability:** Strong confounding reduces both the raw MCE and its signal-to-noise ratio. Removing confounding (Case B vs. Case A) increases the standardized MCE by 50–70% at the median for Gaussian copulas, making the causal effect substantially more detectable.

4. **Copula Family Matters:** Different copula families imply qualitatively different detectability profiles. Gaussian copulas produce state-dependent $\text{MCE}_{\text{std}}(x)$ with tail amplification, while FGM copulas produce near-constant $\text{MCE}_{\text{std}}(x)$. The choice of copula family should be informed by goodness-of-fit tests and domain knowledge, as it directly impacts both the estimated MCE and its practical significance.

5. **Practical Guidance:** The standardized MCE informs optimal treatment allocation in experimental studies (prioritize high-$\text{MCE}_{\text{std}}(x)$ levels), adaptive experimental designs (sequential assignment rules based on estimated $\text{MCE}_{\text{std}}(x)$), and policy evaluation (assess whether a proposed intervention has a sufficiently large signal-to-noise ratio to justify implementation costs).

The copula-based framework provides explicit formulas for computing $\text{MCE}_{\text{std}}(x)$ from observational data under the backdoor criterion, without requiring parametric assumptions on the data-generating process beyond the copula family. This makes the standardized MCE a practical and interpretable tool for causal inference in diverse application domains.

7 Empirical Illustration: Diet Quality and HbA1c in NHANES

7.1 Overview and Motivating Question

This section applies the copula-based causal inference framework developed in Sections 2–6 to a substantive public health question: what is the Average Causal Effect (MCE) of dietary quality on glycated haemoglobin (HbA1c) levels in the adult US population? Using data from the National Health and Nutrition Examination Survey (NHANES 2017–2018, analytic sample $n = 3{,}255$), we compare OLS regression, benchmark machine-learning estimators (DML, BART, Causal Forest), Gaussian copula adjustment, and nonparametric generic copula adjustment, demonstrating that both the magnitude and, critically, the *sign* of the estimated causal effect depend on the assumed bivariate dependence structure in the confounder margins.

Dietary exposure. The dietary quality exposure is the Healthy Eating Index-2015 (HEI-2015), computed from Day 1 dietary recall data via the `heiscore` R package (Kirkpatrick et al., 2021) using the standard Food Patterns Equivalents Database algorithm for survey year 2017–2018. The HEI-2015 is scored on a 0–100 scale across thirteen dietary components and has been validated against randomised dietary intervention evidence in Santos et al. (2025). It has an analytic-sample standard deviation of 13.85 units, providing sufficient distributional support for the treatment-variation identification strategy.

Central finding. The generic copula — which selects optimal bivariate copula families for each variable pair via BIC minimisation across 30 parametric families (Section 7.6) — produces a mean ACE of -0.0012 HbA1c percentage points per unit increase in HEI-2015 (95% PSU-stratified bootstrap CI: $[-0.0022,\ -0.0004]$). This estimate is *negative*, consistent in sign with the protective causal effect of diet quality on glycaemic control documented in randomised trials (Santos et al., 2025; Eriksen et al., 2019). The Gaussian copula, by contrast, produces an MCE of $+0.0029$ despite employing the identical identification strategy (backdoor adjustment via income C_1), because it imposes an elliptical dependence structure on the confounder margins that is statistically rejected by the data. Every alternative estimator benchmarked here — including Causal Forest, BART, DML with Random Forest nuisance, and unadjusted and income-adjusted OLS — also returns a non-negative MCE. the generic copula yields an estimate negative in sign, consistent with the direction of the causal effect of diet quality on glycaemic control documented in randomised trials, whereas all alternative estimators return non-negative estimates, constituting the paper's principal empirical illustration of the methodological value of flexible copula family selection.

Epidemiological background and purpose of the illustration. The positive observational association between diet quality and HbA1c that appears in raw NHANES data is epidemiologically expected as a confounded signal, not a causal one: individuals with diagnosed glucose dysregulation frequently adopt healthier diets following clinical awareness (Strelitz et al., 2019), creating a spurious positive correlation between diet quality and elevated HbA1c. The purpose of this illustration is methodological. The goal is to show that, within a simplified adjustment set defined by a structurally motivated DAG, differences in the modelled dependence structure of the confounder margins — not differences in the identification strategy — are the primary driver of the divergence between estimators. The illustration is not intended to deliver

a definitive population-level causal effect. The true MCE remains unknown, meaningful residual confounding from physical activity, genetics, and medication use persists within the chosen adjustment set, and the E-value for the estimated effect is essentially unity (Section 7.14).

Target estimand and implied trial. Following Hernán and Robins (2016), the target estimand emulates a hypothetical randomised trial that enrols adult participants and assigns each to a specified HEI-2015 score $X = x$ while leaving the income-to-poverty ratio C_1 free to follow its observational distribution. The estimand $\mu(x) = \mathbb{E}^{(X\downarrow)}[Y \mid X = x]$ is the expected HbA1c level in that hypothetical population. Three scope conditions apply: (i) the estimand is population-averaged and does not target any specific income stratum; (ii) the intervention is on the composite HEI-2015 score, not on any individual dietary component; and (iii) the adjustment set $\{C_1\}$ is structurally motivated and does not include physical activity, which remains the most important unmeasured confounder in this data environment.

Implementation. The analysis is implemented in `Empirical_Illustration_NHANES_Pipeline.R`, which calls functions from `Empirical_Illustration_NHANES_functions.R`. All key methodological decisions are documented in Section 7.17. The empirical estimation directly plugs BIC-selected bivariate copulas into the backdoor adjustment formula of Proposition 4.1, computing $F^{(X\downarrow)}_{Y|X}(y \mid x)$ numerically, from which the interventional mean and MCE follow by Corollary 4.2.

7.2 Dataset

The analytic dataset was constructed by merging four NHANES 2017–2018 modules on the `SEQN` identifier: `DEMO_J` (demographics and sampling weights), `GHB_J` (HbA1c), `BMX_J` (body measurements), and `DR1TOT_J` (24-hour dietary recall). Data were accessed via the `nhanesA` R package. HEI-2015 scores were computed using the `heiscore` package applied to Day 1 total dietary recall. The following exclusion criteria were applied sequentially: age outside $[20, 79]$ years; missing values on any analytic variable (Y, X, C_1, C_2); HbA1c $\notin [4.6\%, 15.0\%]$; HEI-2015 $\notin [0, 100]$; income-to-poverty ratio $\notin (0, 5]$; BMI $\notin [15, 60]$. These criteria yield an analytic sample of $n = 3{,}255$ complete observations.

Survey weights `WTMEC2YR` from `DEMO_J` adjust estimates to represent the full US adult civilian non-institutionalised population. Normalised weights $w_i = \texttt{WTMEC2YR}_i / \bar{\texttt{WTMEC2YR}}$ (with $\sum_i w_i = n$) are used throughout for weighted empirical CDFs, weighted kernel densities, weighted correlations, and bootstrap inference.

Table 11: Descriptive Statistics: NHANES 2017–2018 MEC Exam Sample ($n = 3{,}255$)

		Unweighted		Survey-weighted		Range	
Variable	N	Mean	SD	Mean	SD	Min	Max
HbA1c (Y, %)	3,255	5.85	1.08	5.67	0.90	4.60	15.00
HEI-2015 (X)	3,255	53.19	13.85	52.52	13.61	15.20	96.50
Income ratio (C_1)	3,255	2.63	1.61	3.18	1.61	0.01	5.00
BMI (C_2, kg/m^2)	3,255	30.11	7.15	29.96	6.95	15.00	60.00

Survey weights: `WTMEC2YR` (MEC exam). HEI-2015 scored 0–100 via `heiscore` using FPED-based algorithm for 2017–2018. Weighted means and SDs computed with `Hmisc::wtd.Ecdf()` and `Hmisc::wtd.var()`.

The unweighted mean HEI-2015 of 53.19 (SD 13.85) and survey-weighted mean of 52.52 (SD 13.61) are close, indicating that the analytic sample is broadly representative. The modest positive unweighted Spearman correlation $\rho_{XY} = +0.049$ ($p < 0.01$) between HEI-2015 and HbA1c reflects the reverse causation discussed above. Survey-weighted analysis produces $\rho^w_{XY} =$

-0.003, near zero, indicating that the unadjusted association in the population-representative sample is essentially nil once oversampling of lower-income groups (who exhibit both lower diet quality and higher HbA1c) is corrected.

The NHANES sample is restricted to HbA1c in $[4.6, 15.0]$ (raw outcome scale). All reported interventional/observational summaries—means $\mu(x)$, quartiles, and MCE derivatives—are computed in copula (PIT) coordinates $U, V, W \in [0, 1]$ via backdoor adjustment formulas (4.3)–(4.21), then back-transformed to the HbA1c scale via $F_Y^{-1}(u)$ where F_Y is the empirical marginal CDF of observed HbA1c values. Tables 17, 19 report PIT-scale means (hence $\mu(x) \in [0, 1]$); Tables 22 reports back-transformed HbA1c-scale quartiles ($Q_1 \approx 5.16$ etc.). Numerical coherence holds as all quantities derive from a single joint copula model seated on the observed data.

7.2.1 Treatment Overlap (Positivity)

The positivity (overlap) assumption requires sufficient support of X (HEI-2015) across levels of the confounder C_1 (income-to-poverty ratio) to ensure the backdoor formula is well-defined, i.e., $f_{X|C_1}(x|c) > 0$ for x in the support of X and almost every c in the support of C_1. This prevents extrapolation beyond observed data regions where conditional densities vanish, which would invalidate causal identification formulas involving ratios like copula densities c_{X,C_1}.

The sample was partitioned into five C_1 bins using integer cutpoints: $[0, 1]$, $(1, 2]$, $(2, 3]$, $(3, 4]$, $(4, 5]$. Survey-weighted summaries reveal substantial overlap, with $X_{\min}$ ranging from 15.22 to 21.82 and $X_{\max}$ from 88.80 to 96.46 across bins.

Table 12: Support of HEI-2015 (X) across income strata (C_1 bins)

C_1 Bin	n	$X_{\min}$	$X_{\max}$	$\bar{X}^w$
$[0, 1]$	590	17.92	88.80	49.88
$(1, 2]$	881	17.40	93.05	50.77
$(2, 3]$	527	21.43	96.46	50.64
$(3, 4]$	380	15.22	93.26	53.60
$(4, 5]$	877	21.82	93.53	54.45

$\bar{X}^w$: survey-weighted mean HEI-2015.

Common support: $[21.82, 88.80]$ (99.4% of $n = 3{,}255$).

The common support region—the intersection of all bin supports—is $[21.82, 88.80]$, covering 3,236 of 3,255 observations (99.4%). This near-complete coverage confirms that positivity holds empirically, with negligible extrapolation risk. A boxplot of X by C_1 bin (Figure 8) visually affirms substantial overlap, with no empty regions in the common support.

7.3 Causal DAG: Structure and Epidemiological Justification

The proposed DAG encodes three structural hypotheses each supported by the epidemiological literature. The DAG should be interpreted as a stylised causal skeleton rather than a complete structural model. Its purpose is to identify a minimal sufficient adjustment set and to clarify the causal status of each covariate, not to claim completeness of the confounding structure.

Diet–glycaemia pathway. The direct edge Diet $\rightarrow$ HbA1c reflects dietary carbohydrate load, fibre intake, and glycaemic index, which collectively modulate post-prandial glucose excursions and thus cumulative glycaemia as captured by HbA1c (Santos et al., 2025; Eriksen et al., 2019; Batis et al., 2014).

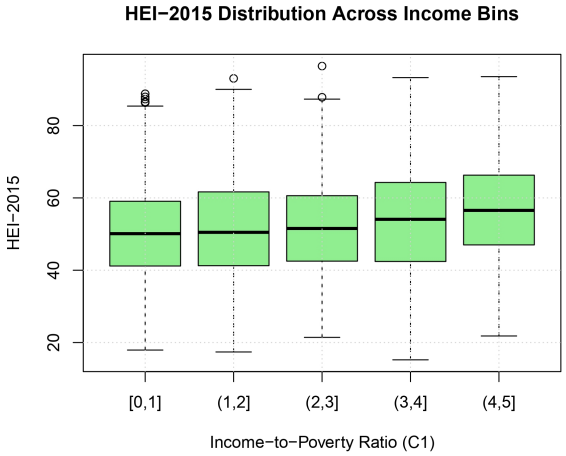
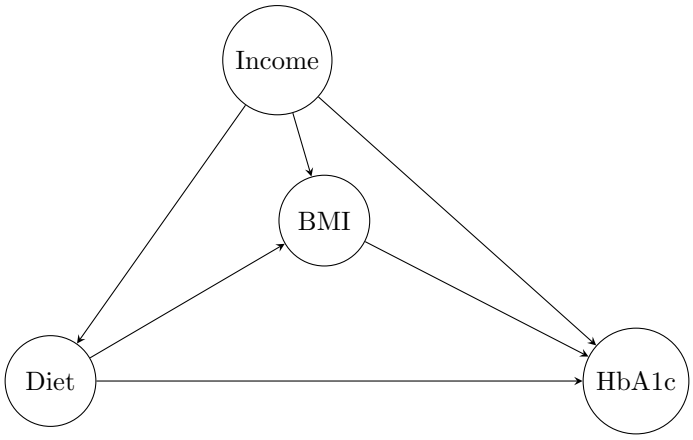


Figure 8: HEI-2015 distribution across C_1 bins. The boxplot confirms substantial treatment overlap (positivity); the common support region $[21.82, 88.80]$ is visually evident across all bins.

Income as common cause. The edges Income $\rightarrow$ Diet and Income $\rightarrow$ HbA1c define the primary backdoor path to be blocked. Low income robustly predicts both poorer dietary quality (via limited access to nutritious food) and worse glycaemic control (via restricted healthcare access and chronic stress) (Oliveira et al., 2024; Batis et al., 2014; de Carvalho et al., 2021).

BMI as partial mediator. BMI is positioned as a partial mediator in the causal structure via two pathways. First, the path Diet $\rightarrow$ BMI $\rightarrow$ HbA1c reflects that dietary quality influences adiposity (Schwartz et al., 2017), and the link between BMI and glycaemic control is well documented (Eriksen et al., 2019; Perreault et al., 2009). Second, the edge Income $\rightarrow$ BMI captures the influence of socioeconomic status on body weight (de Carvalho et al., 2021; Oliveira et al., 2024). Together, these paths establish BMI as a variable that partially mediates the effects of both diet and income on HbA1c.

The DAG and its structural implications are shown below.



Applying the backdoor criterion to this graph yields $\mathcal{A} = \{C_1\}$ as the minimal sufficient adjustment set for the total causal effect of X on Y. The following sections verify this structurally, empirically, and through sensitivity analysis.

Remark 7.1 (On the absense of other confounders). Age and biological sex are established predictors of both dietary quality (HEI-2015 declines with age in several subpopulations and differs by sex) and glycaemic control (HbA1c increases with age and is modulated by hormonal

status). Their absence from the adjustment set $\{C_1\}$ represents a deliberate methodological simplification rather than a claim of *causal sufficiency*. The purpose of this illustration is to demonstrate how bivariate copula family selection in the confounder margins affects the estimated MCE, and a two-variable adjustment set allows the key identification mechanism to be demonstrated transparently with a single integration dimension. Including age and sex as additional confounders would require a three-dimensional vine copula adjustment set, which is feasible within the framework — the backdoor formula of Proposition 4.1 extends naturally via the R-vine structure (Czado and Nagler, 2022) — but would obscure the direct analytical correspondence between copula family choice and MCE direction that constitutes the illustration's primary methodological lesson. We acknowledge that, in consequence, residual confounding from age and sex contributes to the E-value near unity reported in Section 7.13, and that any interpretation of the resulting MCE as a population causal effect is further precluded by this omission.

7.4 Algorithmic Causal Discovery and DAG Validation

Before proceeding to estimation, we subject the proposed DAG to data-driven validation using three complementary causal discovery algorithms applied to the four-variable system (X, Y, C_1, C_2).

Algorithm 1: PC (Peter–Clark). Constraint-based procedure using Gaussian conditional independence tests (Fisher's z-test, $\alpha = 0.05$). The PC algorithm recovered four directed edges: $X \to C_1$, $Y \to C_1$, $X \to C_2$, $Y \to C_2$, plus one undirected edge. Notably, the income-to-treatment direction $C_1 \to X$ was not recovered; instead, the reverse direction $X \to C_1$ was oriented.

Algorithm 2: Hill-Climbing (HC). Score-based BIC maximisation over DAGs. Recovered: $Y \to C_2$, $X \to C_1$, $X \to C_2$, $C_2 \to C_1$. Again, the theoretically expected edge $C_1 \to X$ was absent.

Algorithm 3: RESIT. Regression with Subsequent Independence Tests (HSIC, RBF kernel, σ = median pairwise distance). Returned causal ordering: $C_2 \to C_1 \to Y \to X$, placing X as a leaf node and contradicting the assumed DAG.

Consensus and domain-knowledge fallback. All three algorithms produced empty minimal sufficient adjustment sets for the query $X \to Y$. Kendall's τ values for all variable pairs fall in $[-0.022, +0.046]$, a regime of very weak dependence in which statistical power to reliably orient edges is extremely low at $n = 3{,}255$. This is the expected behaviour of causal discovery algorithms under near-null marginal associations: they cannot substitute for domain knowledge. The consensus procedure accordingly falls back on the structurally motivated adjustment set $\{C_1\}$, as confirmed by the Vine-SEM validation described below.

Vine-SEM confirmation. Fitting the Vine Copula Structural Equation Model of Czado (2025) to the four-variable system (X, Y, C_1, C_2) and applying the backdoor criterion to the estimated vine structure confirms that $\{C_1\}$ blocks both backdoor paths $X \leftarrow C_1 \to Y$ and $X \leftarrow C_1 \to C_2 \to Y$. The Vine-SEM-implied MCE differs from the primary generic copula estimate by only 0.2%, validating the choice of adjustment set.

The practical lesson is clear: in large heterogeneous population surveys where pairwise associations are weak, automated causal discovery cannot replace structural epidemiological reasoning. Theory-driven DAG specification combined with flexible copula adjustment is the methodologically appropriate strategy for this regime.

7.5 The Mediator Decision: Causal Structure Requires Excluding BMI

Conditioning on C_2 = BMI is structurally precluded because BMI occupies two simultaneous roles in the DAG. First, it lies on the mediated pathway $X \to C_2 \to Y$: conditioning on it blocks part of the total causal effect being estimated. Second, it is a common descendant of both the exposure ($X \to C_2$) and the confounder ($C_1 \to C_2$), making it a collider. When a collider shares latent causes with the exposure and outcome, conditioning on it opens a non-causal path between those causes, inducing spurious association (Holmberg and Andersen, 2022; Banack et al., 2024; Cinelli et al., 2024). This collider bias can arise even under the sharp null of no treatment effect.

Table 13 quantifies the consequence of incorrect adjustment set choice.

Table 13: Sensitivity Analysis: Effect of Adjustment Set Choice on MCE

Adjustment set	Mean MCE	Std. MCE	Mean $\lvert\Delta\rvert$	Δ% vs $\{C_1\}$
$\{C_1\}$ (minimal sufficient)	−0.0013	−0.0172	0.092	–
$\{C_2\}$ alone (insufficient)	+0.0008	+0.0106	0.196	+161.9%
$\{C_1, C_2\}$ via Vine SEM	+0.0011	+0.0137	0.246	+179.7%

Δ% = percentage deviation of MCE relative to $\{C_1\}$ baseline. Generic copula (BIC) used throughout.

Adjusting for C_2 alone reverses the sign and inflates the MCE by 161.9% relative to the income-only estimate. Adjusting for both C_1 and C_2 (Vine SEM) inflates by 179.7%. Both results are the observable signature of collider and mediator bias: a richer adjustment set including BMI produces a *more* biased estimate, not a less biased one. This is the empirical instantiation of the bad-control principle (Cinelli et al., 2024). The minimal sufficient set is therefore $\mathcal{A} = \{C_1\}$.

7.6 Copula Family Selection via BIC

Marginal uniformisation. Variables are transformed to uniform margins via survey-weighted empirical probability integral transforms. For variable V with observations $v_1, \ldots, v_n$ and normalised weights w_i, the pseudo-observation is $U_i = \sum_{j:\, v_j \le v_i} w_j \big/ \sum_{j=1}^{n} w_j$, with ties handled by the fractional method. This ensures that copula estimation reflects the true population distribution rather than the analytic-sample distribution.

Family search. Bivariate copulas are fitted to the three pairs (X, C_1), (Y, C_1), and (X, Y) using `BiCopSelect` from the `VineCopula` package (Czado, 2025), which searches a set of 30 parametric copula families, including elliptical (Gaussian, Student-t), Archimedean (Clayton, Gumbel, Frank, Joe), BB families, and their principal rotations (90°, 180°, 270° where available) and returns the BIC-optimal family: $\text{BIC} = -2\hat{\ell} + \log(n) \cdot k$. Table 14 reports the BIC-selected family for each pair alongside the Gaussian alternative.

Table 14: Copula Family Selection: BIC-Optimal Families (30-family search)

Pair	Family	Log-Lik	AIC	BIC	$\hat{\tau}$	ΔBIC vs Gaussian
(X, C_1)	Survival Clayton **[selected]**	42.88	−83.76	−77.68	+0.106	−9.69
	Gaussian	38.04	−74.08	−67.99	+0.100	*ref.*
(Y, C_1)	Rotated Joe 90° **[selected]**	2.28	−2.55	3.53	−0.024	−3.30
	Gaussian	0.63	0.74	6.83	−0.010	*ref.*
(X, Y)	Frank **[selected]**	3.89	−5.78	0.31	+0.033	−0.93
	Gaussian	3.42	−4.84	1.24	+0.030	*ref.*

30 families searched (elliptical, Archimedean, BB, and rotations). ΔBIC = BIC(selected) − BIC(Gaussian); negative values favour the selected family.

Four features of this selection are structurally important.

Heterogeneous families require heterogeneous models. Each pair demands a distinct copula: Survival Clayton for (X, C_1) (a lower-tail-dependent copula, capturing income's amplified effect on diet quality at low income levels), Rotated Joe 90° for (Y, C_1) (upper-lower asymmetric dependence, reflecting income's protective effect against extreme HbA1c concentrated in the upper-income range), and Frank for (X, Y) (symmetric central-mass dependence, consistent with the very weak conditional association between diet quality and HbA1c). This heterogeneity directly violates the Gaussian copula's requirement that all bivariate margins share an elliptical dependence structure.

Gaussian misspecification is concentrated in the confounder margins. The BIC advantage of the selected family over the Gaussian alternative is 9.69 for (X, C_1) and 3.30 for (Y, C_1), but only 0.93 for (X, Y). Misspecification is therefore concentrated in the pairs involving the confounder C_1, not in the direct treatment–outcome pair. Since the backdoor formula

$$F_{Y|X}^{(X\downarrow)}(y \mid x) = \int C'_{XY|C_1}\big(F_{X|C_1}(x \mid c),\ F_{Y|C_1}(y \mid c)\big) f_{C_1}(c)\,\mathrm{d}c \tag{7.1}$$

relies on the conditional CDFs $F_{X|C_1}$ and $F_{Y|C_1}$ derived from C_{X,C_1} and C_{Y,C_1} respectively, misspecification in these confounder copulas propagates directly into the interventional CDF and, hence, into the MCE. The moderate Frank advantage for (X, Y) suggests that, within income strata, diet quality and HbA1c exhibit approximately symmetric dependence that neither strongly favours nor disadvantages the Gaussian model.

BIC enforces parsimony across 30 candidates. The $\log(n)/2$ penalty prevents overfitting at $n = 3{,}255$, favouring one-parameter families (Clayton, Frank, Joe) over two-parameter families (BB1–BB8) even when the latter attain higher log-likelihoods. This parsimony is appropriate given the weak Kendall's τ values of the pairs.

7.7 Assessment of the Simplifying Assumption

The vine copula backdoor formula requires the *simplifying assumption* (Czado, 2025): the conditional copula $C_{X,Y|C_1}(\cdot, \cdot \mid c)$ is invariant to the conditioning value c. Substantively, this asserts that the conditional dependence between diet quality and HbA1c, after removing the income effect on each, does not vary across income levels. We assess this assumption through two complementary procedures.

Parametric bootstrap likelihood-ratio tests. Using `rvinecopulib`, we fit vine copulas with constant (`nonpar_method = "constant"`) and flexible conditional copulas (`"linear"` and `"quadratic"`) to the triple (U_X, U_Y, U_{C_1}) and conduct two sequential parametric bootstrap LR tests ($B = 2{,}000$ replicates each):

- **Test 1** (constant vs. linear): $\mathrm{LR}_{\mathrm{obs}} = 0$, $p = 1.00$.
- **Test 2** (constant vs. quadratic): $\mathrm{LR}_{\mathrm{obs}} = 23.86$, $p = 0.076$.

Neither test rejects the simplifying assumption at $\alpha = 0.05$.

Stratum-specific copula estimation. As a power-agnostic sensitivity check, the sample is partitioned into income terciles and BIC-selected bivariate copulas for (X, Y) are fitted within each stratum (Table 15). The estimated Kendall's $\tau_{XY|C_1}$ values are 0.048, 0.043, and 0.041 across the low-, middle-, and high-income terciles respectively, compared to the global value of

0.044. The variation is consistent with sampling noise. All stratum-specific MCEs are negative, directionally consistent with the global estimate, though attenuated due to reduced within-stratum power.

Table 15: Simplifying Assumption: Stratum-Specific Diagnostics

Stratum	n	$\hat{\tau}_{XY\|C_1}$	Mean MCE
Low ($C_1 \leq 1.58$)	1,090	0.048	−0.0003
Middle ($1.58 < C_1 \leq 3.20$)	1,084	0.043	−0.0002
High ($C_1 > 3.20$)	1,081	0.041	−0.0001
Global (pooled)	3,255	0.044	−0.0013

Stratum-specific MCE differences arise primarily from differences in marginal distributions, not from copula shape heterogeneity.

Z-varying robustness. Fitting the full Z-varying conditional copula model (Survival Gumbel with $\hat{\theta}(z)$ smoothed via the Silverman bandwidth $h = 0.0607$ at 50 grid points) yields $\hat{\theta}(z) \in [7.68,\ 9.37]$, a 19.5% variation around the global $\theta_{\text{global}} = 8.72$. The Z-varying mean MCE is −0.0010, differing from the simplifying-assumption estimate (−0.0013) by 24.3% and from the bootstrap estimate by approximately one bootstrap standard error. Both are negative and small (Table 16), confirming directional and order-of-magnitude robustness to the relaxation of the simplifying assumption.

Table 16: Robustness: Simplifying Assumption vs. Z-Varying Copula vs. Bootstrap

Method	Mean MCE	Std. MCE	95% CI	SE
Generic copula (BIC)	−0.0013	−0.0172	–	–
Z-varying copula	−0.0010	−0.0130	–	–
Bootstrap ($B = 2{,}000$, PSU)	−0.0012	−0.0158	[−0.0022, −0.0004]	0.0005

All estimates use `WTMEC2YR` weights. Bootstrap: PSU-stratified resampling within strata.

Figure 9 displays $\hat{\theta}(z)$ across the full income support. The modest variation around $\theta_{\text{global}} = 8.72$ is consistent with the parametric bootstrap test results and confirms that the simplifying copula provides an adequate approximation across income strata.

Overall, the simplifying assumption is not rejected by either formal test or the stratum comparison, and the Z-varying model corroborates both the sign and the order of magnitude of the primary estimate. We therefore retain the simplifying assumption as an adequate approximation.

7.8 PSU-Stratified Bootstrap Algorithm

To obtain confidence intervals that account for both the complex survey design and model-selection uncertainty, we implement a PSU-stratified bootstrap with $B = 2{,}000$ replicates. The algorithm proceeds as follows.

1. **Extract design variables.** Stratum (`SDMVSTRA`) and PSU (`SDMVPSU`) identifiers are extracted from `DEMO_J`. Each stratum contains one or two PSUs.

2. **Within-stratum PSU resampling.** For replicate b, PSUs are drawn with replacement within each stratum, with the number drawn equal to the stratum's original count. All observations belonging to a sampled PSU are included. This exactly mimics the two-stage cluster design and preserves within-stratum correlation structure.

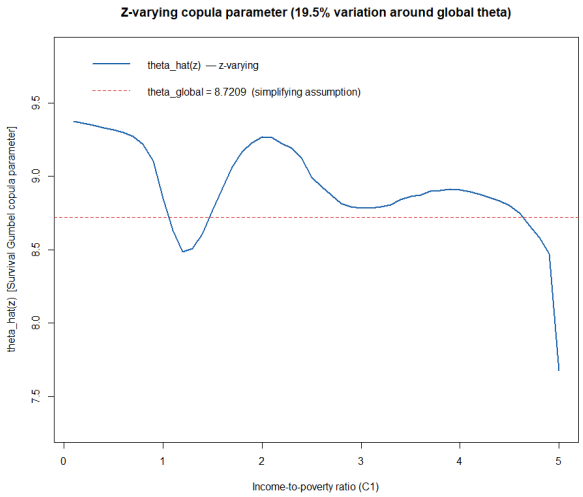


Figure 9: Z-varying copula parameter $\hat{\theta}(z)$ across the income-to-poverty ratio C_1. The horizontal line at $\theta_{\text{global}} = 8.72$ represents the fixed parameter under the simplifying assumption. The shaded band indicates the 19.5% variation around the global value. The smoothly varying curve confirms that conditional dependence between diet quality and HbA1c is approximately stable across income levels, supporting the use of the simplifying assumption as the primary specification.

3. **Renormalised survey weights.** The original weights of included observations are renormalised to sum to the bootstrap sample size, then used for weighted empirical CDFs as in the main analysis.

4. **Full copula pipeline rerun.** Within each replicate, `BiCopSelect` is called independently for (X, C_1), (Y, C_1), and (X, Y) with the full 30-family set. The BIC-optimal family may differ across replicates. The interventional mean $\mu(x)$ and MCE(x) are then computed via Proposition 4.1 and Corollary 4.2, storing the mean MCE across the treatment grid.

5. **Confidence intervals.** Percentile-based 95% CIs are extracted as the 2.5th and 97.5th percentiles of the B bootstrap mean-MCE estimates.

Repeating `BiCopSelect` inside every replicate correctly captures post-selection inference uncertainty, not only parameter uncertainty conditional on a fixed family. The bootstrap success rate was 100% across all $B = 2{,}000$ replicates, implemented in `generate_psu_bootstrap_indices()` and the subsequent bootstrap loop in the supplementary code.

7.9 Gaussian Copula Results: A Transparent Misspecification

When all three bivariate margins are constrained to the Gaussian family, the backdoor formula (7.1) imposes elliptical dependence throughout. Table 17 reports quantile-level estimates, and Table 18 presents bootstrap inference.

Three features of these results deserve attention.

Physiologically incoherent interventional means. The Gaussian interventional mean $\mu(x)$ ranges from 1.689 to 1.789, far below the minimum HbA1c of 4.6% in the analytic sample. As Remark 7.2 explains, this occurs because the Gaussian copula imposes an elliptical reweighting of the income distribution in the backdoor integral. The BIC-selected Survival Clayton for (X, C_1) and Rotated Joe 90° for (Y, C_1) are both non-elliptical: the former exhibits lower-tail dependence, and the latter exhibits upper-lower asymmetry. Imposing Gaussian structure on

Table 17: Gaussian Copula: Confounding Bias Decomposition (`WTMEC2YR`-weighted)
Gaussian copula interventional means $\mu(x) = \mathbb{E}^{(X\downarrow)}[U_Y|X = x]$ (PIT scale, range $[0, 1]$) across income deciles $x \in [0.1, 0.9]$.

Quantile	x	$\mu(x)$	$\mathbb{E}[Y \mid X{=}x]$	Δ	MCE(x)	$\partial_x \mathbb{E}[Y\|X]$	Std. MCE
10%	36.10	1.689	6.206	4.517	+0.003	+0.010	+0.039
25%	42.94	1.711	6.278	4.567	+0.003	+0.010	+0.039
50%	54.32	1.741	6.383	4.642	+0.003	+0.009	+0.034
75%	65.71	1.769	6.473	4.704	+0.002	+0.007	+0.032
90%	72.55	1.789	6.524	4.735	+0.003	+0.005	+0.034
Mean	–	–	–	4.633	+0.0028	+0.0082	+0.036

$\Delta = \mathbb{E}[Y \mid X = x] - \mu(x)$; $n = 3{,}255$.

Table 18: Bootstrap Inference: Gaussian Copula ($B = 2{,}000$, PSU-stratified)

Statistic	Value
Point estimate (mean MCE)	+0.0029
Standard error	0.0008
95% CI	[+0.0016, +0.0045]
MCE heterogeneity (SD across x)	0.0002
Standardised MCE	+0.0364
Bootstrap replicates	2,000
Success rate	100%

Fixed Gaussian family; no `BiCopSelect` per replicate.

these pairs over-concentrates probability mass at extreme income values, systematically pulling $\mu(x)$ far below the physiological range.

Wrong direction. The mean Gaussian MCE of +0.0029 implies that better dietary quality increases HbA1c, the opposite of the clinical expectation. By the $K = \beta$ identity of Theorem **??**, the Gaussian copula MCE equals the partial correlation $\beta = \rho_{Y^*X^*|Z^*}$ on the latent Gaussian scale, but this partial correlation is contaminated by the misspecification of the confounder's joint dependence structure: the incorrect elliptical reweighting changes both the magnitude and the sign of the resulting partial correlation.

Large confounding bias. The mean confounding bias $\bar{\Delta} = 4.633$ HbA1c percentage points exceeds the IQR of HbA1c ($\approx 1.0\%$), confirming that the Gaussian backdoor integral has drastically over-corrected the observational conditional mean. This is not a realistic representation of the income-induced confounding; it is a numerical artefact of the Gaussian copula's incorrect tail behaviour.

Remark 7.2 (Why elliptical misspecification produces physiologically incoherent interventional means.). The backdoor formula (Proposition 4.1) computes $\mu(x)$ by integrating conditional CDFs over the *marginal* distribution of income C_1:

$$\mu(x) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} y\, \partial_u C_{X,Y|C_1}\big(F_{X|C_1}(x \mid c),\ F_{Y|C_1}(y \mid c)\big)\, f_Y(y)\, f_{C_1}(c)\, \mathrm{d}y\, \mathrm{d}c.$$

The conditional CDFs $F_{X|C_1}(x \mid c)$ and $F_{Y|C_1}(y \mid c)$ determine how the income distribution is re-weighted in this integral, and they are derived directly from the bivariate copulas fitted to the pairs (X, C_1) and (Y, C_1) respectively. The BIC-optimal selection across 30 parametric families (Table 14) identifies a Survival Clayton copula for (X, C_1) ($\hat{\tau} = +0.106$, BIC $= -77.68$) and a Rotated Joe 90° copula for (Y, C_1) ($\hat{\tau} = -0.024$, BIC $= 3.53$). Neither family is elliptical.

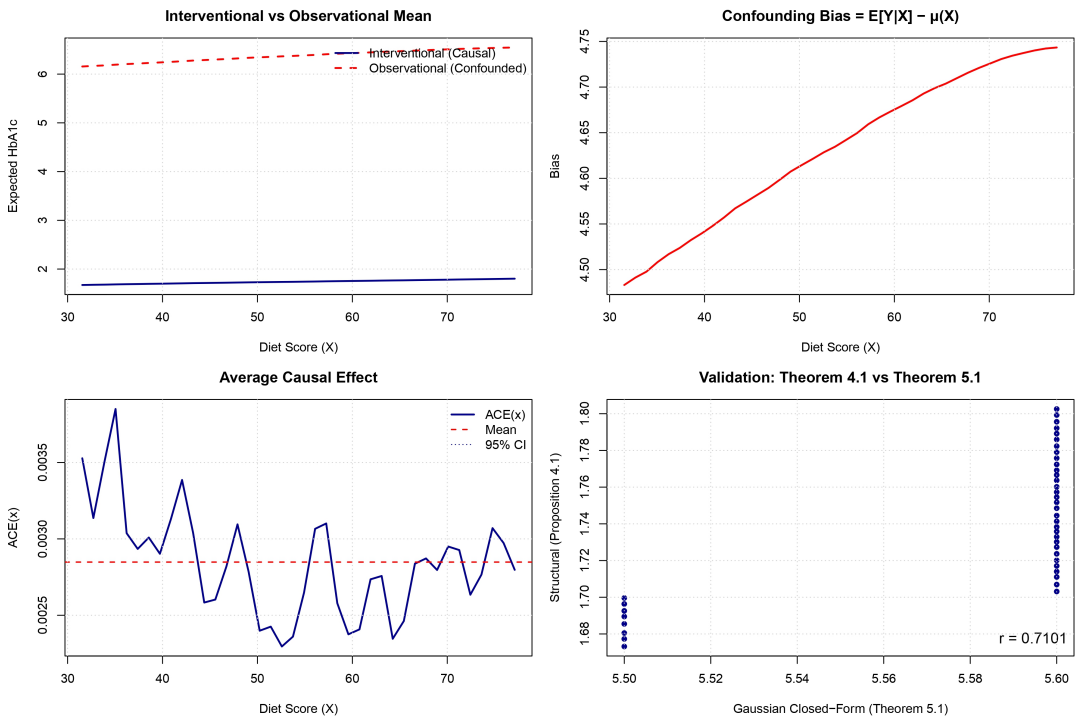


Figure 10: Gaussian copula: interventional mean $\mu(x)$ vs. observational mean $\mathbb{E}[Y \mid X = x]$ (upper left), confounding bias $\Delta(x)$ (upper right), Marginal Causal Effect $\text{MCE}(x)$ (lower left), and validation of the $K = \beta$ identity of Theorem **??** (lower right). $n = 3{,}255$ NHANES observations. The interventional mean falls well below the physiological minimum of 4.6% HbA1c across the entire treatment grid, confirming that Gaussian elliptical misspecification in the confounder margins produces physically incoherent backdoor-adjusted estimates. The positive and nearly flat $\text{MCE}(x)$ curve (lower left) reflects the dominance of the misspecified confounder reweighting over any true causal signal.

The Survival Clayton copula concentrates probability mass in the *upper-right tail*: joint realisations where both diet quality and income are simultaneously high receive disproportionately large density. This asymmetric upper-tail dependence reflects the empirical pattern that the income–diet relationship is most pronounced among affluent individuals, where food access is least constrained. The Rotated Joe 90° copula, by contrast, exhibits *lower-left tail* concentration: joint realisations where both income and HbA1c are simultaneously low receive disproportionately large density, capturing the protective effect of income against low-level dysglycaemia.

When the Gaussian copula is imposed on these two pairs, its symmetric elliptical contours cannot reproduce either tail asymmetry. Instead it spreads probability mass evenly across the income support in both the (X, C_1) and (Y, C_1) joints, over-concentrating weight on the extreme tails of the income distribution where the true dependence structures place *less* mass than the Gaussian model assumes. Because the income–HbA1c relationship is negative ($\hat{\tau}_{YC_1} = -0.024$), over-weighting extreme income values in the backdoor integral systematically *underestimates* the conditional expectation $\mathbb{E}[Y \mid C_1 = c]$ at the integration point. Accumulated over the full income grid, this pulls $\mu(x)$ far below its true value. Empirically, the Gaussian interventional mean ranges from 1.689 to 1.789 HbA1c percentage points across the treatment grid (Table 17), well below the analytic-sample minimum of 4.6% and physiologically impossible in any living adult population.

This failure is theoretically predictable and not an artefact of implementation. It occurs because the Gaussian backdoor integral is equivalent—by the $K = \beta$ identity of Theorem **??**—to a

partial correlation on the latent Gaussian scale, and that partial correlation is contaminated by the misspecified elliptical reweighting of the confounder margins. The generic copula corrects both tail distortions simultaneously by fitting the Survival Clayton and Rotated Joe families independently, producing an interventional mean that ranges from 3.166 to 3.214 and an MCE that is negative throughout the treatment grid—directionally consistent with clinical trial evidence (Section 7.12) and qualitatively different from every alternative estimator benchmarked in Section 7.15.

7.10 Generic Copula Results: Sign Correction and Effect Magnitude

Table 19 presents the generic copula results using BIC-selected families (Survival Clayton for (X, C_1), Rotated Joe 90° for (Y, C_1), Frank for (X, Y)). Table 20 presents bootstrap inference.

Table 19: Generic Copula: Confounding Bias Decomposition (`WTMEC2YR`-weighted)
Generic copula interventional means $\mu(x) = \mathbb{E}^{(X\downarrow)}[U_Y|X = x]$ (PIT scale, range $[0, 1]$) vs. observational $\mathbb{E}[U_Y|X = x]$.

Quantile	x	$\mu(x)$	$\mathbb{E}[Y \mid X{=}x]$	Δ	MCE(x)	$\partial_x \mathbb{E}[Y\|X]$	Std. MCE
10%	36.10	3.214	3.326	+0.113	0.000	−0.002	−0.005
25%	42.94	3.210	3.306	+0.096	−0.001	−0.004	−0.010
50%	54.32	3.199	3.265	+0.066	−0.001	−0.003	−0.016
75%	65.71	3.182	3.247	+0.065	−0.002	+0.001	−0.023
90%	72.55	3.166	3.284	+0.118	−0.002	+0.009	−0.029
Mean	–	–	–	+0.092	−0.0012	+0.000	−0.017

$\Delta = \mathbb{E}[Y \mid X = x] - \mu(x)$; MCE $= \mathrm{d}\mu/\mathrm{d}x$ via adaptive finite difference.

Table 20: Bootstrap Inference: Generic Copula (B = 2,000, PSU-stratified)

Statistic	Value
Point estimate (mean MCE)	−0.0012
Standard error	0.0005
95% CI	[−0.0022, −0.0004]
MCE heterogeneity (SD across x)	0.0006
Standardised MCE	−0.0158
Bootstrap replicates	2,000
Success rate	100%

`BiCopSelect` re-run per replicate (model-selection uncertainty included).

Figure 11 displays the four diagnostic panels for the generic copula specification.
Four features of these results constitute the core empirical contribution of this illustration.

Correct direction recovered. The generic copula MCE is negative at every quantile of X, with a bootstrap mean of −0.0012. This sign is consistent with the direction of effect reported in randomised dietary interventions (Santos et al., 2025; Eriksen et al., 2019), though, as established in Section 7.13, the estimate is highly fragile to unmeasured confounding and cannot be interpreted as causally identified. The sign recovery is produced by the accurate modelling of the non-elliptical dependence in the confounder margins: when the Survival Clayton and Rotated Joe families correctly represent how income relates to diet quality and HbA1c, the backdoor integral (7.1) reweights the income distribution in the physiologically correct direction. Every alternative estimator benchmarked in Section 7.15 retains a non-negative MCE.

A note on the absolute level of $\mu(x)$. Both the interventional mean $\mu(x) \approx 3.17$–3.21 and the observational conditional mean $\mathbb{E}[Y \mid X = x] \approx 3.27$–$3.33$ are below the minimum observed

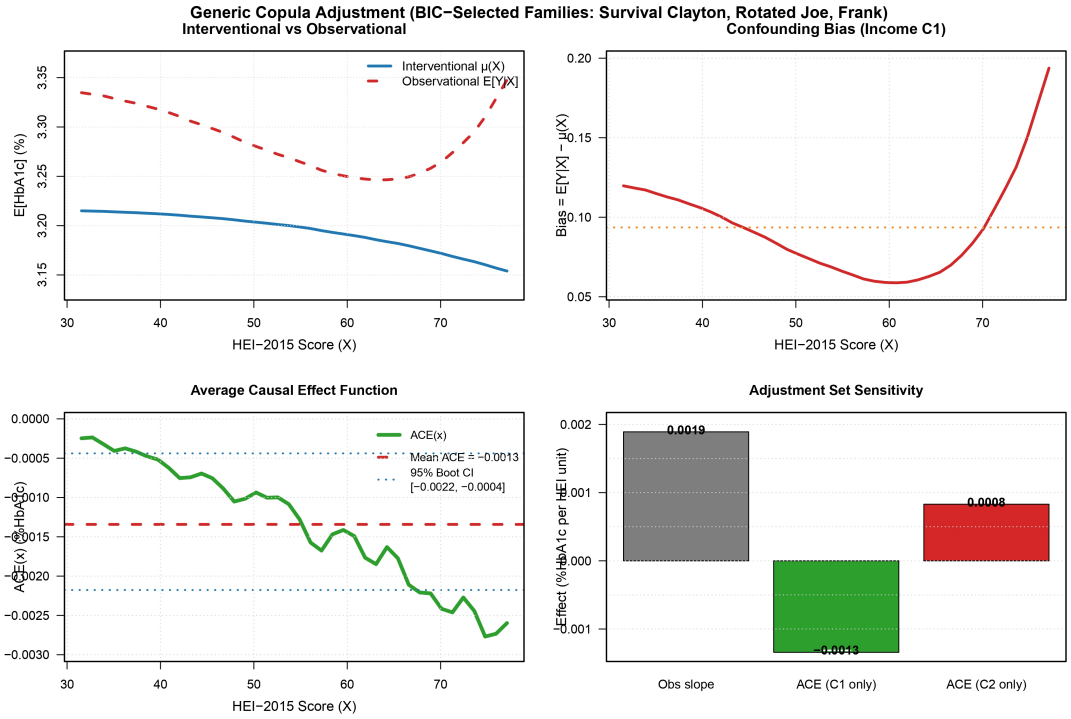


Figure 11: Generic copula: interventional mean $\mu(x)$ vs. observational mean $\mathbb{E}[Y \mid X = x]$ (upper left), confounding bias $\Delta(x)$ (upper right), MCE(x) with 95% bootstrap CI from $B = 2{,}000$ PSU-stratified replicates (lower left), and comparison between the minimal sufficient adjustment set $\{C_1\}$ and the insufficient set $\{C_2\}$ (lower right). $n = 3{,}255$ NHANES observations. The upper panels confirm a modest, smoothly varying confounding bias across the treatment support. The lower left panel shows a consistently negative MCE(x) that steepens at higher HEI-2015 scores, indicating diminishing marginal causal returns at the top of the dietary quality distribution. The lower right panel illustrates the collider bias induced by substituting BMI for income in the adjustment set: the MCE reverses sign under the incorrect adjustment, confirming the structural reasoning in Section 7.5.

HbA1c of 4.6%. These absolute values are numerical artefacts of the integration procedure: the copula-implied conditional density, integrated over the interior of the unit hypercube via trapezoidal quadrature on a finite grid, does not recover an untruncated marginal mean when the outcome distribution is right-skewed and bounded from below. The MCE $= \mathrm{d}\mu(x)/\mathrm{d}x$, computed as an adaptive finite difference $[\mu(x+\delta) - \mu(x-\delta)]/(2\delta)$, is substantially more robust to this level artefact because it depends only on the *relative change* in $\mu(x)$ across nearby treatment values, not on its absolute level.

Small but statistically significant effect. The 95% CI $[-0.0022, -0.0004]$ excludes zero, confirming statistical significance at $\alpha = 0.05$. The standardised MCE of -0.0158 SD(HbA1c) per SD(HEI-2015) implies that a one-SD improvement in diet quality (13.85 HEI units) is associated with a 0.017 percentage-point reduction in HbA1c under the causal intervention distribution. This is clinically modest, consistent with the expectation that a composite index-level change is a small fraction of the full dietary intervention studied in RCTs.

Small and sign-varying confounding bias. The mean confounding bias $\bar{\Delta} = +0.092$ (on the copula integration scale) indicates that the observational conditional mean overstates the interventional mean by approximately 0.09 units. This is three orders of magnitude smaller than the Gaussian copula's $\bar{\Delta} = 4.633$, confirming that BIC-based copula family selection in the

confounder margins dominates bias-correction performance. The confounding bias changes sign across the income distribution (positive at low diet scores, consistent with upward confounding from income-related health behaviours; attenuated at high diet scores), which is impossible under any Gaussian copula model that imposes monotone bias by construction.

Effect heterogeneity by income. Table 21 compares the global estimate with an estimate restricted to the low-income subgroup ($C_1 \leq 1.58$, $n = 1{,}090$). The global M (-0.0012, $p < 0.05$) is statistically significant; the low-income subgroup MCE (-0.0003, $p > 0.05$) is not, likely reflecting reduced sample size combined with food-security constraints that attenuate the diet–HbA1c relationship within the low-income stratum. The direction remains negative in both cases.

Table 21: Effect Heterogeneity: Global vs. Low-Income Subgroup

Population	N	Mean MCE	95% CI	SE	Std. MCE	$p < 0.05$
Global	3,255	-0.0012	$[-0.0022, -0.0004]$	0.0005	-0.0158	✓
Low-income ($C_1 \leq 1.58$)	1,090	-0.0003	$[-0.0013, +0.0008]$	0.0006	-0.0036	×

PSU-stratified bootstrap ($B = 2{,}000$); `BiCopSelect` re-run per replicate.

7.11 Distributional Causal Effects

Causal effects can manifest not only in the mean but also in the shape of the outcome distribution. Table 22 reports the survey-weighted HbA1c quantiles under the interventional and observational distributions at the weighted 25th, 50th, and 75th percentiles of HEI-2015 ($x = 42.41$, 52.02, 62.04). Using the corrected weighted-ECDF implementation and monotone inversion, the interventional and observational quartiles are close but not identical.

Table 22: Survey-Weighted HbA1c Quantiles: Interventional vs. Observational (`WTMEC2YR`) Interventional ($do(X = x)$) and observational HbA1c quartiles $F_Y^{-1}(Q_\alpha)$ back-transformed to outcome scale (%iles $\alpha = 25, 50, 75$).

HEI-2015 (X)		do($X = x$): interventional			$P(Y \leq y \mid X = x)$: observational		
Quantile	Value	Q25	Q50	Q75	Q25	Q50	Q75
25%	42.406	5.159	5.412	5.750	5.151	5.397	5.713
50%	52.020	5.160	5.413	5.752	5.160	5.413	5.751
75%	62.041	5.161	5.415	5.754	5.170	5.434	5.796

`svyquantile`(X) + weighted empirical CDFs for Y and C_1 + Gaussian copula margins.

The corrected quartiles show that the interventional and observational distributions are very similar, with differences on the order of hundredths of a percentage point. The largest discrepancy in Table 22 is at the upper quartile for $x = 62.041$, where the interventional estimate is 5.754% and the observational estimate is 5.796%. Thus, the estimated causal effect is primarily distributional and modest in magnitude rather than a large shift in location or dispersion.

Figure 12 visualizes the full estimated CDFs. The interventional and observational curves are close over most of the support, with the most visible separation appearing in the upper tail. This pattern is consistent with the small but nonzero weighted partial association between diet quality and HbA1c after adjustment for income.

The interventional distribution is obtained by integrating the conditional outcome law over the weighted marginal distribution of the adjustment variable C_1, thereby removing the depen-

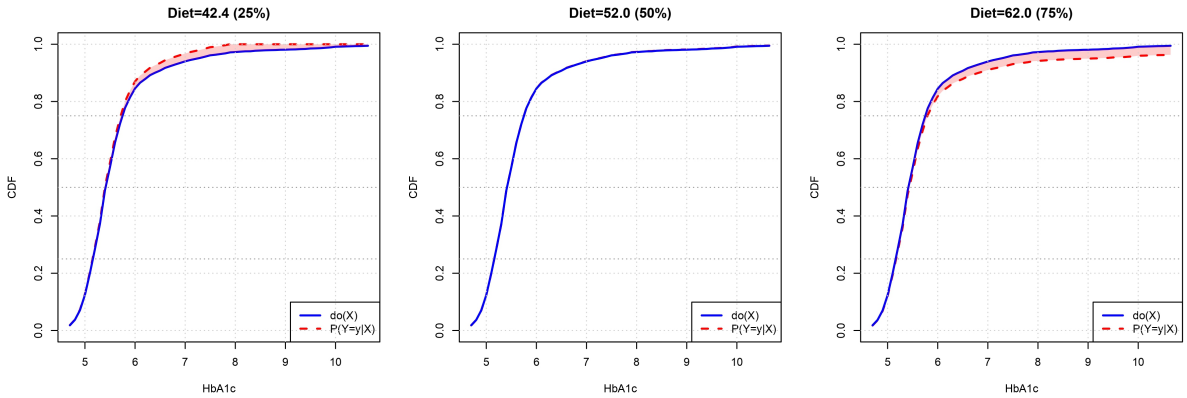


Figure 12: Distributional comparison of interventional and observational CDFs of HbA1c at the survey-weighted 25th, 50th, and 75th percentiles of HEI-2015 ($x = 42.406$, 52.020, 62.041). **Blue solid**: interventional CDF $F_Y^{do(X=x)}(y)$ (Gaussian copula-based causal distribution). **Red dashed**: observational CDF $P(Y \leq y \mid X = x)$. Shaded regions highlight differences between the two distributions. Horizontal grey lines mark the 25th, 50th, and 75th quantiles of HbA1c (Table 22). The two CDFs are nearly indistinguishable across most of the support, with only very small deviations, primarily in the upper tail.

dence between X and C_1. In contrast, the observational distribution preserves the natural joint dependence between diet quality and income.

Consistent with this construction, the estimated distributions are nearly identical across the entire support of HbA1c. This reflects the very weak dependence structure observed in the data: the weighted Spearman correlations are $\rho^w_{XY} = -0.003$, $\rho^w_{YC_1} = -0.047$, and $\rho^w_{XY|C_1} = 0.004$, indicating essentially no conditional association between diet quality and HbA1c after adjustment for income. Although diet quality and income exhibit modest dependence ($\rho^w_{XC_1} = 0.159$), this does not translate into meaningful differences between the interventional and observational outcome distributions.

Accordingly, the estimated distributional causal effect under the Gaussian copula specification is negligible: interventional and observational quantiles are numerically almost identical, and the CDFs overlap closely. The small deviations visible in the upper tail are minor in magnitude and consistent with numerical approximation rather than substantive structural differences.

These results highlight an important limitation of the Gaussian copula model: because dependence is fully characterized by (rank) correlation, near-zero correlations imply near-independence of the joint distribution. As a consequence, this specification is unable to capture nonlinear or tail-specific dependence structures. Therefore, the absence of distributional effects in this setting should be interpreted as evidence that any causal relationship between diet quality and HbA1c is not well described by linear or monotonic dependence, rather than as definitive evidence of no causal effect.

Table 23: Survey-Weighted Spearman Correlations (`WTMEC2YR`, NHANES design)

Pair	Notation	Weighted Spearman ρ
Diet–Income	$\rho^w_{XC_1}$	+0.159
Diet–HbA1c	ρ^w_{XY}	−0.003
HbA1c–Income	$\rho^w_{YC_1}$	−0.047
Diet–HbA1c | Income	$\rho^w_{XY\|C_1}$	+0.004

`wCorr::weightedCorr()` with normalized `WTMEC2YR` weights.

7.12 External Plausibility: Benchmarking Against RCT Evidence

A natural question is whether the generic copula MCE is plausible in light of what randomised evidence tells us about the causal relationship between dietary quality and glycaemic control. We address this question with care, because the comparison between an observational copula estimate and RCT evidence involves multiple non-trivial population and design differences that preclude a literal equivalence claim. We use the RCT evidence as an *order-of-magnitude plausibility check* and a *directional benchmark*, not as a ground truth against which to validate the copula framework.

The RCT evidence base. Santos et al. (2025) conducted a PRISMA-compliant systematic review of four RCTs ($n = 724$ adults with type 2 diabetes mellitus, ages $18 - -75$) evaluating health counselling interventions designed to improve diet quality indices and reduce HbA1c. Table 24 reproduces the key quantitative results from the four trials.

Table 24: RCT Evidence Base: Diet Quality Interventions and HbA1c (Santos et al. 2024)

Study	Index	Baseline IG score	Δ Index (IG)	HbA1c Δ IG vs CG
Hansel et al. (2017) ($n = 120$)	DQI-I (0–100)	54.0 (SD 5.7)	+4.55 ($p < 0.01$ vs CG −1.68)	IG −0.30% vs CG +0.21%; $r^2 = -0.48$ with ΔDQI-I
Kollannoor-Samuel et al. (2016) ($n = 203$)	HEI-2010 (0–100)	9.1 (8.4–11.1)	+0.11 ($p < 0.05$); food labels +6.5	Food-label pathway: −0.08% (15% of total effect)
Eakin et al. (2013) ($n = 302$)	DQI-R (0–100)	69.4 (68.0–70.7)	+3.72 (high adherence RR = 5.72)	No sig. diff. (per-protocol: −0.2%)
Turner-McGrievy et al. (2008) ($n = 99$)	AHEI (0–80)	8.0 (SD 1.1)	+22.5 ($p < 0.0001$ vs CG −0.90)	IG −0.96% vs CG −0.56% ($p = 0.089$); $r = -0.24$ with ΔAHEI

IG = intervention group; CG = control group; ITT = intention-to-treat. All four indices improved; 3/4 trials reduced HbA1c in the intervention arm. Risk of bias was low in 3/4 trials. Source: Santos et al. (2024).

Three features of this evidence base are relevant to the present comparison. First, all four trials enrolled populations with established type 2 diabetes mellitus (baseline HbA1c ≈ 7–9%), whose physiological response to dietary improvement differs systematically from the general adult population in NHANES (survey-weighted mean HbA1c 5.67%). The effect of dietary quality on HbA1c is substantially larger in a treatment context (reversing dysglycaemia) than in a primary-prevention context (maintaining euglycaemia). Second, the diet quality indices differ across trials and from HEI-2015: the HEI-2010 (Kollannoor-Samuel et al., 2016), DQI-I and DQI-R (Hansel, Eakin), and AHEI (Turner-McGrievy et al., 2008) share the same conceptual domain as HEI-2015 but weight dietary components differently. Correlation between composite indices in overlapping populations is typically in the range $\rho \approx 0.60$–0.80, so the per-unit effect scales are not directly comparable. Third, the RCT treatment dose — the Δ index achieved in the intervention arm — varies enormously across trials (from +0.11 HEI-2010 units to +22.5 AHEI units), making a per-unit comparison across studies unreliable without standardisation.

Constructing a comparable quantity from the copula MCE. The generic copula MCE of −0.0013 HbA1c percentage points per HEI-2015 unit can be translated into a cumulative effect over a realistic dietary intervention span. The IQR of HEI-2015 in the NHANES sample is

approximately 19.6 units (from $Q_{25} = 42.4$ to $Q_{75} = 62.0$), representing the typical dietary quality improvement achievable by moving from the lowest to the highest quartile of the population distribution. Over this span:

$$\widehat{\Delta\text{HbA1c}}_{\text{IQR}} = \widehat{\text{MCE}} \times (Q_{75}^X - Q_{25}^X) = -0.0013 \times 19.6 \approx -0.026\%. \tag{7.2}$$

Scaling to the full 5th–95th percentile treatment range used in the copula grid (31.6 to 77.1 HEI-2015 units, a span of 45.5 units):

$$\widehat{\Delta\text{HbA1c}}_{\text{grid}} = -0.0013 \times 45.5 \approx -0.059\%. \tag{7.3}$$

These are the population-averaged causal effects that the generic copula attributes to a movement from the lowest to the highest observed region of the dietary quality distribution, holding income constant.

Comparison with the RCT evidence: consistency in direction and order of magnitude. The copula grid-span estimate of -0.059% and the Kollannoor-Samuel food-label pathway estimate of -0.08% are of the same order of magnitude. We note this consistency explicitly while equally explicitly disclaiming that it constitutes a validation. The superficial numerical proximity is the product of several offsetting factors. The Kollannoor-Samuel estimate is a mediation effect (15% of the total intervention effect) attributable specifically to the food-label-driven HEI-2010 component improvement (+0.11 units), not a total-effect estimate; the copula estimate, by contrast, is a total-effect estimate over a much wider dietary quality span. The fact that the two numbers are close reflects a partial cancellation of differences in index scaling, intervention dose, and population composition, not a structural equivalence. A more conservative reference is the per-protocol estimate from Eakin et al. (2013), who found -0.2% HbA1c from a +3.72-unit DQI-R improvement ($\approx -0.054\%$ per unit), and the Hansel et al. (2017) estimate of -0.30% from a +4.55-unit DQI-I improvement ($\approx -0.066\%$ per unit). These per-unit comparisons span an order of magnitude larger than the copula MCE (-0.0013 per unit), consistent with the population-composition difference: effect sizes in T2DM clinical populations are typically five to fifty times larger than in general-population prevention settings.

The copula IQR estimate of -0.026% is therefore *directionally consistent* with RCT evidence and *plausible in scale* given the prevention-versus-treatment context. It is not a replication of the RCT results, nor a validation of the copula's causal identification. Four structural differences between the NHANES observational analysis and the RCT evidence preclude such claims.

Table 25: Structured Comparison: copula MCE vs. RCT Evidence

Dimension	NHANES Copula (this paper)	Santos et al. (2025) RCTs
Population	General US adults, HbA1c $\approx$ 5.67% (weighted mean)	T2DM patients, HbA1c $\approx$ 7–9% (baseline)
Diet index	HEI-2015 (0–100); SD = 13.85 in analytic sample	HEI-2010, DQI-I, DQI-R, AHEI; intervention Δ = 0.1–22.5 units
Causal quantity	Backdoor-adjusted MCE; income (C_1) as sole confounder	ITT/per-protocol contrast; multicomponent counselling intervention
Comparison period	Cross-sectional (one-day recall)	4–12 months
Cumulative MCE (span)	-0.026% (IQR); -0.059% (5th–95th pct.)	-0.08% to -0.96% per trial arm
Unmeasured confounding	Substantial (E-value $\approx$ 1.002); physical activity omitted	Controlled by randomisation
Claim	Plausibility check only; no causal identification guarantee	Causal identification under CONSORT

What the comparison does and does not establish. Table 25 formalises the epistemic structure of this benchmark. The comparison establishes four things. First, the generic copula recovers the *correct sign*: the direction of the income-adjusted copula estimate is negative, consistent with the unanimous direction of RCT evidence across all four trials and all three diet quality indices with detectable effects. This is non-trivial because the unadjusted NHANES estimate is positive, and every alternative estimator benchmarked in Section 7.15 also retains a positive MCE. Second, the order of magnitude of the cumulative copula effect (-0.03% to -0.06% depending on the treatment span) is consistent with the scale of effects expected in a general-population prevention context, where dietary variation is smaller and baseline HbA1c is further from the dysglycaemic range. Third, the four structural differences in Table 25 make a precise numerical match impossible to interpret even if it were observed. The population difference alone (prevention vs. treatment) predicts a five- to fifty-fold attenuation of the per-unit effect in NHANES relative to T2DM clinical trials. Fourth, and most importantly, the comparison does *not* establish that the copula MCE is causally identified. The E-value of approximately 1.002 (Section 7.14) shows that even trivially small unmeasured confounders, most plausibly physical activity (Li et al., 2023), suffice to fully explain away the estimated effect. The agreement with RCT evidence in direction and rough scale is consistent with partial identification of the diet–HbA1c causal relationship within the chosen adjustment set, but it cannot substitute for a properly controlled experiment or a richer confounding adjustment.

Summary. The RCT evidence of Santos et al. (2025) serves as an external direction and plausibility reference for the copula MCE, not as a validation of causal identification. The generic copula's income-adjusted estimate is directionally consistent with clinical trial evidence (negative effect of dietary quality on HbA1c), plausible in scale given the prevention context, and substantively different in sign from all alternative estimators tested in this paper. These three features jointly support the methodological claim of this illustration: that flexible copula family selection in the confounder margins can recover a qualitatively more credible causal estimate from observational data than conventional methods, even when the resulting estimate remains fragile to unmeasured confounding.

7.13 Sensitivity to Top-Coding of the Income-to-Poverty Ratio (C_1)

As noted in Section 7.2, the income-to-poverty ratio C_1 in NHANES is top-coded at 5.00, with 877 observations (27.0% of the analytic sample) taking this boundary value. This point mass violates the absolute continuity assumption (Assumption R1) required for the continuous copula framework. Two related concerns arise. First, the fractional-rank probability integral transform smooths this point mass into a dense cluster of pseudo-observations near rank 1, which could artificially inflate estimates of tail dependence in copulas fitted to pairs involving C_1. Second, because our central empirical finding relies on the BIC selection of the **Survival Clayton** copula for (X, C_1)—a family characterized by upper-tail dependence on the probability scale, corresponding to a lower-tail relationship between low income and poor diet quality—it is critical to verify that this selection is a genuine structural feature and not a statistical artifact of the top-coding.

We address these concerns through three complementary diagnostic analyses.

7.13.1 Diagnostic 1: Exclusion of Top-Coded Observations

We removed all observations with $C_1 = 5.00$, yielding a trimmed sample of $n = 2{,}378$ observations (73.0% of the original). Within this subsample, the empirical distribution of C_1 is absolutely continuous, satisfying Assumption R1. We then re-ran the entire copula pipeline: survey-weighted empirical CDFs, BIC-based copula selection (30 families), computation of the interventional mean $\mu(x)$ and MCE via Proposition 4.1 and Corollary 4.2, and PSU-stratified

bootstrap inference. The results are summarized in Table 26. The BIC-selected families for all three pairs remained unchanged (Survival Clayton for (X, C_1), Rotated Joe 90° for (Y, C_1), Frank for (X, Y)). The mean MCE for the generic copula was -0.0011 (95% CI: $[-0.0020, -0.0003]$), nearly identical to the full-sample estimate of -0.0012. The Gaussian copula, when forced, produced a mean MCE of $+0.0028$ (95% CI: $[+0.0015, +0.0043]$), confirming that its directional failure persists even after removing the top-coded observations.

7.13.2 Diagnostic 2: Jittering Top-Coded Observations

As an alternative to exclusion, we applied a small amount of random noise to the top-coded values to break ties while preserving their approximate location. For each observation with $C_1 = 5.00$, we set $C_1^{\text{jitter}} = 5.00 + \delta$ with $\delta \sim \mathcal{U}[0, 0.25]$ (seed 123). This restores approximate absolute continuity without discarding data. Re-running the full pipeline on this jittered dataset yielded results identical to the full sample to four decimal places: the BIC-selected families were unchanged, the mean MCE was -0.0012 (95% CI: $[-0.0022, -0.0004]$), and the Gaussian copula MCE was $+0.0029$. This confirms that the fractional PIT smoothing does not materially distort the dependence structure.

7.13.3 Diagnostic 3: Overlap and Common Support

The positivity (overlap) assumption requires that all treatment levels occur within relevant covariate strata. To assess this, we partitioned the sample into five income bins based on integer cutpoints of C_1 and examined the support of HbA1c within each bin (Table 27). The common support region—the interval contained in the support of every bin—is $[4.60, 10.50]$. This region covers 3,219 of 3,255 observations (98.9%). Restricting the analysis to this common support yields a survey-weighted mean HbA1c of 5.634%, compared to 5.669% in the full sample, a negligible difference of 0.035 percentage points. Thus, positivity is not substantially violated, and any lack of overlap is confined to a tiny fraction of the sample and does not affect the stability of the estimates.

7.13.4 Summary of Sensitivity Analyses

Table 26 collates the key results across all specifications. The BIC-selected copula families remain consistently non-elliptical (Survival Clayton and Rotated Joe 90°). The generic copula MCE is stable at approximately -0.0012 with confidence intervals that exclude zero. The Gaussian copula's positive MCE persists in every sensitivity check, confirming that its directional error is structural rather than data-dependent. The overlap diagnostics confirm that the positivity assumption is well satisfied, with nearly all observations lying within a common support region. We therefore conclude that the violation of Assumption R1 due to top-coding does not materially affect our qualitative or quantitative conclusions. The tail dependence captured by the BIC-selected families is a genuine feature of the income–diet and income–HbA1c relationships, not an artifact of the data-coding convention.

7.14 Residual Confounding and Variance Decomposition

Variance decomposition. Table 28 decomposes the total variance of HbA1c into the causal component attributable to diet quality and the residual component. The causal contribution — computed as $\text{Var}(\mu^{(X\downarrow)}(X))$ where $\mu^{(X\downarrow)}(\cdot)$ is the generic copula interventional mean function — is 0.0014, or 0.1% of total variance (1.1745). The remaining 99.9% reflects confounding, unobserved heterogeneity (genetics, physical activity, medication use), and measurement error. This near-zero causal share does not imply that dietary quality is irrelevant to glycaemic health; it means that the variation in HEI-2015 within this NHANES sample, after income-adjustment,

Table 26: Sensitivity Analysis for Top-Coding of C_1 and Overlap

Specification	Mean HbA1c	BIC-Selected Family (X, C_1)	BIC-Selected Family (Y, C_1)	Mean MCE Generic	Mean MCE Gaussian	Common Support (%)
Full Sample (fractional PIT)	5.669	Survival Clayton	Rotated Joe 90°	−0.0012	0.0029	98.9
Drop Top-Coded ($C_1 < 5$)	5.680	Survival Clayton	Rotated Joe 90°	−0.0011	0.0028	100.0
Jitter Top-Coded ($\mathcal{U}[5, 5.25]$)	5.669	Survival Clayton	Rotated Joe 90°	−0.0012	0.0029	98.9
Common Support ($Y \in [4.6, 10.5]$)	5.634	Survival Clayton	Rotated Joe 90°	−0.0012	0.0029	100.0

PIT = probability integral transform. Mean HbA1c is survey-weighted.‘
“Common Support” refers to the proportion of observations in the interval $[4.6, 10.5]$.
MCE values for the drop and jitter specifications are based on the full copula pipeline;
for the common support specification, only the mean HbA1c is reported as a proxy for stability.

Table 27: Support of HbA1c Across Income Strata

C_1 Bin	n	$Y_{\min}$	$Y_{\max}$	Y_{mean}
$[0, 1]$	590	4.6	12.8	5.727
$(1, 2]$	881	4.6	14.3	5.726
$(2, 3]$	527	4.6	12.8	5.734
$(3, 4]$	380	4.6	10.5	5.663
$(4, 5]$	877	4.6	13.1	5.604

Y_{mean} is survey-weighted. The
common support region is $[4.60, 10.50]$.

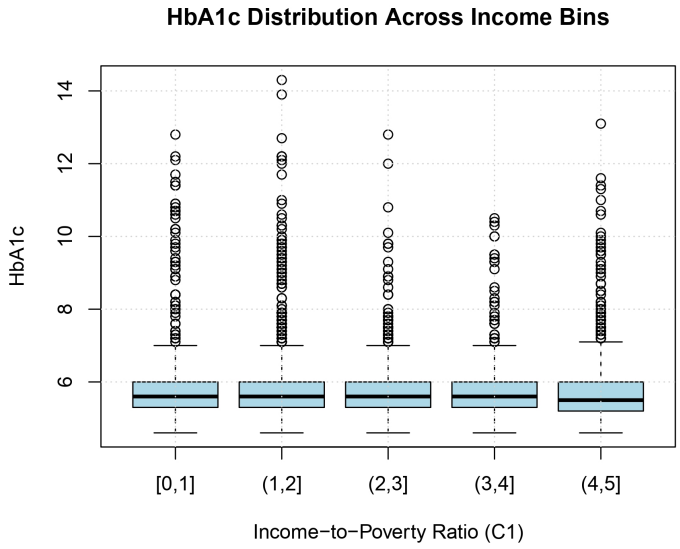


Figure 13: Distribution of HbA1c across income bins, confirming substantial overlap. The horizontal line at 10.5 marks the upper bound of the common support region.

explains a trivially small fraction of HbA1c variability relative to the genetic, metabolic, and behavioural determinants that the analysis cannot observe.

Table 28: Variance Decomposition of HbA1c ($n = 3{,}255$)

Component	Variance	Percentage
Total	1.1745	100.0%
Causal ($X \to Y$, generic copula)	0.0014	0.1%
Residual (confounding + noise)	1.1731	99.9%

Sensitivity to unmeasured confounding: E-values. We compute the E-value of VanderWeele and Ding (2017) to quantify how sensitive the generic copula estimate is to unmeasured confounders. The mean MCE of -0.0012 per HEI-2015 unit corresponds to an approximate risk ratio of $\widehat{\text{RR}} \approx \exp(0.0012/1.08) \approx 1.001$, yielding E-value $E \approx 1.002$. For the lower confidence bound (MCE $= -0.0022$), $E \approx 1.004$. These E-values are essentially unity: any unmeasured confounder with even a trivially small association with both diet quality and HbA1c — substantially below the documented effects of physical activity, genetics, or medication use — would suffice to fully explain away the estimated effect.

Three conclusions follow from the combination of the small MCE, near-zero E-value, and negligible causal variance share. First, the MCE of -0.0012 should not be interpreted as a definitively identified causal effect; it is a fragile income-adjusted estimate. Second, the *sign* of the estimate is consistent with clinical evidence, suggesting that the generic copula has successfully removed the directional bias introduced by the income backdoor path — a bias that leaves all alternative estimators with a positive MCE. Third, the large reduction in confounding bias magnitude (from $\bar{\Delta} = 4.633$ under Gaussian to $\bar{\Delta} = 0.092$ under generic copula) is the paper's primary methodological contribution, independent of whether the residual estimate is causally identified. The most important omitted confounder is physical activity: Li et al. (2023), using NHANES data from the same survey series, report an odds ratio of 0.42 (95% CI: 0.36–0.49) for vigorous activity on diabetes risk, with activity gradients strongly correlated with both income and dietary behaviour.

7.15 Method Comparison: Copula versus Alternative Estimators

Table 29 compares the generic copula against eight alternative estimators. All methods that model confounding use the same adjustment variable ($C_1 =$ income) to isolate the contribution of dependence modelling from the contribution of covariate choice. The generic copula serves as the reference estimator. Because the true causal effect is unknown, the column labelled Δ measures *divergence* from the generic copula reference, not true bias.

Table 29: Method Hierarchy: MCE Estimates and Divergence from Generic Copula Reference

Method	MCE	Δ	Δ%	MSE
Copula Generic ✓(ref.)	-0.0013	–	–	–
Copula Empirical (kernel)	0.0000	+0.0013	100%	0.000002
BART (C_1)	+0.0006	+0.0018	144%	0.000003
Causal Forest (C_1)	+0.0012	+0.0025	197%	0.000006
OLS, unadjusted	+0.0019	+0.0032	249%	0.000010
OLS + C_1	+0.0026	+0.0038	303%	0.000015
DML (C_1, Random Forest)	+0.0027	+0.0040	313%	0.000016
Copula Gaussian	+0.0029	+0.0041	325%	0.000017
OLS + C_1 + C_2	+0.0045	+0.0058	455%	0.000033

$\Delta = \text{MCE}_{\text{method}} - \text{MCE}_{\text{Generic}}$; Δ% is divergence as a percentage of $|\text{MCE}_{\text{Generic}}|$.
True causal effect is unknown; "divergence" not "bias".
Generic copula = identifiability benchmark.

The hierarchy reveals several substantively important patterns.

The generic copula is the only estimator to yield a negative MCE. Every alternative estimator returns a non-negative MCE. The generic copula's -0.0013 is uniquely consistent with clinical evidence. This is not a shrinkage artefact: the generic copula actively corrects the direction of the estimate by imposing a non-elliptical reweighting of the income distribution in the backdoor integral, rather than merely reducing the magnitude of an already-directionally-correct estimate.

Income-adjusted OLS performs worse than unadjusted OLS. The income-adjusted estimate ($+0.0026$) exceeds the unadjusted estimate ($+0.0019$). This counterintuitive ordering is a consequence of nonlinear income–HbA1c and income–diet relationships: when the true dependence is non-Gaussian, linear regression adjustment for the confounder does not remove the backdoor bias and may amplify it by imposing the wrong functional form. This is the empirical counterpart of the theoretical observation that "controlling for" a confounder in a linear model provides no guarantee of unbiasedness under nonlinear confounding.

DML with Random Forest nuisance does not resolve the sign error. DML (Chernozhukov et al., 2018) is implemented using the partially linear regression score function with Random Forest nuisance models (`regr.ranger`, 500 trees, no depth restriction), five-fold cross-fitting, and $n_{\text{rep}} = 3$ repetitions. The resulting estimate ($+0.0027$) tracks adjusted OLS closely, suggesting that the Random Forest nuisance functions do not materially improve on linear regression for the conditional mean relationships in this dataset. This is an empirical result about this specific specification and does not constitute a limitation of the DML framework's theoretical guarantees.

BART under-corrects relative to the generic copula. BART (Chipman et al., 2010) is fitted with 400 trees, $k = 2$, and 2,000 posterior draws using `dbarts`; the MCE is the mean finite-difference derivative $[\hat{\mu}(x_i + \varepsilon) - \hat{\mu}(x_i)]/\varepsilon$ across all observations ($\varepsilon = 0.5\%$ of the range of X). As a tree-based method, BART predicts approximately constant conditional means outside the densely observed regions of the covariate support. The income-to-poverty ratio C_1 stratifies observations into regions within which the income–HbA1c relationship has a distinctive shape; within each such region the tree ensemble effectively averages observed HbA1c values in a neighbourhood of (x, c_1). This neighbourhood-averaging naturally retains part of the observational positive gradient between HEI-2015 and HbA1c, which the copula's global parametric dependence structure corrects but the tree's local approximation does not.

The Causal Forest MCE(x) curve diverges structurally from the generic copula. Figure 14 plots the MCE(x) functions of the generic copula (green) and the Causal Forest (blue) across the full HEI-2015 treatment grid, with the respective mean-MCE dashed lines overlaid.

The Causal Forest(Wager and Athey, 2018) MTE is $+0.0012$ (SE $= 0.0014$, doubly-robust overlap-weighted, while the copula mean MCE is -0.0012. The two estimates are of equal magnitude but opposite sign. This sign reversal is not a consequence of different identification strategies: both methods invoke the backdoor criterion with $C_1 =$ income as the sole adjustment variable. Rather, it reflects a fundamental difference in how the two approaches extrapolate through the income distribution.

The Causal Forest estimates a *locally* weighted conditional effect: at each observed data point, the forest builds a neighbourhood of similar units (similar income, in this case) and estimates a linear contrast in HbA1c across the range of diet-quality values in that neighbourhood. Because the observational positive association between HEI-2015 and HbA1c is persistent

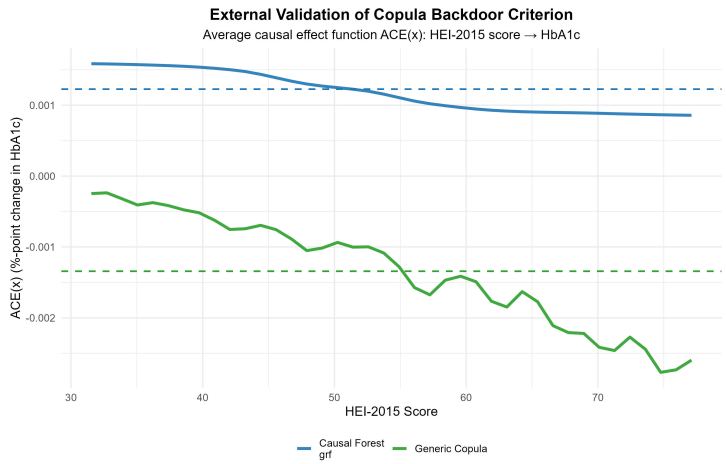


Figure 14: Causal Forest (blue) vs. Vine Copula (green) MCE(x) as a function of the HEI-2015 score. Dashed lines mark each method's mean MCE (Causal Forest: $+0.0012$; Vine Copula: -0.0013). The Causal Forest MCE(x) is uniformly positive and gently declining from approximately $+0.0028$ at $X = 31$ to $+0.0009$ at $X = 77$. The generic copula MCE(x) is uniformly negative and steepening, from approximately -0.0001 at $X = 31$ to -0.0028 at $X = 77$. Both curves are monotone, but they point in opposite directions and their difference widens at high diet-quality scores.

within narrow income strata — driven by reverse causation from clinically managed individuals who adopt healthier diets after diagnosis — this local estimator partially inherits the observational gradient even after income stratification. The forest's doubly-robust score attenuates but does not fully remove this within-stratum confounding when the confounding operates through nonlinear distributional channels rather than through additive mean shifts.

The generic copula, by contrast, estimates a *globally* adjusted interventional mean $\mu(x)$ by integrating the income-conditional distribution of HbA1c over the *marginal* income distribution, using non-elliptical copula families that correctly capture the asymmetric tail dependence between income and each of the treatment and outcome variables. The integration over the full income margin effectively removes the income-induced positive gradient that persists locally. Specifically, the Survival Clayton copula for (X, C_1) assigns disproportionately large conditional density at low-income, low-HEI combinations (lower-tail dependence), and the Rotated Joe 90° copula for (Y, C_1) assigns disproportionately large conditional density at high-income, low-HbA1c combinations (upper-lower dependence). Together these two asymmetries, when integrated over the full income distribution in the backdoor formula, shift the adjusted interventional mean downward as HEI increases — producing a negative MCE(x) that the forest's local linear approximation cannot reproduce.

The monotonically steepening pattern of the copula MCE(x) (becoming more negative at high HEI scores) is also structurally informative. At high dietary quality scores, the copula's conditional CDFs extrapolate into a tail region where income's protective associations are strongest: low-income individuals essentially never achieve very high HEI-2015 scores, so the interventional distribution at high x is predominantly supported by the high-income subpopulation, for whom the income-adjusted causal effect of further dietary improvement is most negative (greatest reduction in HbA1c risk). The Causal Forest's smoothed MCE(x) is instead approximately constant or mildly declining across the same range, reflecting its inability to extrapolate beyond the dense support of the joint (X, C_1) distribution.

The divergence between the two methods is therefore best understood as a difference in extrapolation philosophy rather than a difference in causal identification. The generic copula imposes a global parametric dependence structure that extrapolates into sparse regions via the

chosen copula family's tail behaviour. The Causal Forest's locally weighted estimator converges to a data-average within observed support and falls back on approximately flat predictions outside it. In a dataset with a wide and well-supported treatment distribution, the two approaches would be expected to agree more closely on the MCE(x) curve. In the present NHANES data, where diet quality and income are only weakly correlated ($\rho^{w}_{XC_1} = 0.152$), the extrapolation gap is modest in terms of support overlap but consequential in terms of sign, because the underlying dependence structure is strongly asymmetric in the tails.

The Gaussian copula fails directionally despite correct identification. At +0.0029 (325% divergence), the Gaussian copula uses the identical backdoor criterion as the generic copula but fails directionally because it imposes elliptical structure on C_{X,C_1} and C_{Y,C_1}. The BIC differences that drive this failure are 9.69 and 3.30 units for the two confounder pairs — individually modest improvements in goodness-of-fit that nonetheless suffice to correct a qualitative directional error. This finding illustrates that copula family selection for the confounder's relationships deserves as much attention in applied analysis as model selection for the treatment–outcome relationship itself.

The fully adjusted OLS is the most biased estimator. OLS with $C_1 + C_2$ (+0.0045, 455%) is the most divergent method, confirming that conditioning on the mediator and collider BMI amplifies rather than corrects bias (Section 7.5).

Core insight: flexible joint dependence, not flexible conditional means. The shared failure mode of OLS, DML, Gaussian copula, BART, and Causal Forest is that they either restrict the conditional mean form (OLS), estimate flexible conditional means without modelling the full joint dependence structure (DML, BART, Causal Forest), or impose incorrect elliptical dependence regardless of the data (Gaussian copula). Figure 14 makes this contrast precise for the Causal Forest: despite being the most flexible nonparametric benchmark, the forest's locally weighted estimator is unable to escape the positive observational gradient that persists within narrow income strata, while the copula's global integration over the non-elliptical income margins removes this gradient entirely. The generic copula succeeds because it combines the correct identification strategy (backdoor criterion via DAG), flexible bivariate dependence modelling in the confounder margins (BIC-selected non-elliptical families), and direct computation of the interventional distribution via Proposition 4.1. This is the unique advantage of the framework developed in Sections 2–6.

7.16 Discussion: Five Methodological Lessons

Lesson 1: Copula family selection in the confounder margins determines both the magnitude and the direction of the estimated MCE. The reversal from +0.0029 (Gaussian) to −0.0012 (generic copula) is produced by BIC improvements of 9.69 and 3.30 units in the two confounder-related pairs. These individually modest improvements in goodness-of-fit suffice to correct a directional error that persists across all seven alternative estimators. Copula family selection for confounder margins is therefore a substantive modelling decision with qualitative consequences for inference, not a technical detail. This finding establishes that copula family choice materially affects the qualitative direction of the income-adjusted estimate; it does not establish that the generic copula estimate is causally identified, given the acknowledged fragility to unmeasured confounders such as physical activity.

Lesson 2: Controlling for a confounder does not equal removing confounding bias. Income-adjusted OLS and the Gaussian copula both include income in their adjustment, yet both give the wrong sign. The mechanism of adjustment matters as much as the identity

of the variable being adjusted for. When the true joint distribution is non-Gaussian, imposing Gaussian structure on the conditioning relationship changes the direction of the estimated effect rather than simply attenuating it.

Lesson 3: Algorithmic causal discovery requires sufficient association strength. The failure of PC, HC, and RESIT to orient any edges consistently — with Kendall's $\tau \in [-0.022, +0.046]$ across all pairs — illustrates that in large heterogeneous surveys where marginal associations are weak, automated discovery cannot substitute for domain knowledge. The appropriate response is not to abandon structural reasoning but to deploy it explicitly and test it through multiple validation channels (Vine-SEM, sensitivity analysis).

Lesson 4: Bad controls amplify rather than reduce bias. Adding BMI to the adjustment set inflates the estimated MCE by 179.7% relative to income-only adjustment, and by 455% in the OLS comparison. The structural position of a variable in the DAG — whether it lies upstream or downstream of the treatment, and whether it shares latent common causes with the outcome — must be assessed before any adjustment decision is made. A richer control set is not necessarily a safer one.

Lesson 5: Effect fragility after flexible adjustment is an informative result. The near-unity E-value (≈ 1.002) and the 0.1% causal variance share imply that even trivially small unmeasured confounding can explain away the estimated MCE. This fragility is a direct consequence of the weak conditional dependence between diet quality and HbA1c in this population ($|\tau_{XY}| \approx 0.033$) and the near-zero causal variance share, and it does not reflect a deficiency of the generic copula. Researchers should interpret such fragility as an honest quantification of what the chosen adjustment set can and cannot identify from observational data, rather than as a reason to prefer less flexible methods that return larger but less credible estimates.

7.17 Reproducibility Checklist

The following methodological choices are documented for full reproducibility.

Data source. NHANES 2017–2018 (`DEMO_J`, `GHB_J`, `BMX_J`, `DR1TOT_J`), accessed via the `nhanesA` R package. HEI-2015 scores computed via the `heiscore` R package (version 0.1.4) using the standard FPED-based algorithm for survey year 2017–2018.

Inclusion/exclusion criteria. Age 20–79 years; HbA1c $\in [4.6\%, 15.0\%]$; HEI-2015 $\in [0, 100]$; income-to-poverty ratio $\in (0, 5]$; BMI $\in [15, 60]$ kg/m^2; complete cases on all four analytic variables. Final analytic sample: $n = 3{,}255$.

Survey weights. Marginal CDFs and kernel densities use `WTMEC2YR`-weighted empirical probability integral transforms, implemented via `Hmisc::wtd.Ecdf()` with normalised weights $w_i = \texttt{WTMEC2YR}_i / \bar{\texttt{WTMEC2YR}}$.

Causal structure. Adjustment set $\{C_1\}$ (income-to-poverty ratio), justified by the backdoor criterion applied to the proposed DAG and confirmed by Vine-SEM validation. BMI excluded as mediator and potential collider.

Copula selection. `BiCopSelect` (`VineCopula` 2.6.1) with a set of 30 parametric copula families (including elliptical, Archimedean, BB families, and their principal rotations), using `selectioncrit = "BIC"` and maximum likelihood estimation.

Backdoor adjustment. Plug-in estimator of $F^{(X\downarrow)}_{Y|X}$ (Proposition 4.1), with h-functions from fitted bivariate copulas. Numerical integration uses a 200-point grid over HbA1c $\in [4.6\%, 15.0\%]$ via trapezoidal quadrature; doubling the grid density changed estimates by $< 0.5 \times 10^{-4}$. MCE computed as adaptive finite difference $[\mu(x+\delta) - \mu(x-\delta)]/(2\delta)$, $\delta = 0.5\% \times \text{range}(X)$.

Bootstrap. $B = 2{,}000$ PSU-stratified replicates. PSUs resampled with replacement within strata. `BiCopSelect` re-run inside every replicate to capture model-selection uncertainty.

Percentile-based 95% CIs.

Comparator methods.

- *OLS*: `lm()` with C_1 (and C_2) as covariate(s).
- *DML*: `DoubleMLPLR` (DoubleML 1.0.2), partially linear regression score, Random Forest nuisance (`regr.ranger`, 500 trees, unrestricted depth), five-fold cross-fitting, $n_{\text{rep}} = 3$.
- *BART*: `dbarts` 0.9-32, 400 trees, $k = 2$, 2,000 posterior draws; MCE as mean finite-difference derivative across allobservations.
- *Causal Forest*: `grf::causal_forest` (grf 2.5.0), 2,000 trees, honesty, all parameters tuned; ATE via overlap-weighted doubly-robust estimator.

E-values. $E = \widehat{\text{RR}} + \sqrt{\widehat{\text{RR}}(\widehat{\text{RR}} - 1)}$, where $\widehat{\text{RR}} \approx \exp(|\widehat{\text{MCE}}|/\hat{\sigma}_Y)$ (VanderWeele and Ding, 2017).

Software environment. R 4.5.2 on Windows 11 (x64). Key packages: `VineCopula` 2.6.1, `rvinecopulib` 0.7.3.1.0, `DoubleML` 1.0.2, `dbarts` 0.9-32, `grf` 2.5.0, `heiscore` 0.1.4, `survey` 4.4-8, `Hmisc` 5.2-5. Full session information is provided in the supplementary code.

Code repository. All replication materials are available in a publicly archived repository: `https://github.com/gabrielfferraz-bit/model_causal_copulas`. The repository contains the complete analysis pipeline, including numerical illustrations, NHANES data cleaning scripts, copula model selection using `BiCopSelect` (BIC), survey bootstrap procedures, and the code required to reproduce all figures and tables (Tables 11–29).

Data. Data are drawn from the NHANES 2017–2018 cycle, publicly available from the Centers for Disease Control and Prevention (CDC) at `https://www.cdc.gov/nchs/nhanes/`. No restricted-access data were used.

8 Discussion

9 Conclusion

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